

Forecasting the impact of systematic uncertainties on the dispersion measure redshift relation of fast radio bursts

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Abstract

Fast Radio Bursts (FRBs) are pulses of electromagnetic waves of extragalactic origin. As an FRB propagates through the Universe, its signal gets broadened by the free electrons in the intergalactic medium, leading to a dispersed signal. This dispersion measure (DM) is easily observable via the delayed arrival time which is directly proportional to the column density of free electrons, making the DM signal an excellent probe of baryons in the Universe. It is possible to model the DM signal by modelling the cosmological electron number density on large scales; nonetheless, the DM signal receives contribution from the galaxy that hosts the FRB source in addition to a contribution from the Milky Way (MW) galaxy, which poses as systematic uncertainties in a cosmological analysis. Furthermore, redshift measurements and the telescope's sensitivity also represent measurement uncertainty. Failing to account for these effects can lead to biases in the inferred cosmological parameters.

In this thesis, I explore how measurement and systematic uncertainties affect a cosmological analysis and suggest strategies to mitigate their impact for a survey with 10 000 detected FRBs. I use the halo model and the baryonification correction model to generate lognormal maps of electron density in bins of redshift in which the mock FRBs are placed. After that, I contaminate the catalog with one of the uncertainties and study its impact of this systematic on the DM-redshift relation. To make the simulation pipeline scalable, I use GPUs to calculate the covariance matrices of the mock data.

I find that ignoring redshift uncertainty leads to more than 1σ and up to 3.5σ deviation from the fiducial DM-redshift relation. In addition, ignoring host contribution leads to 1σ and up to 4.6σ deviation. Ignoring the MW contribution leads to up to 1σ deviation, however, by choosing the appropriate galactic cut, the bias can be reduced without adding a term to the covariance that accounts for the additional MW contribution. For FRBs with unlocalised hosts, I find that unlocalised FRBs at high redshifts lead to severe deviation as compared to unlocalised low redshifts for the same number of unlocalised sources. Finally, I find that it is possible to mitigate all these systematic uncertainties by modelling their impact on the level of the covariance and the FRB catalog.

In total, the efficient simulation pipeline presented in this thesis can be used for simulation-based inference where the systematic uncertainties can be easily put in the simulation to quantify the bias in estimated parameters.

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Introduction

The questions on how the universe came to be, why the universe is as it is, and what the universe is heading towards fall upon cosmology to answer. Cosmology is the study of the universe on large scales in an attempt to answer the latter questions [cite maybe Peebles](#). This pursuit started as data-starved research [cite](#) and transformed into a data-rich endeavour as a result of a plethora of imaging surveys such as the LSST Vera Rubin observatory, Euclid, and Nancy Roman [cite](#). The qualitative questions, with which this paragraph started, translate in the statistical sense to be what is the cosmological model that best describes the universe across all length scales and energy scales.

The first attempt to describe the universe as a whole came through the lens of the general theory of relativity where Robertson and Walker [cite](#) built on the work of Friedmann and Lemaître to describe the geometry of the universe through what is now known as the Friedmann-Lemaître-Robertson-Walker (FLRW) metric. We discuss this metric further in [2.2](#). This metric was largely motivated by the observations of Edwin Hubble who saw that galaxies further away from the Earth are moving faster than galaxies closer to Earth Hubble, [1929](#). Figure shows the plot from Hubble's work in 1929.

Further observations of the magnitude of distant galaxies have shown that the universe is expanding [cite](#). The general theory of relativity attributes this expansion to the energy density content of the universe. Therefore, the question of what is driving the expansion of the universe and what are the components that contribute to the energy budget of the universe represent a crucial question in cosmology for which multiple cosmological models have been suggested (Di Valentino et al., [2025](#)). In fact, the cosmological model has been motivated by both cosmological and astrophysical observations. Some of these observations are the measurement of galaxy rotation curves by Vera Rubin [cite](#) that hinted at an additional matter component that does not interact with light and the discovery of the cosmic microwave background (CMB) radiation that hinted at a hot and homogeneous early universe.[cite](#). Furthermore, early observations of clusters of galaxies have shown that the spatial distribution of galaxies deviates from randomness. This was shown by Peebles [cite](#). By calculating the two point correlation function, there was clearly a signal in the way that galaxies are distributed which points out to an underlying mechanism in the formation of these galaxies (Totsuji and Kihara, [1969](#)). The development of observational techniques and instrumentation allowed the launch of the Center for Astrophysics Redshift Survey (M. Davis, Huchra et al., [1982](#)) which showed that galaxy positions are correlated on large scales and that there is a structure in the universe (M. Davis and Peebles, [1983](#)).

In the early 1990s, cosmology experienced a paradigm shift with the launch of the Cosmic Background Explorer (COBE) satellite (Boggess et al., [1992](#)). The COBE mission provided the first detection of

primordial CMB anisotropies and further confirmed that the universe is flat (Smoot, 1999). In the context of the FLRW metric, the flatness of the universe requires a total sum of the relative density parameters to be 1 (within error bounds). Therefore, in addition to the matter component (dark and baryons), there should be a third component that can account for the missing energy density. Using supernova data from the Hubble Space Telescope (Bahcall and Spitzer, 1982), A. G. Riess, Filippenko et al. (1998) found strong evidence for an expanding universe with negative acceleration and supported the findings of the COBE mission for the existence of an additional density component different from matter which is driving the expansion of the universe. Hereafter, the term ‘dark energy’ was coined to refer to this component (Turner, 2001). With this, the Λ CDM model was born, and the efforts to obtain precise estimates of the cosmological parameters intensified (Bennett et al., 2003; Pope, 2004).

While early experiments relied on CMB, galaxy number counts, or supernova to infer cosmological parameters, gravitational lensing was in development as a tool to constrain the cosmological parameters. Gravitational lensing is the distortion of light geodesics by gravity when light travels from a distant source through the inhomogeneous universe towards an observer on the Earth (P. Schneider, Ehlers and Falco, 1992). On the level of a galaxy, gravitational lensing effects manifest themselves as a distortion of the shape of the galaxy and a change in the angular position with respect to a particular observer. Kaiser and Squires (1993) showed that these distortions could be used to obtain a map of the integrated mass density along the line of sight. The use of galaxy shape distortions for cosmology was a direct result of the Kaiser-Squires inversion formula where Seitz and P. Schneider (1996) realised that the expectation value of galaxy shape is an unbiased estimate of the shear in the weak field limit. In principle, it is possible to relate the power spectrum of galaxy shapes with the matter power spectrum, offering a concrete method of probing cosmology by studying the correlations between galaxy shapes (Bartelmann and P. Schneider, 2001). This new probe of the large scale structure is called cosmic shear and was used by Maoli et al. (2001) to obtain constraints on the cosmological parameters. These constraints were not as tight as the ones obtained by COBE, however, it paved the road towards using cosmic shear to constrain the cosmological model. The advantage of cosmic shear as an integrated quantity along the line of sight means that it is sensitive to structure growth in the late universe. Therefore, it represents a very good tool to constrain the equation of state for dark energy (Hu and Tegmark, 1999; Jain and Taylor, 2003).

After that, multiple galaxy surveys used cosmic shear for cosmological parameter inference, and numerous statistical tools were developed to deal with the increasing data size (P. Schneider, Van Waerbeke et al., 1998; P. Schneider, Eifler and Krause, 2010). Likewise, with increasing data size, the noise level in a cosmological signal increases and the need to mitigate the impact of the systematic and measurement uncertainties is imperative. In this work, we focus on tackling one of the most crucial systematics, namely baryonic feedback on the large scale structure.

Baryonic feedback are the set of baryonic physics processes that impact the distribution of matter (Rudd, Zentner and Kravtsov, 2008). These processes include, but are not limited to, supernova feedback, active galactic nuclei feedback, and star formation. By analysing hydrodynamical simulations, Daalen et al. (2011) realised that baryonic feedback affects the matter distribution in the universe at different scales, meaning that their impact should be modelled and included in a theory of structure formation. However, the theoretical formalism of structure growth, and therefore the matter power spectrum, assumes that structure growth is only driven by gravitational effects, ignoring baryonic feedback (Dodelson and Schmidt, 2026). Modelling baryonic feedback analytically is a cumbersome task, especially since the physics of galaxy formation and evolution is highly uncertain where different suites of hydrodynamical simulations have different feedback models, and each one of these models yields a totally different

description of the matter power spectrum suppression (Daalen et al., 2011). The mismodelling of the matter power spectrum tunnels down to the modelled cosmic shear signal and yields biased parameter estimates in a cosmological analysis.

On a conquest for precision cosmology, multiple statistical and modelling methods attempted to mitigate the impact of baryonic feedback on the large scale structure. For example, Krause et al. (2021) imposed scale cuts on the measured shear correlation function and only considered larger scales where baryonic feedback is minimal. However, the cosmic shear signal derives most of its sensitivity from non-linear scales, i.e. small scales (Piccirilli et al., 2025). Therefore, imposing a scale cut leads to loss of information. Other statistical techniques involve the use of principal component analysis to compress the parameter space of baryonic feedback parameters from hydrodynamical simulations and marginalising over these parameters in a cosmological analysis (Eifler et al., 2015). Another statistical method involves introducing an additive term to the covariance matrix called the theoretical error covariance (Moreira et al., 2021; Maraio, Hall and Taylor, 2025) which describes the variation of different baryonic feedback scenarios among all available hydrodynamical simulations. Analyses involving all the aforementioned results yield unbiased estimates of the cosmological parameters, however, it comes at a cost of introducing additional scatter due to the existing nuisance parameters. Finally, A. Schneider and Teyssier (2015) introduced the baryonification correction model, a model that parametrizes the displacement of dark matter particles in the existence of baryonic feedback. This model and its developments (A. Schneider, Kovač et al., 2025) were calibrated on hydrodynamical simulations and were able to emulate baryonic feedback in other hydrodynamical simulations that they were not calibrated on. Having this model, it is possible to obtain a prediction for any probe of baryons in the universe. In particular, this model was used to constrain baryonic feedback parameters with X-ray and the kinematic Sunyaev-Zel'dovich effect (Kovač et al., 2025) since it is possible to obtain a prediction of gas power spectra from which a simulated signal of the corresponding baryon tracer could be obtained. Though these probes are promising, they rely on some assumptions with respect to the structure of the galaxy clusters such as having hydrodynamic equilibrium and require many convolutions from a prediction of the gas power spectrum up to a predicted signal. Luckily, fast radio bursts provide a direct probe of the electron density in the universe through the information that their signal encodes.

Fast radio bursts (FRBs) are pulses of electromagnetic waves of unknown extragalactic origin. First discovered in 2007 (Lorimer et al., 2007), the most probable candidate for the sources of FRBs is the magnetar which is a star remnant that underwent a supernova. The signal of FRBs gets dispersed while travelling through the universe, meaning that the wave packets of lower frequency will arrive later than wave packets of higher frequency. This difference in arrival time is directly proportional to the integrated electron density along the line of sight where the dispersion measure signal could be easily simulated with minimal assumptions about the relationship of the dispersion measure and the electron number density in the universe. The grand benefit of the dispersion measure signal lies in its constraining power of baryonic feedback parameters since it is a direct tracer of the baryon distribution in the universe. Even though the source of these bursts is unknown (Petroff, J. W. Hessels and Lorimer, 2019), it is still possible to exploit the dispersion measure signal in a cosmological context, given that FRBs are hosted within galaxies whose redshift can be identified.

Simulated likelihood analyses showed that the cross-correlation of FRBs with other probes of structure yields better constraints on some cosmological parameters. In particular, calibrating baryonic feedback with FRBs and weak lensing led to one order of magnitude better constraints on the baryonic feedback and 50% improvements on constraints of the sum of neutrino masses $\sum m_\nu$ and the baryonic fraction in the universe Ω_b (Reischke, Neumann et al., 2023). Furthermore, combining FRBs with galaxy number

counts in a 3x2-point analysis provided better constraints on baryonic feedback parameters as compared to galaxy number counts (Sharma, Krause et al., 2026). In practice, we can aim to use FRBs alone as a probe to impose tight constraints on the baryonic feedback parameters and use these constraints as priors in a cosmological analysis involving probes of structure, allowing the information content of LSS data to constrain the cosmological parameters and delegating the task of constraining baryonic feedback to FRBs.

The promising potential of FRBs in constraining baryonic feedback could be hindered by multiple modelling and measurement uncertainties. First, FRBs are observed using radio telescopes with only limited follow-up optical observations that are necessary to calculate the redshifts of galaxies and in turn quantify the associated mean DM at that redshift. Second, measurements of these redshifts are highly uncertain especially for sources at high redshifts. Third, the measured dispersion measure signal receives three main contributions: a contribution from the Milky Way, a contribution from the host galaxy, and a contribution from the circumgalactic medium which is the quantity of cosmological interest (Reischke and Hagstotz, 2025). The models for both the Milky Way and the host galaxy contribution are both uncertain, leading to biases in the parameter constraints if not dealt with correctly. Fourth, the theoretical formalism of fast radio bursts assumes that the radio signal travels in a homogeneous universe, ignoring any gravitational lensing effect due to large scale structure. There are contradicting results on the impact of gravitational lensing; Alonso (2021) showed that first order contributions from general relativistic effects to the DM angular power spectrum are subdominant compared to the contribution from the electron column density, meanwhile, Takahashi (2024) showed that the second order contribution due to lensing could lead to a 1% deviation in the luminosity distance, resulting in an underestimation of the luminosity distance.

Therefore, the purpose of this thesis is to model, forecast, and explore mitigation strategies of systematic uncertainties and describe their importance with varying sample size. With upcoming radio surveys, a plethora of data will be available such that an accurate modelling of the dispersion measure signal is crucial to extract meaningful information from the signal (Luke Hoffmann et al., 2024). I scale up previous work by Konar et al. (2025) by using simulation-based inference (SBI) to infer the impact of systematic uncertainties on the baryonic parameter constraints using fast radio bursts only. Using SBI in this context has multiple benefits. First, it facilitates integrating different uncertainty effects on the level of the dispersion measure signal either on the level of the dispersion measure or the level of the redshift. Second, it removes the necessity to approximate the likelihood of the data given the parameters, unlike traditional Bayesian analyses. Results from ray tracing in hydrodynamical simulations (Konietzka et al., 2025) suggest that a Gaussian or lognormal approximation to the probability distribution function breaks down for large numbers of FRBs, and an analytic model of the dispersion measure is needed to simulate the PDF of FRBs (Torkamani et al., 2026). More details on SBI and the method used in this thesis can be found in 3.5

This thesis is structured as follows: I review the theoretical background in 2 including the general theory of relativity (2.1), the FLRW metric (2.2), the cosmological model (2.3), weak gravitational lensing (2.7), baryonic feedback on the large scale structure (2.8), the basics of fast radio bursts (2.9), and the systematics uncertainties in modelling the DM signal (2.10). I describe the methodology in 3 including modelling the electron power spectrum (3.1), simulating the dispersion measure signal (3.2), describing the systematic uncertainties (3.3), and explaining simulation-based inference (3.5). In 4, I describe the results. Finally, I discuss the results in 5

Theoretical Background

This chapter is mainly based on 2 textbooks and 1 lecture notes. The first book is *Modern Cosmology* by Dodelson and Schmidt (2026) and the second is *Spacetime and Geometry: An Introduction to General Relativity* by Carroll (2024). The first lecture notes comes from the *Advanced Topics in Cosmology* course given in the Summer semester 2024 by Prof. Dr. Cristiano Poriciani and Jun-Prof. Andrina Nicola. In this chapter, I will introduce the principles of cosmology, cosmic shear, and fast radio bursts (FRBs).

2.1 The general theory of relativity

The general theory of relativity introduces a geometric interpretation of gravity, resting on a set of geometric assumptions and principles evolved from the older classical Newtonian theory of gravity and relativity. The first of these is the **weak equivalence principle** which states that '**the inertial mass and gravitational mass of any object are equal**'. Einstein reformulated this statement to fit the language of the special theory of relativity by stating that '**In small enough regions of spacetime, the laws of physics reduce to those of special relativity; it is impossible to detect the existence of a gravitational field by means of local experiments.**' The consequences of this principle are that freely falling frame of reference cannot recognize that freely falling test particles in the vicinity of the frame are being accelerated (or under the influence of gravity). The consequences of this principle invites the formulation of a geometric theory of gravity where gravity manifests itself as curvature.

To have a mathematical theory where observation could be interpreted, Einstein used differentiable manifolds to describe the four dimensional spacetime. Differentiable manifolds allow a definition of physical quantities such as position and momentum along with a continuous derivative of these quantities on the manifolds. Having a differentiable quantity allows also to introduce conservation laws which are a direct results of symmetries in a particular problem (Noether, 1983). Therefore, to describe the motion of particles in a curved spacetime, the metric tensor is defined on a non-Riemannian manifold. The metric tensor is a solution for the Einstein field equations which are second order differential equations on $\mathbb{R}^{4 \times 4}$.

The metric tensor $g_{\nu\mu}$ (in Einstein index notation) is a 4×4 symmetric tensor that describes the physical length of an infinitesimal line given a particular system of coordinates such that

$$ds^2 = g_{\nu\mu} dx^\nu dx^\mu, \quad (2.1)$$

where $\{dx^\mu\}_{0,1,2,3}$ is a basis for the contangent bundle of the manifold. In classical terms, $\{dx^\mu\}_{0,1,2,3}$ is the differential along a coordinate axis $\{x^\mu\}_{0,1,2,3}$. For a generalized space time, the path of a free particle is described by the geodesic equation. If the coordinates of a particle in spacetime x^μ is parameterized with an affine parameter λ , then the coordinates $x^\mu(\lambda)$ is constrained by an ordinary differential equation, namely the geodesic equation given as:

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0, \quad (2.2)$$

where $\Gamma_{\rho\sigma}^\mu$ are the Christoffel symbols given in terms the metric tensor such that

$$\Gamma_{\mu\nu}^\sigma = \frac{1}{2} g^{\sigma\rho} \left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu} \right). \quad (2.3)$$

In the Newtonian limit, the gravitational potential Φ is related to the mass density ρ in the following equation

$$\nabla^2 \Phi = 4\pi G \rho, \quad (2.4)$$

with G being the gravitaional constant. To formulate a theory which reduces to Eq. 2.4, we define the energy-momentum tensor $T_{\mu\nu}$ which is a 4×4 symmetric tensor that describes the energy density and shear in the four-dimensional spacetime. Since the gravitational field manifest itself as curvature, the analogy to Eq. 2.4 motivate a proportionality relationship between the second derivative of the metric tensor $g_{\mu\nu}$ and the stress-energy tensor $T_{\mu\nu}$

$$\left[\nabla^2 g \right]_{\mu\nu} \propto T_{\mu\nu}. \quad (2.5)$$

A geometric quantity that is composed of the second derivatives of the metric tensor is the Ricci tensor $R_{\mu\nu}$ expressed in terms of the Christoffel symbols

$$R_{\mu\nu} = \partial_\lambda \Gamma_{\nu\mu}^\lambda - \partial_\nu \Gamma_{\lambda\mu}^\lambda + \Gamma_{\lambda\rho}^\lambda \Gamma_{\nu\mu}^\rho - \Gamma_{\nu\rho}^\lambda \Gamma_{\lambda\mu}^\rho. \quad (2.6)$$

Meanwhile, the Ricci scale is defined as the contraction of the Ricci tensor

$$R = R^\mu_\mu = g^{\mu\nu} R_{\mu\nu}. \quad (2.7)$$

Therefore, a possible relationship between the stress-energy tensor and the Ricci tensor is

$$R_{\mu\nu} = \kappa T_{\mu\nu}. \quad (2.8)$$

To find κ , the conservation of energy is invoked as

$$\nabla^\mu T_{\mu\nu} = 0, \quad (2.9)$$

where ∇^μ is the covariant derivative. Following Eq. 2.8 and using Eq. 2.6 and Eq. 2.7, the covariant

derivative of the Ricci tensor is

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla^\mu R g_{\mu\nu}. \quad (2.10)$$

Therefore, we can define another tensor $G_{\mu\nu}$ with a vanishing covariant derivative

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \quad (2.11)$$

$G_{\mu\nu}$ is called the Einstein tensor, and the Einstein's field equation regardless

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (2.12)$$

where I used $\kappa = 8\pi G/c^4$ which can be obtained by reducing Eq. 2.12 to the Newtonian limit in Eq. 2.4. Einstein modified Eq. 2.12 to account for a vacuum solution by adding a term to represent the vacuum energy $\Lambda g_{\mu\nu}$ where Λ is called the cosmological constant, and the final equation becomes

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (2.13)$$

Finally, the 4-velocity of a particle in general relativity is defined as

$$U^\mu = \frac{dx^\mu}{d\lambda}. \quad (2.14)$$

Therefore, the 4-velocity would follow the normalization

$$g_{\mu\nu} U^\mu U^\nu = -1, \quad (2.15)$$

and the 4-momentum for a particle of mass m is defined as

$$P^\mu = m U^\mu, \quad (2.16)$$

and the normalization of momentum is

$$g_{\mu\nu} P^\mu P^\nu = -m^2. \quad (2.17)$$

Notice that either of the previous normalization is invariant under coordinate transformation. We define the observed energy of a particle of 4-momentum P^μ with respect to an observer with 4-velocity U^μ such that it reduces to the special relativistic limit if the particle has a vanishing spatial velocity and the observer

$$E_{obs} = -g_{\mu\nu} U^\mu P^\nu. \quad (2.18)$$

The Einstein theory of general relativity can explain cosmological observations such as cosmological redshift and the expansion of the universe where the expansion is driven by the energy content in the universe. Though the FLRW metric (explained in 2.2) is phenomenologically motivated, the variation of the scale factor with conformal time a is described by the Einstein's field equation.

2.2 The Friedmann-Robertson-Lemaître-Walker metric

To describe the entire universe using one metric tensor requires educated simplifications and assumptions. The first guiding principle is the Copernican principle (Bondi and Roxburgh, 2010) which states that **'we do not occupy a special place in space'**. The Copernican principle motivates the cosmological principle which states that 'the universe is homogeneous and isotropic'. Early galaxy surveys have shown that the universe is only homogeneous on large scales, therefore, the full cosmological principle is stated as **'the universe is homogeneous and symmetric on large scales'**. An additional assumption that simplifies that narrows down the possible formula for a metric is Weyl's postulate which states that **'the world lines of the galaxies form a 3-sheaf of non-intersecting geodesics orthogonal to layers of spacelike hypersurfaces'**. Weyl's postulate, coupled with the cosmological principle, allows a better definition of where homogeneity and isotropy apply, namely that the universe is 'spatially homogeneous and isotropic'. Therefore, a maximally symmetric metric that follows both principles takes the form

$$ds^2 = -c^2 dt^2 + g_{ij} dx^i dx^j, \quad (2.19)$$

where t is cosmic time. A two-dimensional schematic diagram of this metric is shown in 2.1.

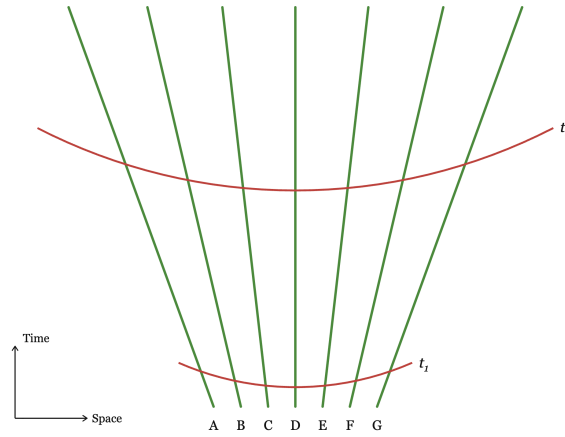


Figure 2.1: A schematic representation of spacetime that follows Weyl's postulate. The green lines labelled with letters are world lines, the path of freely falling objects in the spacetime (or geodesics). The redlines are spacelike hypersurfaces labelled with the same cosmic time t_i . These hypersurfaces are homogeneous and isotropic.

Using the tools of differential geometry, a maximally symmetric metric that also accounts for a homogeneous expansion of the universe is

$$ds^2 = -c^2 dt^2 + a^2(t) ds_3^2, \quad (2.20)$$

where $a(t)$ is the scale factor that couples the spatial part with time and the spatial part, c is the speed of light, ds_3^2 represent a volume element in a maximally symmetric space that takes the following form in hyperspherical coordinates

$$ds_3^2 = d\chi^2 + S_k(\chi)^2 (d\theta^2 + \sin^2 \theta d\phi^2), \quad (2.21)$$

the function $S_k(\chi)$ takes one of three forms by virtue of the number k

$$S_k(\chi) = \begin{cases} \frac{1}{\sqrt{k}} \sin(\sqrt{k}\chi) & \text{if } k > 0, \text{ for a closed universe} \\ \chi & \text{if } k = 0, \text{ for a flat universe} \\ \frac{1}{\sqrt{|k|}} \sinh(\sqrt{|k|}\chi) & \text{if } k < 0, \text{ for an open universe} \end{cases}. \quad (2.22)$$

The case when $k = 0$ represents a flat universe, where the spatial part reduces to the usual spherical coordinates. The flat FLRW metric reads

$$ds^2 = -c^2 dt^2 + a^2(t) \left[d\chi^2 + \chi^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right]. \quad (2.23)$$

The coordinate χ could be understood by considering two space like points in spacetime, one with $\chi = 0$ and another with $\chi = \chi_c$ leading To

$$s = a(t)\chi_c, \quad (2.24)$$

which means that the physical distance changes with time for a given χ_c . We call χ_c the comoving distance. Taking the derivative with respect to cosmic time in Eq. 2.24, we can retrieve the empirical law that Hubble (1929) discovered by recording the relationship between the distance to galaxy and the speed of the galaxy

$$\frac{ds}{dt} = V = \frac{\dot{a}}{a}s, \quad (2.25)$$

with $\dot{a} = da/dt$. We define the Hubble parameter at a cosmological time t such that

$$H(t) = \frac{\dot{a}}{a}. \quad (2.26)$$

Another distance indicator is gravitational redshift, similar to doppler redshift, gravitational redshift is the modulation of light frequency as light travels across the universe. If a fundamental observer at cosmic time t_0 emits a burst of light, then the energy of this light particle is given by Eq. 2.18. If another fundamental observer observes this light at a later time t , the ratio between the two energies is defined as

$$1 + z = \frac{E_e}{E_o}, \quad (2.27)$$

where E_e and E_o are the energy of light measured by the emitter and the observer respectively, and z is called the cosmological redshift. z is related to the wavelength of light in the limit where each of the emitter and the observer are fundamental observers, meaning that their 4-velocity is $U^\mu = (1, 0, 0, 0)$. Therefore, the light energy could be expressed using Planck's law (Planck, 1900), $E = h/\lambda$, with $h = 6.626 \times 10^{-34}$ J·s is Planck's constant, and λ is the wavelength of light. The final relationship describing the modulation in the observed wavelength of light is

$$1 + z = \frac{a(t_o)}{a(t_e)} = \frac{\lambda_o}{\lambda_e}, \quad (2.28)$$

with λ_e and λ_o are the light wavelength measured by the emitter and the observer respectively. This relationship could be exploited by quantifying the modulation of the wavelength and its relationship to

the scale factor. In a nutshell, a light emitted at a scale factor $a(t_e)$ travels through an expanding universe until measured by another observer at time $t_o > t_e$ corresponding to $a(t_o) > a(t_e)$, therefore $\lambda_o > \lambda_e$, which explains why distant sources are redshifted. In practice, by measuring the spectrum of light of a distant galaxy and convolving it with a standard spectrum (the expected spectrum of galaxy in the nearby universe), it is possible to obtain the redshift of the distant galaxy. If we choose $a = 1$ in the current epoch, the relationship between redshift and the scale factor reduces to

$$a = \frac{1}{1+z}, \quad (2.29)$$

where $z = 0$ refers to the current epoch. Since light moves along a null geodesic, it is possible to obtain the comoving distance to an emitter given the redshift of the emitter. Starting from Eq. 2.23, and working in a coordinate system with the light travelling along $\theta = 0$ and $\phi = 0$, the null geodesic dictates that

$$cdt = a(t)d\chi, \quad (2.30)$$

which could be further arranged using Eq. 2.26 and Eq. 2.29 into

$$\chi = \int_1^{z_e} \frac{c}{H(z)} dz. \quad (2.31)$$

From the perspective of Eq. 2.29, the cosmological redshift can be used as proxy of time where larger value of redshift corresponds to smaller value of the scale factor a . Meanwhile from the perspective of Eq. 2.31, the cosmological redshift is proxy for distance where sources of larger redshift correspond to larger distance. In a cosmological galaxy survey, measuring distances from the Earth towards galaxies is not a trivial task, however, the cosmological redshift can be measured by measuring the deviation the electromagnetic spectrum of a galaxy from the expected spectrum of a galaxy at redshift $z = 0$ ¹.

Another distance scale of interest is the Hubble scale. The Hubble scale is the distance at which the recession velocity of a galaxy equals the speed of light. Using Hubble's law, the Hubble scale is given by

$$s_H = \frac{c}{H}. \quad (2.32)$$

The Hubble scale is a distance scale that is used to define the size of the observable universe. It has a significant consequences on the growth of perturbations in the universe, as will be discussed in 2.8.

Throughout this chapter, a question is lurking behind all the kinematics of the universe that the FLRW metric proposes: how does the scale factor a change with cosmic time? In principle, this requires using the Einstein's field equation Eq. 2.13 along with assumptions about the form of the stress energy tensor $T^{\mu\nu}$. Assuming that the energy density in the universe is that of a perfect fluid, solving the Einstein's field equation for a yields

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{\Lambda c^2}{3} \quad (2.33a)$$

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3} \rho + \frac{\Lambda c^2}{3} - \frac{kc^2}{a^2}, \quad (2.33b)$$

¹ Of course, the process of measuring redshifts is cumbersome in practice, and this statement is meant to explain the basic principle behind measuring the cosmological redshift.

with p being the isotropic pressure of the cosmic fluid and ρ being the density². These equations are known as the Friedmann equations, describing the evolution of the scale factor a with respect to cosmic time t . To solve these equations, a relationship between the pressure p and the energy density is needed, known as the equation of state. The possible components that contribute to the energy density ρ have been a process of trial and error starting de Sitter universe (Sitter, 1917) which is a universe with no matter up to the recently speculated model with dynamical dark energy (Adame et al., 2025). Eqs. 2.33 were derived with the assumption of a homogeneous and isotropic universe on all scales; this is not the case when studying the inhomogeneous universe with large-scale structure. Therefore, the physical quantities in Eqs. 2.33 are only a functions of time (the total density $\rho(t)$ and the isotropic pressure $p(t)$). In 2.3, I briefly present the case of inhomogeneous universe with perturbations

$$\frac{\rho(t, \mathbf{x}) - \bar{\rho}(t)}{\bar{\rho}(t)} \ll 1, \quad (2.34)$$

where $\bar{\rho}(t)$ is the average density at a cosmic time t , and $\rho(t, \mathbf{x})$ is the density at a comoving position $\mathbf{x}$ and cosmic time t . In this thesis, we consider the Λ CDM cosmological model and describe its components in 2.3.

2.3 The cosmological model

In this section, we consider the Λ CDM cosmological model and explain its components. As noted by Eqs. 2.33, the time evolution of the scale factor a depends on the density and its relation to the pressure, known as the equation of state. If we refer to the current day value of the Hubble parameter as H_0 and the current day value of the density ρ_0 , Eq. 2.33b can be written in the current epoch

$$1 = \frac{8\pi G}{3H_0^2} \rho_0 + \frac{\Lambda c^2}{3H_0^2} - \frac{kc^2}{H_0^2}. \quad (2.35)$$

Notice that the quantity $\frac{3H_0^2}{8\pi G}$ has the dimensionality of a density, and we call it the critical density ρ_{cr} . It is possible to express the density fractions of the different components in terms of the current day value of the component's density and the critical density.

$$\Omega_0 = \frac{\rho_0}{\rho_{cr}}, \quad (2.36a)$$

$$\Omega_\Lambda = \frac{\Lambda c^2}{3H_0^2}, \quad (2.36b)$$

$$\Omega_k = -\frac{kc^2}{H_0^2}. \quad (2.36c)$$

We list below the constituents of the Λ CDM model of cosmology. It is common to express the density of a constituent as a fraction of the critical density, such that the sum of these fractions equals 1. In

² Since energy and mass are equivalent up to the squared of the speed of light, we may refer to the different constituents of the universe as 'contributing to the energy budget of the universe'.

practice, the density fraction of a constituent X in the current epoch Ω_X is expressed as

$$\Omega_X = \frac{\rho_{0,X}}{\rho_{cr}}, \quad (2.37)$$

with $\rho_{0,X}$ being the density of a constituent X at the current epoch.

Measurements from the cosmic microwave background has shown that we live in a flat universe (Smoot, 1999), meaning that $\Omega_k = 0$ (within error bounds). **For the rest of this work, I assume a flat universe.** The two remaining components are the density fraction Ω_0 and the fraction Ω_Λ . The mechanism of how these components were formed into existence will be omitted from this thesis. It is worth noting that the Hubble parameter H_0 is conventionally given in term of a dimensionless number h such that

$$H_0 = 100 h \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (2.38)$$

and many other quantities are expressed in terms of h ; for example, the density ratios Ω_X could be given as $\Omega_X h^2$. Measurements of the CMB with the Planck satellite (Aghanim et al., 2020) reported a result of

$$H_0 = 67.66 \pm 0.42 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

meanwhile measurements using supernovae from the SH0ES collaboration (A. G. Riess, Yuan et al., 2022) report a value of

$$H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

The discrepancy of H_0 measurements between the early universe with the CMB and late universe measurements with supernovae is called the Hubble tension (Verde, Schöneberg and Gil-Marín, 2024).

The equation of state of each component in the Λ CDM model differs between the early universe with large energy scales and the late universe with very low energy scales. We list below each component and its role in the cosmic evolution of the universe. In general, the equation of state $p_X(\rho_X)$ expresses the relationship between pressure p_X and density ρ_X of a constituent X . Motivated by dimensional analysis³, we can conclude that the pressure and density are related

$$p_X = w \rho_X c^2. \quad (2.39)$$

The relationship between the scale factor a and the density ρ_X could be found by invoking the first law of thermodynamics and solving the following ordinary differential equation,

$$d(\rho_X c^2 a^3) = -p_X d(a^3). \quad (2.40)$$

In total, the Λ CDM universe contains the following constituents:

- **Radiation:** Radiation is mostly composed of freely propagating relativistic particles: photons and neutrinos. Photons have been propagating in the universe since recombination. Recombination is the epoch when free electrons combine with nuclei to form neutral atoms, after which the photon mean free path becomes effectively infinite and photons decouple from matter; these photons are observed today as the cosmic microwave background (CMB) radiation. CMB anisotropies contain

³ Dimensional analysis is the method of comparing the units of two physical quantities to conclude the relationship between them up to a proportionality constant.

a wealth of information about the density fractions of the other components of the Universe and its spatial curvature. The temperature of the photon background scales as $T_r \propto a^{-1}$. With equation of state

$$p_r = \frac{1}{3}\rho_r c^2, \quad (2.41)$$

and the radiation energy density scales as $\rho_r \propto a^{-4}$.

$$\rho_r = \rho_{r,0} a^{-4}. \quad (2.42)$$

In addition to photon, neutrinos contribute to the radiation budget in the universe. At early times, neutrinos are in equilibrium with other particles such as electrons and photons, however, as the universe expands and the temperature drops, electrons no longer interact with neutrinos, therefore, neutrinos decouple from the rest of the components in the universe in a process known as freeze-out. The density of neutrinos scales to the scale factor in a similar way to photons, however, the present day temperature of neutrinos is different from that of photons

- **Baryonic matter:** Baryonic matter is composed of the particles that make up the standard model of particle physics. The particles that make up this component interact with light and with itself through long range pressure. In the early universe, baryonic matter is coupled to photons through absorption, emission, and scattering processes. Due to the expansion of the universe, the absorption cross-section approaches zero, therefore, the photons emitted from recombining electrons with nuclei have less energy than needed to ionize a similar atom, leaving the photon free. The process of forming neutral atoms by combining free electrons with nuclei is called recombination. This process continues until all the atoms are neutral, and then, photons propagate freely in the universe, with a very low probability of interacting with Baryons. After an epoch of structure formation (with the universe being transparent), neutral gas clouds in dark matter halos collapse under gravity to form stars and trigger fusion reactions⁴. Photons emitted from stars reionize the medium around stars, and baryons interact again with photons (Draine, 2011). The fraction of baryonic matter in the universe is measured to be $\Omega_b h^2 = 0.0225 \pm 0.0003$. In the early universe, the equation of state of baryonic matter is derived with relativistic consideration, however, after recombination, the short range pressure of baryons is negligible and the equation of state for baryons is that of dark matter on large scales. On small scale, baryonic processes in the late universe are highly non-linear and cumbersome to model which impose a limitation on the modelling of the cosmic shear signal. **This is the exact problem that this thesis is trying to tackle.**
- **Dark matter:** Dark matter is matter that does not interact with light, however, its effect is observed gravitationally. Recent results from different cosmological probes including CMB, galaxy clustering and weak lensing favour a cold dark matter component, meaning that the temperature of dark matter is much less than its mass. As a consequence, dark matter could be modelled as a pressureless fluid on large scale (on small scales, the fluid approximation breaks down). Using this simple picture of cold dark matter (CDM), it is possible to solve the fluid equation of dark matter to linear order and obtain a good approximation of the linear density

⁴ A fusion reaction is a nuclear reaction when two low mass atoms combine to form a heavier atom, emitting an amount of energy.

evolution with cosmic time (I show this derivation below). The equation of state of dark matter, therefore, is simply $p = 0$. By solving Eq. 2.40 for dark matter, the density of dark matter depends on the scale factor a such that

$$\rho_c = \rho_{c,0} a^{-3}. \quad (2.43)$$

The present day value of the dark matter density fraction is measured to be $\Omega_m h^2 = 0.1431 \pm 0.0025$ by the Planck collaboration (Aghanim et al., 2020). Dark matter is the main driver of matter clustering in the universe since dark matter can form virialized region of space called dark matter halos which accretes baryons and leads to galaxy formation as a consequence (Wechsler and J. L. Tinker, 2018).

- **Dark energy:** dark energy is a term coined by Michael S. Turner (Turner, 2001) to describe the contribution to the energy budget that is not in a form of particles. From Eqs. 2.36, dark energy is the component with density fraction Ω_Λ . Early on, this component was considered to be smooth and homogeneous, meaning that its corresponding density $\rho_\Lambda = \Omega_\Lambda \rho_{cr}$ only varies with cosmic time. In its simplest form, the dark energy pressure is related to the dark energy density as follows

$$p_\Lambda = w_\Lambda \rho_\Lambda c^2, \quad (2.44)$$

with the simplest case $w_\Lambda = -1$. The equation of state of dark energy is highly debated with many modification suggested to explain tensions in cosmology such as the Hubble tension (Verde, Schöneberg and Gil-Marín, 2024). In this work, we assume an equation of state of dark energy with $w_\Lambda = -1$. The Planck collaboration (Aghanim et al., 2020) reported that $\Omega_\Lambda = 0.6889 \pm 0.0056$.

Figure 2.2 shows a schematic representation of the components discussed above. In general, the matter density fraction Ω_m is the sum of the baryonic matter density fraction Ω_b and the dark matter density fraction Ω_c . The total energy budget of the universe is the sum of the matter density fraction Ω_m , the dark energy density fraction Ω_Λ , and the radiation density fraction Ω_r . Since the radiation density fraction is negligible in the current epoch, we have $\Omega_m + \Omega_\Lambda \approx 1$. With that, the evolution of the expansion rate of the universe in terms of the scale factor a is given by

$$H^2(a) = H_0^2 \left[\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda \right], \quad (2.45)$$

where the different additive terms dominate the evolution at different epochs. In particular, dark energy is the dominant component in the current epoch. To constrain the equation of state of dark energy, cosmic shear stands as a powerful probe of late structure formation since it is an integrated quantity that includes significant contribution from the late evolution of the universe, and therefore sensitive to dark energy. The basic principles of cosmic shear and its relationship to the matter density field are explained in 2.7. This thesis focuses on using FRBs to constrain baryonic feedback on the large scale structure and therefore improve the modelling of cosmic shear on small scales, which in turn improves the constraining power of cosmic shear on dark energy.

So far, I have only introduced the universe as a homogeneous entity, ignoring the fact the universe is inhomogeneous on small scales. As a physical system, the initial conditions of the universe are scale-dependent Gaussian perturbations which grow to be highly non-linear (and non-Gaussian) large perturbations (large relative to the mean density of the universe). Since the initial perturbations were generated via a stochastic process, the resultant perturbations are called random fields. In particular, the

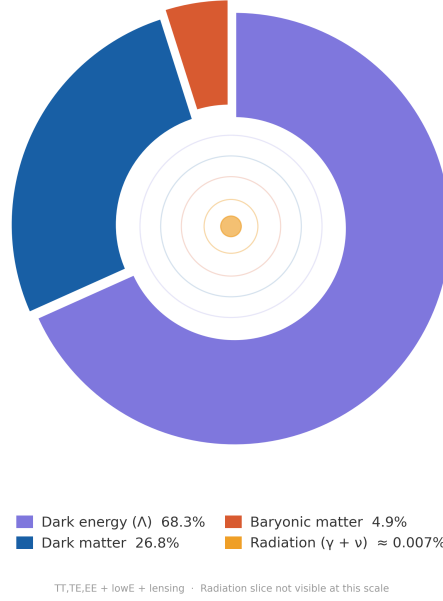


Figure 2.2: A schematic representation of density fractions of the different components in the Λ CDM model. The density fractions are measured by the Planck collaboration (Aghanim et al., 2020) using measurements of the CMB anisotropies.

matter density field $\rho_m(\mathbf{x}, t)$ is the random field that describes the matter density at a comoving position $\mathbf{x}$ and cosmic time t . For simplicity, the matter perturbations are often expressed in terms of the matter density contrast δ_m such that

$$\delta_m(\mathbf{x}, t) = \frac{\rho_m(\mathbf{x}, t) - \bar{\rho}_m(t)}{\bar{\rho}_m(t)}, \quad (2.46)$$

where $\bar{\rho}_m(t)$ is the average matter density at a cosmic time t . In addition, similar definitions apply to any other density field such as the baryonic density contrast or the electron density contrast, which will be important in the study of FRBs. By construction, any density contrast field is a random field with a mean of zero i.e. $\langle \delta_m \rangle = 0$. Higher order statistics of a random field also encode information about the random field. This thesis only uses the two-point statistics of the matter density contrast and its Fourier space counterpart, namely the power spectrum. We discuss these statistics below. Since dark matter is the dominant component of matter in the universe, it is the component responsible for the observed structure in the universe. In addition, the pressure induced by baryonic processes is negligible on large scales, meaning that the evolution of matter density contrast on large scales is well approximated by the evolution of dark matter density contrast. Therefore, the evolution of matter density contrast δ_m is well approximated by the evolution of dark matter density contrast δ_c . I present below the Einstein-Boltzmann equation that describes the phase-space evolution of a particle species in the universe.

2.4 The Boltzmann equation & the linear perturbation theory

This section is partly based on a review on the effective field theory of large scale structure (EFTofLSS) by Tobias Baldauf (Baldauf, 2020). The Boltzmann equation describes the evolution of the phase-space distribution function of a particle species in the universe. This function is labelled as $f(\mathbf{x}, \mathbf{p}, t)$, where $\mathbf{x}$ is the position, $\mathbf{p}$ is the momentum, and t is the cosmic time. Different observables could be obtained from the distribution function f including the density $\rho(\mathbf{x}, t)$ and the velocity field $\mathbf{v}(\mathbf{x}, t)$. The Boltzmann equation for a collisionless and pressureless dark matter

$$\begin{aligned} \frac{df}{dt} &= \frac{\partial f}{\partial t} + \frac{d\mathbf{x}}{dt} \cdot \frac{\partial f}{\partial \mathbf{x}} + \frac{d\mathbf{p}}{dt} \frac{\partial f}{\partial \mathbf{p}} \\ &= \frac{\partial f}{\partial t} + \frac{\mathbf{p}}{ma} \cdot \frac{\partial f}{\partial \mathbf{x}} - am \nabla \phi \cdot \frac{\partial f}{\partial \mathbf{p}} = 0, \end{aligned} \quad (2.47)$$

with m being the mass of the dark matter particle. If dark matter behaves as a fluid, the density and the bulk velocity of the fluid could be obtained by taking the first two moments of the distribution function f such that

$$\rho_m(\mathbf{x}, t) = \frac{m}{a^3} \int d^3p f(\mathbf{x}, \mathbf{p}, t), \quad (2.48)$$

$$v_i(\mathbf{x}, t) = \int d^3p \frac{p_i}{am} f(\mathbf{x}, \mathbf{p}, t) \bigg/ \int d^3p f(\mathbf{x}, \mathbf{p}, t), \quad (2.49)$$

$$\sigma_{ij}(\mathbf{x}, t) = \int d^3p \frac{p_i p_j}{am am} f(\mathbf{x}, \mathbf{p}, t) \bigg/ \int d^3p f(\mathbf{x}, \mathbf{p}, t) - v_i(\mathbf{x}) v_j(\mathbf{x}), \quad (2.50)$$

with $\sigma_{ij}(\mathbf{x}, t)$ representing the velocity dispersion tensor. The distribution function f has higher order moments, however, the anisotropic stress tensor is second order in momentum, therefore, it can be postulated that it is negligible at linear order and at early times where the velocity dispersion is small⁵. In this thesis, I am only interested in the evolution of the matter density contrast. Using Eq. 2.47, the evolution of the matter density contrast and the bulk velocity are governed by the continuity and the Euler equations. The equations up to linear order are as follows

$$\frac{\partial \delta_m}{\partial \tau} + \frac{1}{a} \nabla \cdot \mathbf{v} = 0, \quad (2.51a)$$

$$\frac{\partial \mathbf{v}}{\partial \tau} + \frac{1}{a} (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{a} \nabla \phi, \quad (2.51b)$$

with τ being the conformal time defined as $d\tau = dt/a$. Eqs. 2.51 show that at a linear order the matter density contrast is coupled to the velocity field, and the velocity field is coupled to the gravitational field ϕ . The growth of structure is, therefore, sourced by gravity.

To solve the equations in real space is a cumbersome task, therefore, we take the following steps in order to obtain the evolution of the matter density contrast.

- Apply the Fourier transform to Eqs. 2.51 to obtain the equations in Fourier space. The operator ∇

⁵ The usual practice in standard perturbation theory is to truncate the Boltzmann hierarchy even for higher order perturbations

reduces to a multiplication by $i\mathbf{k}$, with $\mathbf{k}$ being the Fourier space wavevector.

- The velocity field is a vector field, therefore, it can be decomposed into a curl-free component and a divergence-free component. The curl-free component is called the velocity divergence θ and is defined as $\theta = \nabla \cdot \mathbf{v}$, while the divergence-free component is called the vorticity ω and is defined as $\omega = \nabla \times \mathbf{v}$. At linear order, the vorticity decays with time, therefore, it can be neglected at linear order.
- The peculiar gravitational potential ϕ can be related to the matter density contrast by the relationship known as the Poisson equation such that

$$\nabla^2 \phi = \frac{3}{2} \Omega_m H_0^2 \frac{\delta_m}{a}. \quad (2.52)$$

With these steps, the equation that governs the evolution of the matter density field at a linear level is a second order ordinary differential equation with respect to the conformal time τ such that

$$\frac{d^2 \delta_m}{d\tau^2} + \mathcal{H} \frac{d\delta_m}{d\tau} - \frac{3}{2} \Omega_m H_0^2 \frac{\delta_m}{a} = 0, \quad (2.53)$$

with $\mathcal{H} = aH$ being the conformal Hubble parameter. A solution for the matter density contrast δ^6 at linear order in Fourier space as follows

$$\delta(\mathbf{k}, t) = D_+(\tau) \delta_{+,0}(\mathbf{k}), \quad (2.54)$$

where $\delta_{+,0}$ is the initial conditions of the matter density contrast, and $D_+(\tau)$ is the growth factor whose exact expression depends on the cosmological model considered. For an Einstein-de Sitter universe, which is a universe with $\Omega_m = 1$ and $\Omega_\Lambda = 0$, the growth factor is simply $D_+(\tau) = a(\tau)$. The importance of the result 2.54 lies in two observations: the first is that the growth of each Fourier mode of the matter density contrast is independent from other modes (modes of different length scales are not coupled), and the second is that the time evolution of the matter density contrast is separable from its spatial dependence.

The solution to Eqs. 2.51 is given assuming that the universe is way into matter dominated epoch, where the impact of radiation is negligible. In the early universe, the radiation component is dominant and initiates pressure that only acts on scales smaller than the horizon scale, defined in Eq. 2.32, since pressure is a causal process that can only propagate at a speed less than the speed of light. Therefore, $\delta_{+,0}(\mathbf{k})$ should encode all the physics of the early universe including the impact of radiation on the matter density contrast and the initial conditions. In practice, comparing the evolved density field with observation will not give a conclusive answer on whether the theory presented here is correct or not, since the current universe is one realisation of the stochastic process that generated the initial conditions, therefore, estimating the statistical properties of the observed matter density field and comparing it with the theoretical prediction is the way to test the theory. In the next section, I present the power spectrum as a statistic that encodes the statistical properties of the matter density field.

At early times, the amplitude of the matter density contrast is small, therefore, the linear solution in Eq. 2.54 is a good approximation to the evolution of the matter density contrast. As time goes on, the

⁶ δ refers to the matter density contrast δ_m hereafter, and the subscript is omitted for ease of writing.

amplitude of the matter density contrast grows and the linear approximation breaks down. Furthermore, for regions in the universe where the average density within this specific region is large compared to the average density of the universe, dark matter particles collapse under the influence of gravity and virialised regions of space form. These virialised regions are called dark matter halos. The appendix include an explanation of the toy model for the collapse. [add the number of the appendix](#). Baryons fall in the deep potential well of the dark matter halo and accretes to form galaxies, meaning that galaxies traces the position of dark matter halos, and in turn dark matter.

2.5 The 2-point correlation function and the power spectrum

Since $\delta_0(\mathbf{k})$ is a Gaussian random field, its statistical properties are completely determined by its mean and variance. Starting from $\delta_0(\mathbf{k})$, the matter density contrast in real space reads

$$\delta(\mathbf{x}) = \frac{1}{2\pi} \int \delta(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}} d\mathbf{k}. \quad (2.55)$$

The mean of $\delta_0(\mathbf{k})$ is zero by construction, therefore, the variance is the only quantity that encodes the statistical properties of $\delta_0(\mathbf{k})$. The variance of $\delta_0(\mathbf{k})$ is called the power spectrum and is defined as

$$\langle \delta_0(\mathbf{k}) \delta_0^*(\mathbf{k}') \rangle = (2\pi)^3 P(k) \delta_D(\mathbf{k} - \mathbf{k}'), \quad (2.56)$$

where δ_D is the Dirac delta function. Notice that the power spectrum is only a function of the magnitude of the Fourier space wavevector $k = |\mathbf{k}|$ due to the statistical isotropy of the universe. Fig. 2.3 shows the matter power spectrum obtain using the CLASS Boltzmann solver (Lesgourgues, 2011) with varying fraction of baryonic matter Ω_b and varying Hubble parameter H_0 at redshift $z = 0$.

The real space counterpart of the power spectrum is called the 2-point correlation function $\xi(r)$, and is defined assume

$$\xi(r) = \langle \delta(\mathbf{x}) \delta(\mathbf{x} - \mathbf{r}) \rangle \quad (2.57)$$

where ξ is a function of the comoving length scale r and is independent of direction. In practice, ξ represent the excess probability that two points separated with r have density contrast of $\delta(\mathbf{x})$ and $\delta(\mathbf{x} - \mathbf{r})$ respectively.

2.5.1 Dark matter tracers

Though the matter power spectrum is a powerful statistic that encodes the statistical properties of the matter density field, it is not directly observable. In order to relate the matter field to observables, it is important to understand the physical mechanisms by which the observables are related to the matter field. For galaxies, galaxy formation theories suggest that baryonic matter falls into the potential well of a dark matter halo and accretes to form a galaxy. Fig. 2.4 illustrates a snapshot of hydrodynamical simulation with a dark matter halo hosting a galaxy.

The detailed mechanism of how this happens is highly non-linear and complicated. In addition, running comprehensive galaxy formation simulations is computationally expensive in order to cover the large parameter space of galaxy formation physics (Somerville and Davé, 2015). Therefore, it is common to use a phenomenological model to relate the galaxy field to the matter field. The simplest model is the linear bias model which assumes that the galaxy number density contrast δ_g is linearly

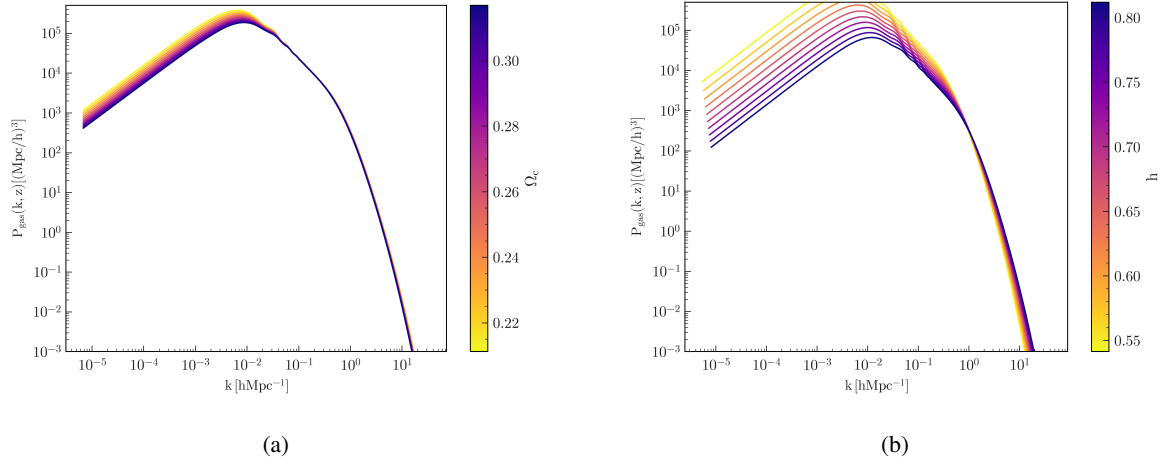


Figure 2.3: A plot of the matter power spectrum at redshift $z = 0$ with varying fraction of cold dark matter density Ω_c and varying Hubble parameter h . The power spectrum is obtained using the CLASS Boltzmann solver (Lesgourgues, 2011). Notice the the power spectrum peaks at a scale equivalent to the Hubble scale at the matter-radiation equality. Furthermore, the wiggles in the power spectrum are due to the sound waves that propagates in the baryon-photon fluid due the contrasting force of the fluids's pressure and gravity.

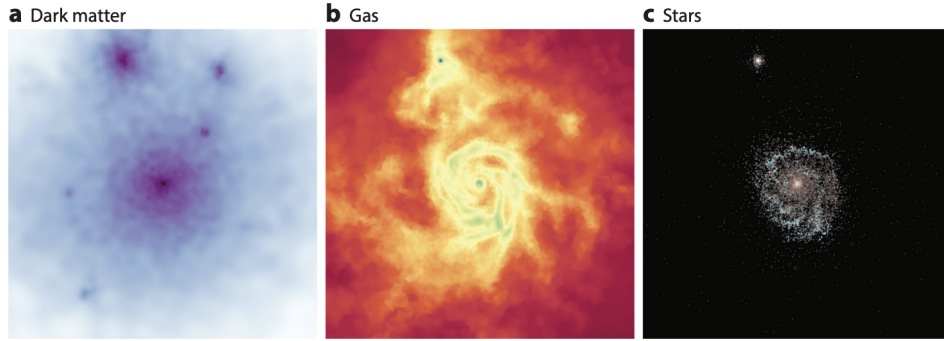


Figure 2.4: A 2D plot from the IllustrisTNG simulation (Pillepich et al., 2018), adopted from Crain and Voort (2023). It is a snapshot of hydrodynamical simulation showing how a 'dancer than average' region of dark matter hosts a galaxy (stars make up the 'skeleton' of a galaxy).

related to the matter density contrast δ such that

$$\delta_g(\mathbf{x}, t) = b(t)\delta(\mathbf{x}, t), \quad (2.58)$$

with b being the linear bias parameter. The linear bias model is a good approximation on large scales, however, on small scales, the relationship between the galaxy field and the matter field is highly non-linear and the linear bias model breaks down (Desjacques, Jeong and Schmidt, 2018). Notice that the galaxy number density contrast is only dependent on the local matter density field and does not depend on the environment surrounding the galaxy. This approximation breaks down on small scales where the environment of a galaxy can have a significant impact of the galaxy formation process. Galaxies are called 'biased tracers of dark matter' since they trace the underlying matter field. For any tracer u of dark

matter, I define the power spectrum of the density contrast of this tracer δ_u in a similar fashion to Eq. 2.56

$$\langle \delta_u(\mathbf{k}) \delta_u^*(\mathbf{k}') \rangle = (2\pi)^3 P_{uu}(k) \delta_D(\mathbf{k} - \mathbf{k}'). \quad (2.59)$$

It is possible to use the same formalism of galaxy bias to relate the matter density contrast with other tracers of dark matter such as dark matter halos. In practice, the bias of dark matter halos is related to the underlying matter field via the following relationship

$$\delta_h(\mathbf{x}, t) = b_h(t) \delta(\mathbf{x}, t), \quad (2.60)$$

with b_h being the dark matter halo bias parameter. Owing to Eq. 2.60, the halo power spectrum can be expressed in terms of the matter power spectrum defined in Eq. 2.56 such that

$$P_{hh}(k, t) = b_h^2(t) P_{mm}(k, t). \quad (2.61)$$

For dark matter halos, it is possible to go beyond a linear bias model by fitting an empirical model of halo bias to the results of an N-body gravitational simulation. In this thesis, I use the fitting formula of J. L. Tinker et al. (2010) to model the halo bias. In this recipe, the halo bias is a function of the halo mass and its redshift $b_h(M, z)$. As a consequence, the halo power spectrum is related to the matter power spectrum with the relationship

$$P_{hh}(M_1, M_2, k) = b(M_1) b(M_2) P_{mm}^{\text{lin}}(k) [1 + \beta^{\text{nl}}(M_1, M_2, k)]. \quad (2.62)$$

In the next section, I present the halo model, a non-perturbative approximation to obtain the non-linear power spectrum of matter or a tracer of matter. In particular, this thesis is concerned with the electron power spectrum in order to model the cosmological contribution of the dispersion measure of fast radio bursts.

2.5.2 The angular power spectrum

The definition of the power spectrum in Eq. 2.56 assumes that the observed field is sampled in three-dimensional space, therefore, the density field in real space can be written in using the Fourier space exponent as in Eq. 2.55 which represent a full orthogonal basis. When observing cosmological fields on the sky, a different orthogonal basis is needed to decompose the cosmological field since the values measured live on the unit sphere and not on a flat geometry. For this purpose, the a cosmological field f in the direction θ can be written in terms of the spherical harmonic functions $Y_\ell(\theta)$, where ℓ is the Fourier space counterpart of θ . Therefore, if f_ℓ is the spherical harmonic coefficients and $\ell = (\ell, m)$ ⁷ for a moment m and a multipole ℓ , the spherical harmonic transform and its inverse are given as

$$f(\theta) = \sum_{\ell m} Y_{\ell m}(\theta) f_{\ell m} \quad \leftrightarrow \quad f_{\ell m} = \int d\theta f(\theta) Y_{\ell m}^*(\theta). \quad (2.63)$$

The angular power spectrum C_ℓ is then defined in terms of the spherical harmonic coefficients

$$\langle \hat{f}_{\ell m} \hat{f}_{\ell' m'}^* \rangle = \delta_{\ell \ell'} \delta_{m m'} C_\ell, \quad (2.64)$$

⁷ more on the spherical harmonics can be found in the appendix

where δ here is the Kronecker delta. Notice how the power spectrum is not dependent on m by virtue of the cosmological principle.

The angular power spectrum of a cosmological field can be obtained from the power spectrum of the same cosmological field or an integral of it. I show how to get the angular power spectrum from the electron power spectrum in 2.9.

2.6 The halo model & the electron power spectrum

The details in this section are based on a review of the halo model by Asgari, Mead and Heymans (2023). The halo model assumes that matter in the universe are entirely contained in dark matter halos. Therefore, any cosmological field δ_u at a comoving coordinate $\mathbf{x}$ can be expressed in terms of the halo density profile $W_{u,i}$ such that⁸

$$\delta_u(\mathbf{x}) = \sum_i N_i W_{u,i}(|\mathbf{x} - \mathbf{x}_i|), \quad (2.65)$$

where the sum runs over all volume elements and $N_i = \{0, 1\}$ determines whether there is a halo centre in that volume element. $W_{u,i}$ could express the halo density profile for any component that traces halos such as electrons. In N-body simulations, the halo profile $W_{u,i}$ of a particular dark matter halo i differs from one halo to another as a function of the halo mass M . To express this scatter, the halo-mass distribution function is introduced. The halo mass distribution function $n(M)$ is defined as the number of dark matter halos with $M_i \in [M, M + dM]$ is $n(M)dM$. Averaging over all possible halo-masses, the expectation value of a field becomes

$$\langle \delta_u(\mathbf{x}) \rangle = \left\langle \sum_i N_i W_u(M_i, |\mathbf{x} - \mathbf{x}_i|) \right\rangle = \int_0^\infty dM n(M) \int d^3x' W_u(M, |\mathbf{x} - \mathbf{x}'|) = \int_0^\infty dM W_u(M) n(M), \quad (2.66)$$

where $W_u(M, \mathbf{x})$ can be split into two multiplicative component: $W_u(M)$ which encodes the amplitude of the halo profile and $U_u(M, \mathbf{x})$ which encodes information about the shape of the profile, and the mass-dependent halo profile can be written as

$$W_u(M, x) = W_u(M) U_u(M, x), \quad (2.67)$$

with $W_u(M)$ appearing in the last equality, and $U_u(M, x)$ is rigorously defined as

$$\int d^3x U_u(M, x) = 1. \quad (2.68)$$

The physical intuition behind the function $W_u(M, x)$ can be understood by going to Fourier space. First, Eq. 2.68 indicates that the Fourier space counterpart of the function $U_u(M, x)$, namely $\hat{U}_u(M, k)$ has the property $\hat{U}_u(M, k \rightarrow 0) = 1$ which implies that $\hat{W}_u(M, k \rightarrow 0) = \hat{W}_u(M)$. This result is understood that at large scales, dark matter halos act as point masses (similar to a Dirac-delta function) which translates to a constant in Fourier space.

With this formalism in place, the ingredients of the halo model allows an estimation for the power spectrum of a species u (in the case of this thesis, the halo model formalism will be applied to obtain a

⁸ notice that the time dependency is omitted for ease of writing.

prediction of the electron density in the universe). Using the definition of the power spectrum in Eq. 2.56, the power spectrum of the field δ_u as described in the halo model is

$$\langle \hat{\delta}_u(\mathbf{k}) \hat{\delta}_u^*(\mathbf{k}') \rangle = \left\langle \sum_{i,j} e^{-i\mathbf{k} \cdot \mathbf{x}_i} e^{i\mathbf{k}' \cdot \mathbf{x}_j} N_i N_j \hat{W}_{u,i}(M, k) \hat{W}_{u,j}(M, k') \right\rangle. \quad (2.69)$$

The sum can be further divided into two contributions:

- a contribution with $i = j$ which is the power resultant from the correlation of dark matter particles within one dark matter halo given by

$$P_{uu}^{1h}(k) = \int_0^\infty \hat{W}_u(M, k) \hat{W}_u(M, k) n(M) dM, \quad (2.70)$$

and this contribution is termed as **the one-halo term**.

- a contribution with $i \neq j$ which is the power resultant from the correlation of dark matter particles in two different halos given as

$$P_{uu}^{2h}(k) = \int_0^\infty \int_0^\infty dM_1 dM_2 P_{hh}(M_1, M_2, k) \hat{W}_u(M_1, k) \hat{W}_u(M_2, k) n(M_1) n(M_2), \quad (2.71)$$

and this contribution is termed as **the two-halo contribution**. P_{hh} is the halo power spectrum defined in Eq. 2.62.

The non-linear power spectrum within the halo model of a species u is given by

$$P_{uu}(k) = P_{uu}^{1h}(k) + P_{uu}^{2h}(k). \quad (2.72)$$

Fig. 2.5 shows an example of the non-linear power spectrum with the different contributions from the one-halo and the two-halo terms. Since P_{uu}^{1h} describes the **inter-halo** clustering of a species u , it dominates the power spectrum on small scales. Meanwhile,

Naturally, the halo-model is a recipe for obtaining an approximation of the non-linear power spectrum. By knowing the halo profile of a species u , it is possible to obtain the non-linear power spectrum of this species. In this thesis, we use the halo model to obtain a theoretical prediction of the electron power spectrum. Section 3.1 describes the procedure of obtaining the electron power spectrum using the baryonification correction model, and section 2.8 describes how electron profiles are obtained using a model for baryonic feedback.

Nonetheless, the halo model has its shortcoming (Mead et al., 2021). The power spectrum estimation via the halo model agrees with the power spectrum estimation from N-body gravitational simulations on very large and very small scales. Meanwhile, it fails on intermediate scales, the scales where the transition from the two- to the one-halo term. However, for the use case specific to this thesis, the halo model is useful since it is able to model the small scales which are mostly impacted by baryonic feedback accurately.

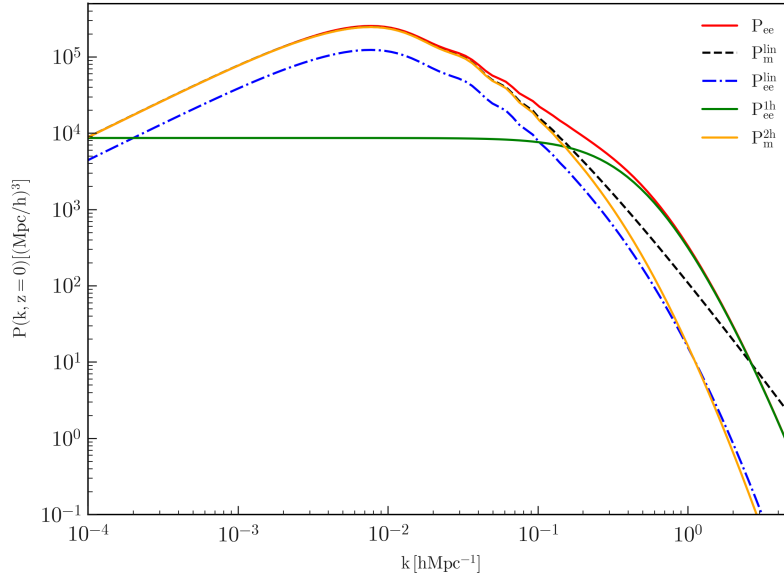


Figure 2.5: A plot showing the gas linear power spectrum, the one-halo, the two-halo, and the non-linear power spectrum, and the linear matter power spectrum. For small k , the two-halo term dominates the power spectrum, while for large k , the one-halo term dominates the power spectrum. In calculating the non-linear power spectrum, the one-halo term is suppressed on larger large scales to remove the contribution of shot noise arising from assuming that halos are infinitely small point on larger scales. The power spectrum is obtained using the halo model formalism described in this section, and the halo profile is obtained using the baryonification correction model described in section 2.8. I use the cosmological parameters listed in table 3.1. The gas power spectrum converges to the linear matter power spectrum on large scales is modified on small scales due to the impact of baryonic feedback.

2.7 Weak gravitational Lensing & cosmic shear

This section is based on the lecture notes of the *Gravitational Lensing* course taught by Prof. Dr. Peter Schneider at the University of Bonn.

Gravitational lensing, in the language of general relativity, is the modification of light (null) geodesics by virtue of the existence of massive objects. In the language of special relativity, it is the bending of light by virtue of the gravitational force of a massive object on a photon. As a result, the image of a bright object (galaxy or a quasar for example) gets magnified and sheared as compared to a situation where there are no lensing effects. In this thesis, I am only interested with the weak field limit of the gravitational lensing where only shear effects are observable. Fig. 2.6 shows a schematic representation of shearing and magnification effects.

In the weak field limit, the shearing effects of gravitational lensing are dominant over the magnification effects. Since the matter density contrast in the present epoch is of order $\delta \approx 1$, it is possible to invoke the weak field approximation to study the shearing effects of distant galaxies' images by the large scale structure. This effect is called **cosmic shear**.

To understand the lensing of a galaxy image, the shape of a galaxy, or the ellipticity, is represented as

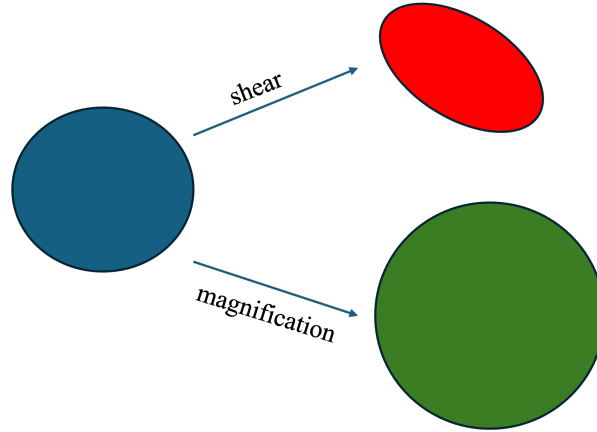


Figure 2.6: A schematic representation of the impact of gravitational lensing on a source with a circular shape (blue). Lensing has two main effect: shearing (red) and magnification (green). In the weak field limit, the magnification effects are negligible and shearing effects dominate.

a spin-2 field with two components⁹ on the unit sphere.

In this section, the symbol θ is used to indicate a solid angle where $\theta = \theta_1 + i\theta_2$ in the source plane. Fig. 2.7 shows how In practice, θ is related to a displacement vector ζ via the relation

$$\zeta = \chi\theta. \quad (2.73)$$

This is clear from Fig. 2.7

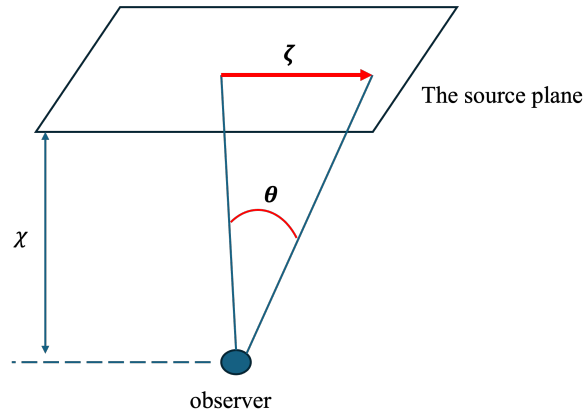


Figure 2.7: A schematic representation showing how the angle θ is related

The source galaxy shape α is observed with a lensed shape β via a transformation $\mathcal{A}$ such that

$$\beta = \mathcal{A}\alpha, \quad (2.74)$$

with $\mathcal{A}$ being a 2×2 matrix. $\mathcal{A}$ can be written in terms of two components: the shear γ and the

⁹ The image of a galaxy has an elliptical shape, therefore, it is possible to describe the shape with two principal components representing the minor axis and the major axis.

convergence κ such that

$$\mathcal{A} = \begin{pmatrix} 1 - \kappa - \gamma_1 & -\gamma_2 \\ -\gamma_2 & 1 - \kappa + \gamma_1 \end{pmatrix}, \quad (2.75)$$

where $\gamma = \gamma_1 + i\gamma_2$ in the the plain of complex numbers. Each of the convergence and the shear has both a geometric and a physical meaning. The trace of the $\mathcal{A}$ is $\text{tr}(\mathcal{A}) = 2(1 - \kappa)$, meaning that the magnification effects are controlled by the quatity κ . Meanwhile, γ is the qunatity that controls the shearing of an image. In this formalism, THe convergence κ is actullay a field, since it is related to the matter densitiy contrast δ via the relationship

$$\begin{aligned} \kappa(\boldsymbol{\theta}) &= \int d\chi p_\chi(\chi) \kappa(\boldsymbol{\theta}, \chi) \\ &= \frac{3H_0^2 \Omega_m}{2c^2} \int_0^{\chi_h} d\chi g(\chi) f_K(\chi) \frac{\delta(f_K(\chi)\boldsymbol{\theta}, \chi)}{a(\chi)}, \end{aligned} \quad (2.76)$$

where χ is the comoving distance, $f_K(\chi) = \chi$ for a flat universe, and $g(\chi)$ is a geometric factor. Furthermore, $p_\chi(\chi)$ is the source distribution i.e. the probability of findingg a galaxy at cosmoving distance χ .

The shear in turn, is a spin-2 field which is related to the convergence via the relationship

$$\begin{aligned} \gamma(\boldsymbol{\theta}) &= \frac{1}{\pi} \int_{\mathbb{R}^2} d^2\theta' \mathcal{D}(\boldsymbol{\theta} - \boldsymbol{\theta}') \kappa(\boldsymbol{\theta}'), \quad \text{with} \\ \mathcal{D}(\boldsymbol{\theta}) &\equiv -\frac{\theta_1^2 - \theta_2^2 + 2i\theta_1\theta_2}{|\boldsymbol{\theta}|^4} = \frac{-1}{(\theta_1 - i\theta_2)^2}. \end{aligned} \quad (2.77)$$

As can be seen from Eq. 2.77, the shear is a convolution of κ with the kernel $\mathcal{D}$, or in other words $\mathcal{D}$ is the shear generated by a point mass (if κ is a delta function then the shear is equal to $\mathcal{D}$ after the integral). This relation can be inverted to find that

$$\begin{aligned} \kappa(\boldsymbol{\theta}) - \kappa_0 &= \frac{1}{\pi} \int_{\mathbb{R}^2} d^2\theta' \mathcal{D}^*(\boldsymbol{\theta} - \boldsymbol{\theta}') \gamma(\boldsymbol{\theta}') \\ &= \frac{1}{\pi} \int_{\mathbb{R}^2} d^2\theta' \text{Re} [\mathcal{D}^*(\boldsymbol{\theta} - \boldsymbol{\theta}') \gamma(\boldsymbol{\theta}')] , \end{aligned} \quad (2.78)$$

where κ_0 is a constant that arises since $\mathcal{D}$ is not defined at $\boldsymbol{\theta} = 0$. In simple words, shear and convergence are complementary quantities. Since the convergence is related to the matter density along a certain direction $\boldsymbol{\theta}$, so as the shear. Therefore, if the impact of shearing on galaxy images is studied, it is possible to constrain the matter density in the universe.

To relate the shear to the shape of galaxy there is one important assumptions. Galaxies in the universe are randomly oriented, however by virtue of the cosmological principles, the expectation value of galaxy shapes should be zero. If ϵ_s is the intrinsic shape of galaxy, therefore

$$\langle \epsilon_s \rangle = 0. \quad (2.79)$$

However, the observed shape of a galaxy ϵ is subject to lensing effects. In the weak field limit, the

expectation value of the observed galaxy shapes is an unbiased estimator of the shear, meaning

$$\langle \epsilon \rangle = \gamma. \quad (2.80)$$

This is an extremely important result which signals the possibility of constraining the matter distribution in the universe via cosmic shear. To use cosmic shear to infer the cosmological parameter, the relationship between the matter power spectrum and the convergence power spectrum is used. This relation is given in the Limber approximation

$$P_{\kappa\kappa}(\ell) = \frac{9H_0^4\Omega_m^2}{4c^4} \int_0^{\chi_h} d\chi \frac{g^2(\chi)}{a^2(\chi)} P\left(\frac{\ell}{f_K(\chi)}, \chi\right), \quad (2.81)$$

where $P = P(k, \chi)$ is the matter power spectrum at scale k and cosmological distance χ . Furthermore, the shear power spectrum and the convergence power spectrum are the same, and by virtue of Eq. 2.80, the same as the galaxy shapes power spectrum,

$$P_{\kappa\kappa} = P_{\gamma\gamma} = P_{\epsilon\epsilon}. \quad (2.82)$$

Eq. 2.81 shows that any approximation to obtain a theoretical prediction of the matter power spectrum $P(k, \chi)$ will also influence the shear power spectrum $P_{\gamma\gamma}$. One of the main assumptions in this calculation is that structure growth in the late universe is only gravitationally driven. However, baryonic feedback on the large scales structure can influence the distribution of matter in the universe on different scales (Daalen et al., 2011). In section 2.8, I discuss how baryonic feedback influences the power spectrum.

2.8 The impact of baryonic feedback on the large-scale structure

This section is a qualitative discussion of baryonic feedback on the large scales structure. It is mainly based on Daalen et al. (2011), one of the first papers to compare and contrast the impact of baryonic feedback on the matter power spectrum. Baryonic feedback are the set of baryonic physics processes that impact the distribution of matter (Rudd, Zentner and Kravtsov, 2008). The real impact of these processes on the matter distribution in the universe is realised by analyzing hydrodynamical simulations. I list below of the feedback mechanism.

- **Active galactic nuclei (AGN) feedback:** AGN feedback originates when gas accretes on a black hole at the center of galaxy and falls into the black hole (Ishibashi and Fabian, 2012). The blackhole then injects energy back into the gas that is accreting, warming up the gas and results in hot jets that can be observed up to scales of $\sim 1\text{Mpc}$. When gas gets pushed outside the perimeter of a galaxy, it creates a potential well that matter follows, reducing the clustering of matter inside a galaxy. Therefore, the impact of AGN feedback is manifested as suppression of power at scales of order $k \sim 10h\text{Mpc}^{-1}$ compared to a scenario where only gravity influences structure formation in the universe (Martin-Alvarez et al., 2025).
- **Supernova feedback:** supernova feedback is a sudden collapse large-mass stars under the influence of gravity, leading to the release of energies in form of photons and charged particles on the order

of $\sim 10^{51} \text{ erg}^{10}$ (S. Huang et al., 2020). The energy released warms up the gas within a galaxy and enriches the interstellar medium and the circumgalactic medium with heavy metals that formed throughout the life time of the star. The exact impact on the matter power spectrum is not known a priori since each scenario of supernova feedback leads to different results and affect a range of (small) scales, and there is not reason to believe one scenario over the other. Nonetheless, by studying the different possible scenario, the supernova feedback on cosmological scales leads to $\sim 1 - 10\%$ suppression in the matter power spectrum on scales between $k \sim 1 - 10 \text{ h.Mpc}^{-1}$ (Daalen et al., 2011).

- Star formation feedback: Due to multiple cooling mechanisms the gas in galaxy collapses under the influence of gravity to form stars. Star formation leads to increase of power up to $\sim 10\%$ on scales $k \gtrsim 10 \text{ h.Mpc}^{-1}$ (Daalen et al., 2011).

Fig. 2.8 shows a plot comparing the ratio of the power spectrum of dark matter only scenario P_{δ}^{DMO} to the matter power spectrum in different hydrodynamical simulations $P_{\delta}^{\text{hydro}}$. Each of the given hydrodynamical simulations has different implementation of small scale physics and is parameterized by a set of nuisance parameters of no particular cosmological. Using astrophysical observations, it is not possible to probe one of these scenarios over the other. Therefore, to mitigate the impact of baryonic feedback on the large scale structure, multiple statistical tools have been introduced, as discussed in the introduction. Other analytical approaches involved parametrizing baryonic feedback with analytical models calibrated on hydrodynamical simulations (Mead et al., 2021; A. Schneider and Teyssier, 2015). These models have two shortcomings. First, the hydrodynamical simulation used for calibration is not guaranteed to agree with the true scenario of baryonic feedback in the universe. Second, the parameters that govern baryonic feedback have uniform and wide prior range, meaning that the constraining power of cosmological data will be absorbed by these nuisance parameters, leaving the cosmological parameters with no constraining power (A. Schneider, Kovač et al., 2025).

To obtain tight priors on the baryonic feedback parameter, it is possible to use probes of gas distribution in the Universe which in turns traces the baryonic matter distribution. This can be done by observing the kinematic Sunyaev–Zeldovich (kSZ) effect from CMB maps (Sunyaev and Zeldovich, 1972; Liu et al., 2025) or the X-ray observation of galaxy clusters. The kSZ is the inverse scattering of CMB photons when interacting with free energetic free electrons due to their bulk motion. This effect is observed as a secondary anisotropy in the observed maps of CMB. It is possible to relate the observed anisotropy ΔT_{kSZ} due to kSZ with the electron number density along the line of sight

$$\frac{\Delta T_{\text{kSZ}}}{T_{\text{CMB}}} = \frac{\sigma_{\text{T}}}{c} \int_{\text{los}} dl e^{-\tau} n_e v_p, \quad (2.83)$$

where $T_{\text{CMB}} = 2.725 \text{ K}$ the blackbody temperature of the CMB, σ_{T} the Thomson cross-section, τ the optical depth, l is the co-moving distance along the line of sight, and v_p is the peculiar velocity of gas. Eq. 2.83 indicates that it is possible to probe the distribution of baryons in the universe by extracting the secondary anisotropies in the CMB that are caused by the kSZ effect. Combining the measurements of kSZ with constraints on the hot gas mass fraction from X-ray observations of galaxy clusters can impose constraints on the baryonic feedback parameters. These constraints can be used as priors in a cosmological analysis of cosmic shear data yielding tighter constraints on the cosmological parameter as compared to having wide priors on these parameters.

¹⁰ 1 Joule = 10^7 erg

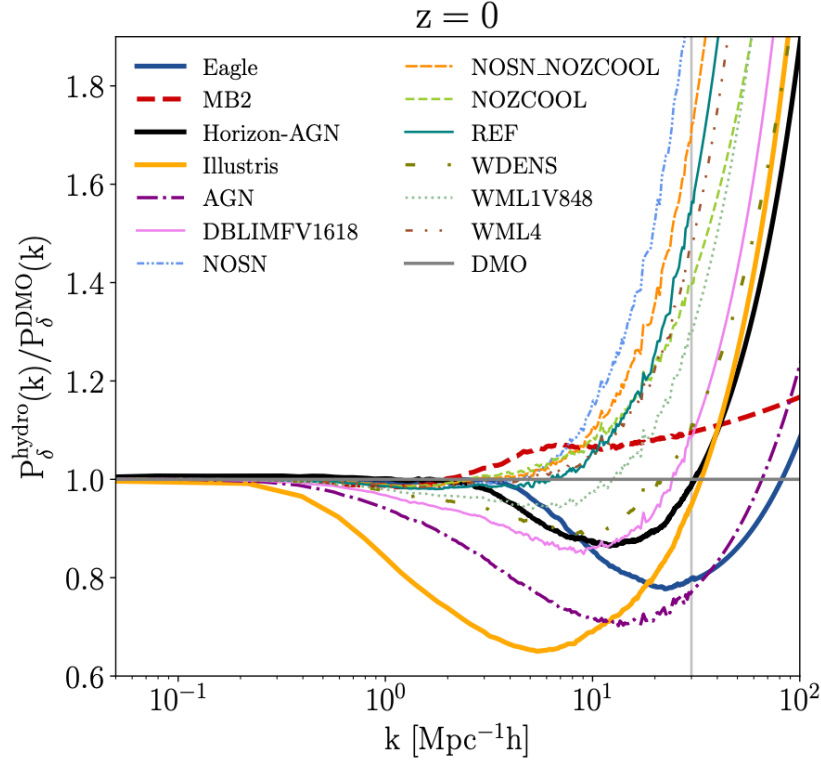


Figure 2.8: A plot adopted from H.-J. Huang et al. (2019) showing the suppression of the power spectrum for different hydrodynamical simulations with different models for baryonic feedback. For simulations where supernova feedback and AGN feedback are present, the power is suppressed on the scales where the gas pressure from these two feedback mechanisms influences matter distribution. The ratio rises for larger for smaller scales since gas cooling leads to star formation, boosting the power on small scales.

Though kSZ is a promising tool, extracting its signal from CMB observation requires assumptions about the temperature and ionization of the gas in galaxy clusters and the bulk velocity of the clusters (Carlstrom, Holder and Reese, 2002). Thankfully, the measured DM of FRBs is a direct probe of the electron number density in the universe and therefore baryonic feedback. In section 2.9, I describe the use of FRBs in cosmology in addition to the systematic and measurement uncertainties which impacts the modelling and measurements of the DM signal.

2.9 Fast radio bursts

This section is based on Alonso (2021), Reischke and Hagstotz (2023) and Petroff, J. W. Hessels and Lorimer (2019). For ease of writing, I consider that $\hbar = c = 1$ in this section. As mentioned in the introduction, the DM of FRBs is proportional to the difference in arrival time between high frequency wave and small frequency wave, and I show how this relationship arises below. Since the FRBs pass through the intergalactic medium (IGM), it interacts with the plasma in this medium. The plasma is made up out of charged electrons. Therefore, when a light wave of frequency ω and wave vector $\mathbf{k}$, the

plasma follows the dispersion relation

$$\omega^2 - |\mathbf{k}|^2 = \omega_e^2, \quad \omega_e \equiv \frac{4\pi e^2 n_e}{m_e}, \quad (2.84)$$

where n_e is the electron number density as measured by the plasma's frame. The velocity of the electromagnetic wave packet in the plasma's frame of reference v_g , is equivalent to the velocity of a massive particle with mass ω_e and energy ω

$$v_g = \sqrt{1 - \frac{\omega_e^2}{\omega^2}}. \quad (2.85)$$

Since the FRBs are in the radio regime, the ratio $\frac{\omega_e^2}{\omega^2} \ll 1$, and the group velocity can be approximated assume

$$v_g \approx \left(1 - \frac{1}{2} \frac{\omega_e^2}{\omega^2}\right). \quad (2.86)$$

To calculate the time delay due to scattering of the free electrons, I turn the problem into calculating the time delay between the arrival of a pulse with frequency ν_0 and a pulse of infinite frequency. I call these pulses a normal and infinite pulses respectively. Once I have an expression for the time delay, it possible to calculate the time delay between pulses of a high frequency ν_{high} and a low frequency ν_{low} . Theoretically, plasma does not affect the infinite pulse since the pulse has infinite momentum. Therefore, the group velocity of the infinite pulse is 1. I use the setup in Fig. 2.9 to find an expression of the time delay. The setup shows freely falling frame of references along the trajectory of the normal pulse. The time delay between the arrival of the normal pulse and the arrival of the infinite pulse (the two events are marked by an x on the yellow lines) can be given by

$$d\tau = \frac{dl}{v_g} - dl, \quad (2.87)$$

with τ being the proper time measured by the plasma (notice that I used the group velocity relative to the plasma) and dl the proper distance between the two stations. The corresponding time delay measured by the comoving observer dt is given by

$$dt = (1 + z)d\tau. \quad (2.88)$$

If the normal pulse has a 4-momentum defined as $P^\mu = dx^\mu/d\lambda$ (see Eq. 2.14), where λ is an affine parameter, the proper length can be written as $dl = P^\mu u_{e,\mu}$, where $u_{e,\mu}$ is the station's 4-velocity as measured by the observed. Thus to lowest order in ω_e/ω , the measured time dt is written

$$dt = \frac{(1 + z)\omega_e^2}{2\omega^2} \left(P^\mu u_{e,\mu}\right) d\lambda. \quad (2.89)$$

Finally, due to cosmological redshift, the station observes a frequency ω different from the what the observers detects, as shown by Eq. 2.28. We relate the two frequencies via the relation $\omega = 2\pi\nu_0(1 + z)$,

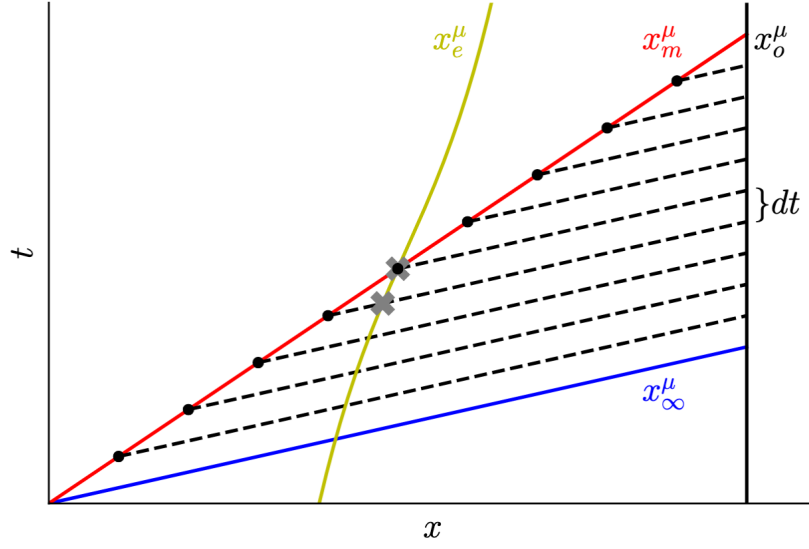


Figure 2.9: Spacetime diagram showing the evaluation of time delay between a pulse of frequency ν_0 (world line in red) and a pulse of infinite frequency (world lines in blue and dashed). The plasma particles follow world lines given in yellow color. The figure is adopted from Alonso (2021)

making the final expression for the time delay

$$\Delta t = \frac{e^2}{2\pi m_e \nu_o^2} \text{DM}, \quad (2.90)$$

where DM is the dispersion measure given as an integral

$$\text{DM} \equiv \int_{\lambda_e}^{\lambda_o} d\lambda \left(n_e \frac{P^\mu u_{e,\mu}}{1+z} \right) \Big|_\lambda, \quad (2.91)$$

where the subscript λ is a reminder that the integrand is evaluated along the pulse's trajectory. In a realistic situation, the plasma has a peculiar velocity and the normal pulse does not follow a null geodesic. I consider a situation where the stations are comoving observer with 4-velocity $u_e^\mu = (1, 0, 0, 0)$, and the pulse moves along a null geodesic in an unperturbed universe with momentum $P^\mu = (A/a, A/a^2, 0, 0)$ ¹¹, where A is a constant. The inner product of momentum and velocity reduces to

$$P^\mu u_{e,\mu} = g_{tt} (A/a)(1) = -A/a. \quad (2.92)$$

¹¹ I omit the proof to obtain this results. The proof can be done by solving the geodesic equation (Eq. 2.2) and setting the coordinate origin such that the photon follows a geodesic with $\phi = 0$ and $\theta = 0$.

Since λ parametrizes the geodesic of the pulse, it is possible to change variables in Eq. 2.91 using the expression for the 4-momentum where $dt/d\lambda = A/a$. The integral becomes

$$DM \equiv \int_{t_o}^{t_e} dt a n_e, \quad (2.93)$$

where I have changed the integral pounds. Remeber that n_e is the physical density as measured by the plasma. It can be written in terms of the comoving density as $n_e = n_{e,c} a^{-3}$. Therefore, the correponding comoving density scales with Using the relationship between cosmic time t and the sacle factor a as in Eq. 2.26 and the relationship between the a and z in Eq. 2.29, the integral can be written as

$$DM \equiv \int_0^{z_e} dz' \frac{1+z'}{H(z')} n_{e,c}(z'). \quad (2.94)$$

Hereafter, I work with comiving qunatities and will label the comoving electorn density at a redshift z with n_e . Eq. 2.94 is the dispersion measure-redshift relation DM_z which I use in this thesis to study the impact of systematic uncertainties. Eq. 2.84 includes an expression of the comoving electron number density $n_{e,c}$ this quantity can be written in terms of the baryon density fraciton Ω_b given some consideration to the ratio of free electron in the universe that disperses the signal. In general, the expression should look like

$$n_e = \frac{\bar{\rho}_b}{m_p} F(z), \quad (2.95)$$

where $\bar{\rho}_b$ is the average of the baryonic density in the universe, m_p is the mass of proton, and $F(z)$ is the fraciotn of free electrons in the intergalactic medium. To find $F(z)$, I make the following considerations

- If the universe has only helium and hydrogen atom, then the fraction of free electrons in the universe is $\chi_e = Y_H X_{e,H}(z) + \frac{1}{2} Y_{He} X_{e,He}(z)$, with Y_H , Y_{He} , $X_{e,H}(z)$, and $X_{e,He}(z)$ the mass fraction of hydrogen, the mass fraction of helium, the ionized fraction of hydrogen, and the ionized fraction of helium respectively. I multiply Y_{He} with a factor 1/2 since the helium atom has two the Using big bang nucleosyntheis, I assume that molecules in the unvierse are in the form of either hydron or helium with the fraction of hydrogen $Y_H = 0.75$ and the fraction of helium is $Y_{He} = 0.25$.
- Hydrogn and helium are fully ionized for $z \lesssim 3$ (Meiksin, 2009), meaning that the ionized fraction of Hydrogen $X_{e,H}(z)$ and $X_{e,He}(z)$ are both equal to 1.
- The fraction of free electrons in the intergalactic medium f_{IGM} is constant for the range of redshifts considered in this problem [find a citations for this](#).

By dropping the redshift dependence on the ionized fraction, I get an expression of $F(z)$ that is constant in redshift for the range of redshifts considered in this thesis,

$$F(z) = f_{IGM} \left[Y_H X_{e,H} + \frac{1}{2} Y_{He} X_{e,He} \right], \quad (2.96)$$

Writing $\bar{\rho}_b = \Omega_b \rho_{cr}$, The final form of Eq. 2.94 is given by

$$DM_{LSS}(z) = \mathcal{A} \int_0^z \frac{1+z'}{E(z')} dz', \quad (2.97)$$

with $\mathcal{A} := \frac{3c\Omega_{\text{b}0}H_0}{8\pi Gm_{\text{p}}}\chi_{\text{e}}f_{\text{IGM}}$, and $E(z) = H(z)/H_0$. This is the expression for the DM that rises from free electrons that follows the large scales structure, hence, I will label the DM that rises from the cosmic electron density with DM_{LSS} hereafter. As seen from 2.97, DM_{LSS} is highly sensitive to the baryon density in the universe Ω_{b} and the Hubble constant H_0 .

Eq. 2.97 expresses the mean DM at a redshift z . For a perturbed universe, the electron number density field is also a function of the angular direction $\hat{\mathbf{x}}$. I defined the electron number density contrast $\delta_{\text{e}}(\hat{\mathbf{x}}, z)$ in terms of the mean electron density $n_{\text{e}}(z)$ and the electron density in a perturbed universe $n_{\text{e}}(\hat{\mathbf{x}}, z)$

$$\delta_{\text{e}}(\hat{\mathbf{x}}, z) = \frac{n_{\text{e}}(\hat{\mathbf{x}}, z) - n_{\text{e}}(z)}{n_{\text{e}}(z)}. \quad (2.98)$$

The corresponding DM_{LSS} is therefore

$$\text{DM}_{\text{LSS}}(\hat{\mathbf{x}}, z) = \mathcal{A} \int_0^z \frac{1+z'}{E(z')} (1 + \delta_{\text{e}}(\hat{\mathbf{x}}, z')) dz'. \quad (2.99)$$

It is clear that $\text{DM}_{\text{LSS}}(\hat{\mathbf{x}}, z)$ is a random field since it is related to $\delta_{\text{e}}(\hat{\mathbf{x}}, z')$ which in turn is related to the matter density contrast δ .

Using the formalism introduced in sections 2.5 and 2.6, it is possible to use the electron power spectrum P_{ee} to express the correlation between two measured DM_{LSS} at angular positions $(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_j)$ and at redshifts (z_i, z_j) , namely the covariance matrix. (Reischke and Hagstotz, 2023) showed that the cosmological covariance of DM_{LSS} is derived by first writing the correlator bracket

$$\text{cov}_{ij} = \langle \text{DM}_{\text{LSS}}(\hat{\mathbf{x}}_i, z_i) \text{DM}_{\text{LSS}}(\hat{\mathbf{x}}_j, z_j) \rangle - \text{DM}_{\text{LSS}}(z_i) \text{DM}_{\text{LSS}}(z_j), \quad (2.100)$$

Simplifying, the expression reduced to

$$\text{cov}_{ij} = \int_0^{z_i} dz'_i W_{\text{DM}}^i(z'_i) \int_0^{z_j} dz'_j W_{\text{DM}}^j(z'_j) \langle \delta_{\text{e}}(\hat{\mathbf{x}}_i, z'_i) \delta_{\text{e}}(\hat{\mathbf{x}}_j, z'_j) \rangle, \quad (2.101)$$

where I have defined the weight $W_{\text{DM}}(z)$ as

$$W_{\text{DM}}(z) = \begin{cases} \mathcal{A} \frac{1+z}{E(z)} & z < z_{\text{FRB}} \\ 0 & z > z_{\text{FRB}} \end{cases}, \quad (2.102)$$

with $z_{\text{FRB}} = z_i, z_j$ the redshift of an FRB. The electron correlator bracket in real space can be written in harmonic space such that

$$\langle \delta_{\text{e}}(\hat{\mathbf{x}}_i, z_i) \delta_{\text{e}}(\hat{\mathbf{x}}_j, z_j) \rangle = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k} \cdot (\mathbf{x}_i - \mathbf{x}_j)} P_{\text{ee}}(k, z_i, z_j), \quad (2.103)$$

where I define the power spectrum in Fourier space in similar fashion to 2.59 as

$$\langle \delta_{\text{e}}(\mathbf{k}, z_i) \delta_{\text{e}}^*(\mathbf{k}', z_j) \rangle = (2\pi)^3 P_{\text{ee}}(k, z_i, z_j) \delta_D(\mathbf{k} - \mathbf{k}'), \quad (2.104)$$

and $\mathbf{x}$ is the comoving spatial position and related to the comoving distance as $\mathbf{x} = (\chi \hat{\mathbf{x}}, \chi)$.

To perform the integral in 2.103, I work in spherical coordinate sin the k -space, and decompose the integral into the radiu k and a solid angle Ω_k . In addition, I expand the exponential $e^{ik \cdot (\mathbf{x}_i - \mathbf{x}_j)}$ into plane waves (As explained in the appendix)

$$\begin{aligned}
 & \langle \delta_e(\hat{\mathbf{x}}_i, z_i) \delta_e(\hat{\mathbf{x}}_j, z_j) \rangle \\
 &= \frac{2}{\pi} \int k^2 dk \int d\Omega_k P_{ee}(k, z_i, z_j) \\
 & \quad \times \sum_{\ell, \ell'} \sum_{m, m'} i^\ell (-i)^{\ell'} Y_{\ell m}(\hat{\mathbf{k}}) Y_{\ell m}^*(\hat{\mathbf{x}}_i) j_\ell(k\chi_i) Y_{\ell' m'}^*(\hat{\mathbf{k}}) Y_{\ell' m'}(\hat{\mathbf{x}}_j) j_{\ell'}(k\chi_j) \\
 &= \frac{2}{\pi} \int k^2 dk P_{ee}(k, z_i, z_j) \sum_{\ell} \sum_m Y_{\ell m}^*(\hat{\mathbf{x}}_i) j_\ell(k\chi_i) Y_{\ell m}(\hat{\mathbf{x}}_j) j_\ell(k\chi_j) \\
 &= \frac{1}{2\pi^2} \sum_{\ell} (2\ell + 1) \int k^2 dk P_{ee}(k, z_i, z_j) j_\ell(k\chi_i) j_\ell(k\chi_j) P_\ell(\cos \theta), \tag{2.105}
 \end{aligned}$$

with $Y_{\ell m}$ and j_ℓ being the spherical harmonic function and the the spherical Bessel function of the first kind respectively. To simplify the sum, I use the property

$$\sum_m Y_{\ell m}(\hat{\mathbf{x}}_i) Y_{\ell m}^*(\hat{\mathbf{x}}_j) = \frac{2\ell + 1}{4\pi} P_\ell(\cos \theta), \tag{2.106}$$

with $\cos \theta_{ij} = \hat{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_j$ and P_ℓ is the Legendre polynomial of degree ℓ (defined in the appendix). In addition, the power spectrum at two as a functon of two redshifts z_i and z_j can be replaced by its geometric mean such that $P_{ee}(k, z_i, z_j) = \sqrt{P_{ee}(k, z_i) P_{ee}(k, z_j)}$ (Kitching and Heavens, 2017). The final expression for the cosmological covariance of DM_{LSS} is

$$\begin{aligned}
 \text{cov}_{ij}(\cos \theta, z_i, z_j) &= \frac{1}{2\pi^2} \sum_{\ell} (2\ell + 1) P_\ell(\hat{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_j) \int k^2 dk \\
 & \quad \times \int_0^{z_i} dz'_i W_{\text{DM}}^i(z'_i) \sqrt{P_{ee}(k, z'_i)} j_\ell(k\chi_i) \\
 & \quad \times \int_0^{z_j} dz'_j W_{\text{DM}}^j(z'_j) \sqrt{P_{ee}(k, z'_j)} j_\ell(k\chi_j) \\
 &= \sum_{\ell} \frac{2\ell + 1}{4\pi} P_\ell(\cos \theta_{ij}) C_{ij}(\ell). \tag{2.107}
 \end{aligned}$$

The expression contains two integral where one integral goes over the sacle k and two integrals to include the DM_{LSS} contribution from redshift $z = 0$ up to the redshift of the FRB $z = z_i, z_j$. In this thesis, fast calculation of the covariance matrix is essential since I use a compression step to pass one simulation instance in the simulation based inference pipeline. Therefore, I use the Limber approximation (Limber, 1953) to accelerate the calculation of the covaraince. I show a comparison between the Limber

approximation and the non-Limber approximation in section 3.1.3. The Limber approximation assumes that the spherical bessel function j_ℓ can be effectively replaced by a Dirac-Delta distribution such that

$$j_\ell(k\chi) \rightarrow \sqrt{\frac{\pi}{2(\ell + \frac{1}{2})}} \delta_D((\ell + \frac{1}{2}) - k\chi). \quad (2.108)$$

The word *effectively* is important in this context since the j_ℓ by itself is **not** approximated by the expression in Eq. 2.108, however, the two resultant integrals are equivalent in some regimes of ℓ . Invoking, the approximation in the full expression of the covariance and changing the integration variable into the comoving radius χ , I obtain the final expression

$$C_{ij}(\ell) = \int_0^{\min(\chi_i, \chi_j)} \frac{d\chi}{\chi^2} \frac{dz}{d\chi} \frac{dz}{d\chi} W_{\text{DM}}^j(z(\chi)) W_{\text{DM}}^i(z(\chi)) P_{\text{ee}}\left(\frac{\ell + 1/2}{\chi}, z'(\chi)\right). \quad (2.109)$$

The choice of the Limber approximation is valid when the terms multiplied by j_ℓ are slowly changing relative to j_ℓ . The calculation via the Limber approximation agrees with the brute force calculation for large ℓ i.e the regime where j_ℓ oscillates very quickly. This is evident by noticing that error on the Limber approximation scales with $\mathcal{O}(\ell^{\frac{1}{2}})$.

Eq. 2.107 has another bottleneck. The sum over the Legendre polynomials is another computationally expensive step that scales as $\mathcal{O}(\ell)$. At small enough angular separations, it is possible to approximate the spherical coordinates as flat, in what is called the flat sky approximations (Kaiser, 1992). In this approximation, the relationship between the angular power spectrum is derived by taking the Fourier transform in two-dimensions of the field instead of the spherical harmonic transform. Starting from the last line of Eq. 2.107, the property in Eq. 2.106, and the definition of the angular power spectrum in Eq. 2.64, the covariance can be re-written as¹²

$$\text{cov}_{ij} = \langle \text{DM}(\hat{\mathbf{x}}_i, z_i) \text{DM}(\hat{\mathbf{x}}_j, z_j) \rangle, \quad (2.110)$$

where I used

$$\langle \text{DM}_{\ell m}(z_i) \text{DM}_{\ell' m'}(z_j) \rangle = \delta_{\ell \ell'} \delta_{m m'} C_{ij}(\ell), \quad (2.111)$$

and

$$\text{DM}_{\ell m}(z_i) = \sum_{\ell m} Y_{\ell m}(\hat{\mathbf{x}}_i) \text{DM}_{\ell m}(z_i). \quad (2.112)$$

To invoke the flat sky approximation, I invoke the Fourier transform of $\text{DM}(\hat{\mathbf{x}}_i, z_i)$ in Eq. 2.110 and assume that in the two-dimensional Fourier space

$$\langle \text{DM}(\ell, z_i) \text{DM}(\ell', z_j) \rangle = (2\pi)^2 \delta^D(\ell - \ell') C_{ij}(\ell), \quad (2.113)$$

I end up with the following expression

$$\text{cov}_{ij} = \int d\ell e^{-i\ell \cdot \theta_{ij}} C_{ij}(\ell), \quad (2.114)$$

with volume element can be written as $d\ell = \ell d\ell d\alpha$, and the integral can be performed in the angular

¹² I omit the subscript from DM_{LSS} hereafter, where the symbol DM refers to the cosmological contribution to the DM.

position of the Fourier space α . I use the properties of the Bessel function of the first kind

$$J_0(\ell\theta_{ij}) = \frac{1}{2\pi} \int_0^{2\pi} e^{i\ell\theta_{ij}\cos\alpha} d\alpha, \quad (2.115)$$

The covariance matrix in the flat sky approximation is

$$\text{cov}_{ij} = \int \frac{\ell d\ell}{2\pi} J_0(\ell\theta_{ij}) C_{ij}(\ell). \quad (2.116)$$

This result is very useful to further accelerate the calculation of the covariance matrix. Though only valid on small angular separation, this approximation is widely used for forecasting purposes. Simulating the dispersion to fulfill the purpose of this thesis will be described in section 3.2.

Now that an expression of the covariance is obtained, I focus on deriving an expression for the variance of the dispersion measure signal at a certain redshift z . The variance of the DM signal is defined as

$$\sigma^2 [\text{DM}(\hat{\mathbf{x}}, z)] = \langle \left((\text{DM}(\hat{\mathbf{x}}, z) - \text{DM}(z))^2 \right) \rangle. \quad (2.117)$$

The expression can be easily derived from Eq. 2.107 by setting $\hat{\mathbf{x}}_i = \hat{\mathbf{x}}_j$ and $z_i = z_j$ writing $C_{ij}(\ell) = C(\ell, z)$, yielding

$$\sigma^2 [\text{DM}(\hat{\mathbf{x}}, z)] = \sum_{\ell} \frac{2\ell + 1}{4\pi} C(\ell, z), \quad (2.118)$$

and if I use the flat sky approximation, the expression changes to

$$\sigma^2 [\text{DM}(\hat{\mathbf{x}}, z)] = \int \frac{\ell d\ell}{2\pi} C(\ell, z). \quad (2.119)$$

In total, the variance of the DM field is an important quantity for constraining baryonic feedback. As you can see from Eq. 2.97, the mean DM- z relation is strongly sensitive to the value of the baryonic density ratio Ω_b and the Hubble constant H_0 and dependent on the other density ratios through the rate of expansion $E(z)$. Therefore, the dependence on the baryonic feedback parameters is entirely encoded in $\sigma^2 [\text{DM}(\hat{\mathbf{x}}, z)]$ and the electron density contrast δ_e . I will explain how to model the electron power spectrum using an analytical model of the baryonic feedback in section 3.1.1. Fig. 2.10 shows a plot of the dispersion measure redshift relation for different values of Ω_b and different values for baryonic feedback parameters.

Figure 2.10: A plot showing the impact of varying Ω_b on the mean DM- z relation and the impact of varying $\log_{10} M_c$ one of the baryonic feedback parameters on the variance of the DM- z relation.

So far, I have only introduced the cosmological DM. In reality, the total measured DM_{tot} receives contributions from the Milkyway galaxy and the host galaxy contribution which are not included in Eq. 2.97. We label each of these contributions with DM_{MW} and DM_{host} respectively. We discuss these two contributions in section 2.10. The total DM measured as measured on Earth is

$$\text{DM}_{\text{tot}} = \text{DM}_{\text{LSS}} + \text{DM}_{\text{MW}} + \text{DM}_{\text{host}}. \quad (2.120)$$

describe how modelling the PDF is a hard thing to do To forward model observations, we convolve the $DM - z$ represented by $P(DM|z)$ with measurement and systematic uncertainties, in addition to other observational effects. A catalog of detected FRBs is expected to have observed dispersion measure DM_{tot}^i , galactic coordinates, and redshifts for a subset of and observations z^i . We assume that each detected FRB was emitted at redshift z^i where the difference between identified and unidentified host can be integrated in how $P(z^i)$ is modeled. Assume that we observe N_{FRB} fast radio bursts signal. We express the observations as the probability of detecting an FRB DM_{tot} given a set of estimated redshifts $\{z^i\}$, can be decomposed using the Bayes rule such that

$$P(DM_{\text{tot}}|\{z^i\}) \propto P(\{z^i\}|DM_{\text{tot}})P(DM_{\text{tot}}). \quad (2.121)$$

Furthermore, if we assume that each observation z_i is independent of other observations we may use the Bayes rule to decompose $P(\{z^i\}|DM_{\text{tot}})$ such that

$$P(\{z^i\}|DM_{\text{tot}}) = \prod_i P(z^i|DM_{\text{tot}}). \quad (2.122)$$

Furthermore, the individual terms in Eq. 2.122 could be derived using the joint probability distribution of the observed and the dispersion measure contribution from the large scale structure, namely DM_{tot} and DM such that

$$\begin{aligned} P(z^i|DM_{\text{tot}}) &\propto P(DM_{\text{tot}}|z^i)P(z^i) \\ &= P(z^i) \int P(DM_{\text{tot}}, DM|z^i) dDM \\ &= P(z^i) \int P(DM_{\text{tot}}|z^i, DM)P(DM|z^i) dDM. \end{aligned} \quad (2.123)$$

The final expression that relates the probability distribution function of an observation to other modelled terms is

$$P(DM_{\text{tot}}|\{z^i\}) \propto P(DM_{\text{tot}}) \prod_i P(z^i) \int P(DM_{\text{tot}}|z^i, DM)P(DM|z^i) dDM. \quad (2.124)$$

Furthermore, to model each component of the resulting expression, we start with $P(DM_{\text{tot}})$. There are three components that contribute additively to DM_{tot} : the large scale structure contribution DM_{LSS} , the Milkyway contribution DM_{MW} , and the host galaxy contribution DM_{host} , and the observed dispersion measure is given as

$$DM_{\text{tot}} = DM_{\text{LSS}} + DM_{\text{MW}} + DM_{\text{host}}. \quad (2.125)$$

For simplicity, we indicate to the contribution from the large-scale structure simply with DM which is the same in Equation Eq. 2.124. Consider the joint probability distribution $P(DM, DM_{\text{MW}}, DM_{\text{host}})$, this can be related to $P(DM_{\text{tot}}) = P(DM + DM_{\text{MW}} + DM_{\text{host}})$ via the constrained integral

$$\begin{aligned}
 P(DM_{\text{tot}}) &= P(DM + DM_{\text{MW}} + DM_{\text{host}}) \\
 &= \int_C P(DM, DM_{\text{MW}}, DM_{\text{host}}) dDM dDM_{\text{MW}} dDM_{\text{tot}} \\
 &= \int_C P(DM)P(DM_{\text{MW}})P(DM_{\text{host}}) dDM dDM_{\text{MW}} dDM_{\text{tot}},
 \end{aligned} \tag{2.126}$$

Where $C = \{DM, DM_{\text{MW}}, DM_{\text{host}}; DM_{\text{tot}} = DM + DM_{\text{MW}} + DM_{\text{host}}\}$. Similarly, I can model $P(DM_{\text{tot}}|z^i, DM)$ as

$$\begin{aligned}
 P(DM_{\text{tot}}|z^i, DM) &= \int_C P(DM|z^i, DM)P(DM_{\text{MW}}|z^i, DM) \\
 &\quad P(DM_{\text{host}}|z^i, DM) dDM dDM_{\text{MW}} dDM_{\text{tot}},
 \end{aligned} \tag{2.127}$$

Each of the terms making up the integral could be further reduced

$$P(DM|z^i, DM) \propto P(DM|z^i)P(DM) \tag{2.128}$$

$$P(DM_{\text{MW}}|z^i, DM) = P(DM_{\text{MW}}) \tag{2.129}$$

$$P(DM_{\text{host}}|z^i, DM) \propto P(DM_{\text{host}}|z^i)P(DM|z^i). \tag{2.130}$$

Keeping in mind, that DM_{host} that we use in Eq. 2.124 is flat since we do not have any information of the

discuss the motivation for the SBI method.

2.10 The systematic uncertainties

In this section, I describe the modelling of systematic uncertainties on the dispersion measure.

- **Milkyway contribution:** The Milkyway (MW) galaxy, just like any disc galaxy, has free electrons in the interstellar medium (ISM). As the FRB pulse pass through the interstellar medium, it gets further dispersed by the free electron in the interstellar medium, adding an additional contribution to the measured DM_{tot} . There are currently no analytical model that can accurately describe the electron number density in the MW, however I use the NPE2001 model which is an empirical model that is calibrated on quasars data, and which we will use in this thesis (Ocker and Cordes, 2024). Nonetheless, the model is highly uncertain near the galactic plane meaning that subtracting DM_{MW} obtained from this model from DM_{tot} leads to large measurement error if the FRB source is located near the galactic plane. Fig 2.11 shows a map of the MW contribution to DM as predicted by the NE2001 model.
- **Host galaxy contribution:** FRB picks up a contribution from the galaxy that hosts the FRB source, namely the host galaxy. To model the host galaxy contribution, N-body simulation provide the best information to date. Using the CAMELS simulation (Villaescusa-Navarro et al., 2021),

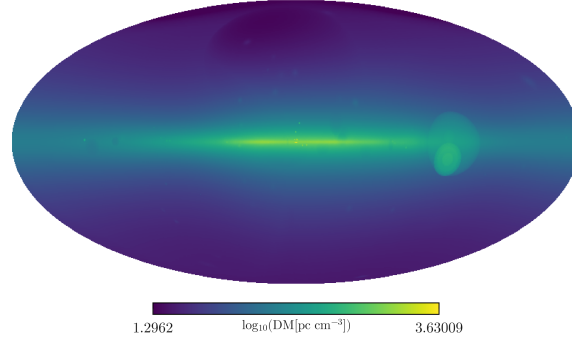


Figure 2.11: A map of DM_{MW} as predicted by the NPE2001 model. Notice how the contribution decreases with increasing latitude since the MW galaxy is a disk galaxy where stars are arranged in a disk-like shape.

Theis et al. (2024) showed that the variability of the host contribution with redshift is mainly driven by the variability of star formation with redshift while the placement of the FRB source in the host galaxy does not impact the mean of the host contribution across different redshifts¹³. Nonetheless, varying the feedback mechanism does impact the distribution of the host contribution. Therefore, Reischke, Kovač et al. (2024) provided an analytic model of the DM host contribution using the baryonification correction model (A. Schneider, Kovač et al., 2025; A. Schneider and Teyssier, 2015). This model was able to reproduce the probability distribution of the host contribution given redshift and halo mass as found from the CAMELS simulation suite. However, integrating this impact in a cosmological analysis will double the dimensionality of the parameter space since another copy of baryonic feedback parameters are needed to account for the host contribution beside the cosmological contribution of the dispersion measure DM_{LSS} . In addition, the analytical model requires simplistic assumption regarding the distribution of stars in halos.

In this work, I assume that the host contribution follows a lognormal distribution since it fits the analytical prediction from Reischke, Kovač et al. (2024) and the simulation results from Theis et al. (2024) for a sample size of 10^4 FRBs, which is what we consider in this thesis. In general, if the lognormal distribution is parameterized by μ and σ_{LN}^2 , then the probability distribution function of the host contribution is given as

$$P_{\text{host}}(DM_{\text{host}}; \mu, \sigma_{LN}) = \frac{1}{DM_{\text{host}} \sigma_{LN} \sqrt{2\pi}} \exp\left(-\frac{(\ln DM_{\text{host}} - \mu)^2}{2\sigma_{LN}^2}\right), \quad (2.131)$$

where $\exp(\mu)$ and $\exp(2\mu + \sigma_{LN}^2) [\exp(\sigma_{LN}^2) - 1]$ are the median and the variance of the distribution.

- **Redshift uncertainty:** Follow up optical observations of FRBs allows spectroscopic estimation of the redshifts of its host galaxy. The measurement uncertainty on the redshift can induce additional scatter in the estimated cosmological covariance matrix and modify the source redshift distribution. The latter effect is especially important when cross-correlating FRBs with cosmic

¹³ Sources that lie at the further end of the galaxy along the line of sight will pick up higher contribution as compared to a source at the near end of a galaxy, however, on average, the mean of the distribution does not change.

shear as performed by Reischke, Neumann et al. (2023). A faster way to localize FRBs is to use wide field spectroscopic or photometric surveys by deriving an astrometric solution, yielding a set of possible candidates of the host with a probability for each candidate (García et al., 2025). This method is, nonetheless, highly uncertain. Therefore, the redshift uncertainty is an important systematic uncertainty to mitigate in an analysis involving FRBs.

In this work, I assume that the measured that a redshift measurement z_i is sampled from a Gaussian distribution with mean z_i and standard deviation $\sigma_i = \eta z_i$, where η_i is a number between 0 and 1.

- **Selection effects:** The time resolution of radio telescope and its ability to detect a wide range of DM values is limited by the instrument. In other words, radio telescopes are only able to detect DM values up to a certain upper limit. This effect originates from dispersive smearing in the different radio frequency channels in the telescope, leading to an artificial broadening of the signal and low signal-to-noise ratio (Hoffmann et al., 2025). In general, this affects the normalization of the PDF of measured DM and not the shape. Therefore, the search efforts are restricted to a $DM_{\max}$, depending on the instrument. Table 2.1 shows some examples for radio surveys and their corresponding $DM_{\max}$. Since DM is an integrated value, a lower value of $DM_{\max}$ restricts the search to include low redshifts only.

Survey	$DM_{\max}$ (pc cm ⁻³)
DSA-110	1500
FAST/CRAFTS	5000
FAST/GPPS	3700

Table 2.1: Maximum dispersion measure search limits for FRB surveys, the values are adopted from Hoffmann et al. (2025)

The impact of selection was studied by Sharma, Ravi et al. (2026). In this work, the authors use a probabilistic model to describe the DM with all the possible selection effect. Beside the selection based on $DM_{\max}$, the authors consider the redshift source distribution by modeling with 3 free parameters and fitting this model to data along with the cosmological and baryonic feedback parameters. Furthermore, this work concludes that selection effects lead to 3σ bias in the inferred parameters for 10^4 FRBs.

In this thesis, I fix the redshift source distribution to coincide with that expected from the Square Kilometer Array Observatory (SKAO) (Braun et al., 2015), and impose a smooth cut on the measured DM by multiplying the PDF $P(DM^i; \{z_{\text{FRB}}^i\})$ with a sigmoid kernel (Shannon et al., 2018) parameterized by $DM_{\max}$ and the smoothness of the cut ϵ

$$\sigma(DM; DM_{\max}, \epsilon) = \frac{1}{1 + \exp\left(\frac{DM - DM_{\max}}{\epsilon}\right)} \quad (2.132)$$

Therefore, for a simulated catalog, each value of DM , is weighed with $\sigma(DM; DM_{\max}, \epsilon)$.

- **Unlocalised hosts:** So far, the number of detected FRBs is 3000¹⁴, and only 100 of them has identified host (Reischke and Hagstotz, 2025) i.e only the redshifts of 100 FRBs is known. This

¹⁴ CHIME/FRB Catalog 2.

poses as an obstacle to use all FRBs for cosmological inference since redshifts are important to calculate data covariance and constrain the DM- z relationship. Since follow-up optical observation to identify the host galaxy is both time and resource consuming, it is important to realise the best strategy to distribute these resources. In this thesis, I explore whether constraining the DM- z relation is possible if only a subset of the observed FRBs has an identified host. I explore the following cases

- A random subset of FRBs has no identified hosts.
- A subset of high redshift FRBs has no identified hosts.
- A subset of low redshift FRBs has no identified hosts.

Retrospectively, FRBs with high redshift have higher signal-to-noise since they pick up higher contribution from the electron density along their path to Earth, however, the measurement uncertainty of their redshift is high. Meanwhile, FRBs with low redshifts have lower signal-to-noise ratio since the path they travel is less as compared to high redshift sources, however, their measured redshifts have lower uncertainties. The results will show whether observing high- or low- redshift sources will be the best strategy to follow for follow-up optical observations.

Methodology

In this section, I explain the methodology used in this thesis. The end purpose is to show how the DM - z relation deviates from what is expected by contaminating mock FRB catalogs with systematic uncertainties through simulating FRB catalogs and studying the impact of systematic uncertainties when contaminating catalogs with these systematic uncertainties. Fig. 3.1 shows a flowchart of how to generate a mock catalog. I explain the steps in detail below. I perform the analysis assuming the detection of $N_{\text{FRBs}} = 10^4$. This is the expected number of detected FRBs per year by future FRB surveys such as the SKAO (Macquart et al., 2015). Future work should focus on scaling up to larger sample sizes since the impact of systematic uncertainties changes with sample size.

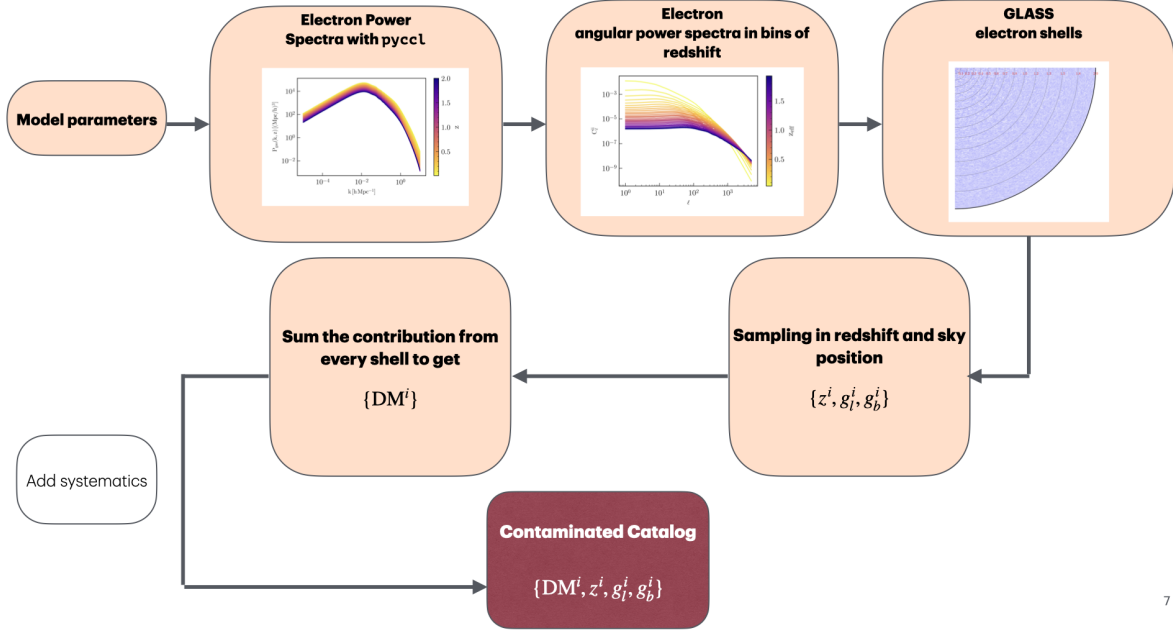
Table 3.1 shows the cosmological and baryonic feedback parameters which are used to generate mock catalogs. These parameters are used to obtain an expectation of the electron power spectrum.

To model the dispersion measure signal, I use the Generator for Large-Scale Structure (GLASS) (Tessore et al., 2023) package. I describe this package in further detail in section 3.2.1. Briefly, the GLASS package generates lognormal maps of a cosmological field starting from a given angular power spectrum. In contrast to the commonly used `healpy` package, it assumes that the cosmological field follows a lognormal distribution. Though this is not the case for a real cosmological field, the GLASS package preserves the two-point statistics of the cosmological field, which is enough for the sample size I am considering in this thesis. Higher-order statistics such as the bispectrum (3-point) and the trispectrum (4-point) require a larger sample size for detection. Using the electron power spectrum, I project the spectra on the unit sphere to obtain an expectation of the angular power spectra in bins of redshifts. After that, I feed the resultant power spectra to GLASS in order to obtain maps of electron density in bins of redshift. The next step consists of sampling randomly on the sky and according to the source redshift distribution in redshift then summing the contribution from different shells. In general, the simulation specifications are shown in Table 3.2.

In this work, I make considerable use of the following python packages: `numpy`, `scipy`, `matplotlib`, `jax`, `joblib`, `multiprocessing`, `pandas`, and `pytorch`.

3.1 Modelling the electron power spectrum

To model the electron power spectrum, I use the halo model (introduced in section 2.6), where I obtain halo profiles using the Baryonification Model (BCM) (A. Schneider and Teyssier, 2015; A. Schneider,



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Figure 3.1: A flowchart of the simulation pipeline introduced in this section. Starting from model parameters I obtain the non-linear electron power spectrum via the halo model. Then, I perform the Limber integral to obtain the electron angular power spectrum in each redshift bin. After that, I use the GLASS package to obtain lognormal maps of electrons in bins of redshifts. Then, I sample in angular positions and redshifts and sum the contribution of each shell to obtain an uncontaminated catalog. Eventually, the systematic uncertainties are then added.

Parameter	Description	Fiducial value	Prior
Ω_b	Baryon density parameter	0.0493	-
Ω_c	Cold dark matter density parameter	0.264	-
h	Dimensionless Hubble parameter	0.6766	-
n_s	Spectral index of primordial power spectrum	0.9665	-
σ_8	Amplitude of matter fluctuations	0.8102	-

Table 3.1: A table with the cosmological parameters used in this thesis. The values are adopted from Planck 2018 results (Aghanim et al., 2020).

Kovač et al., 2025).

3.1.1 The baryonification model

This section is based on the review of the baryonification correction model presented in Torkamani et al. (2026).

The purpose of the baryonification model (BFC) is to modify the profiles of dark matter halos from N-body gravitational simulations to those which involve baryonic feedback. The change is expressed as a shift in the position of dark matter particles which is parameterized by a set of baryonic feedback parameters. The displacement for dark matter (dm) and baryonic matter (bar) is performed along the

Property	Description	Value
$z_{\min}$	Minimum redshift	0.01
$z_{\max}$	Maximum redshift	2.0
n_{shells}	Number of shells	29
$\ell_{\max}$	Maximum multipole	4100
N_{side}	Map resolution (HEALPix)	4096
$b_h(M)$	Bias model as a function of halo mass M	J. L. Tinker et al. (2010)
$n(M)$	Halo mass function	J. Tinker et al. (2008)

Table 3.2: The specifications of FRB catalog simulations. HEALPix (Gorski et al., 2005) is an algorithm to pixellize an image with a certain resolution. The resolution parameter N_{side} can be understood as the inverse of the diameter of one pixel on the map $\theta_{\min} \approx 1/N_{\text{side}}$.

radius of a dark matter halo. In practice, the dm and the bar profiles are modified differently when baryonic feedback is taken into consideration. In this thesis, I only focus on the case of the modified bar profile. In a scenario without baryonic feedback, the baryons follow the distribution of dark matter. The baryon halo profile $\rho_{\text{bar},i}$ can be written in terms of the underlying dark matter profile ρ_{NFW} ¹ on small scales in addition to a large-scale contribution from the matter outside of the collapsed halo region which I label as ρ_{2h} ². In total, the final expression reads

$$\rho_{\text{bar},i}(r) = f_{\text{bar}} [\rho_{\text{NFW}}(r) + \rho_{2h}(r)], \quad (3.1)$$

with $f_{\text{bar}} = \Omega_b/\Omega_c$ and ρ_{2h} given in terms of the linear-order 2-point correlation function ξ_{lin}

$$\rho_{2h}(r) = [1 + b(\nu)\xi_{\text{lin}}(r)] \Omega_b \rho_{\text{crit}}, \quad (3.2)$$

with $b(\nu)$ the halo bias taken from J. L. Tinker et al. (2010). ν in this context refers to the ratio between the critical density contrast for spherical collapse at linear order $\delta_{c,\text{lin}}$ and the standard deviation of halo mass $\sigma(M)$.

Since baryonic feedback affects the distribution of matter on small scales, the component ρ_{NFW} ³ in the modified baryon profile should be modified. Empirically, this means that on small scales, the contribution to the baryon profile comes primarily from gas and stars within a dark matter halo. The modified component $\rho_{\text{bar},f}$ becomes

$$\rho_{\text{bar},f}(r) = \rho_{\text{gas}}(r) + \rho_{\text{star}}(r) + f_{\text{bar}}\rho_{2h}(r). \quad (3.3)$$

Each of the densities in the previous equation is parameterized with a set of cosmological and baryonic feedback parameters. For any species u , the cumulative mass up to a radius r is given by

$$M_u(R) = 4\pi \int_0^R r^2 \rho_u dr = f_u Y_u(R), \quad (3.4)$$

¹ ρ_{NFW} is an analytical expression of dark matter halo density profiles, predicted empirically by fitting the expression to N-body gravitational simulations (Navarro, Frenk and White, 1997)

² This contribution is labelled similarly to the 2-halo term of the power spectrum since it dominates the contribution on large scales similar to the 2-halo term of the power spectrum

³ ρ_{NFW}

where f_u is the abundance of a component (relative to the total matter) and Y_u is normalized such that $Y_u(\infty) = M_{\text{tot}}$. To quantify the deviation $d(r)$ in a particle's position between its initial (gravitational effects only) position $r_{\text{bar},i}$ to its final (gravitational effects and baryonic feedback) position $r_{\text{bar},f}$, the radius r_{bar} as a function of the cumulative mass M_{bar} is obtained

$$r_{\text{bar}} = \text{inv} [M_{\text{bar}}(r)], \quad (3.5)$$

and the resultant deviation is written as

$$d(r) = r_{\text{bar},f} - r_{\text{bar},i}. \quad (3.6)$$

The mapping from $r_{\text{bar},i}$ to $r_{\text{bar},f}$ is parameterized with the baryonic feedback parameters shown in Table 3.3. In addition, Fig. 3.2 shows how the BFC model changes the profiles of dark matter halos.

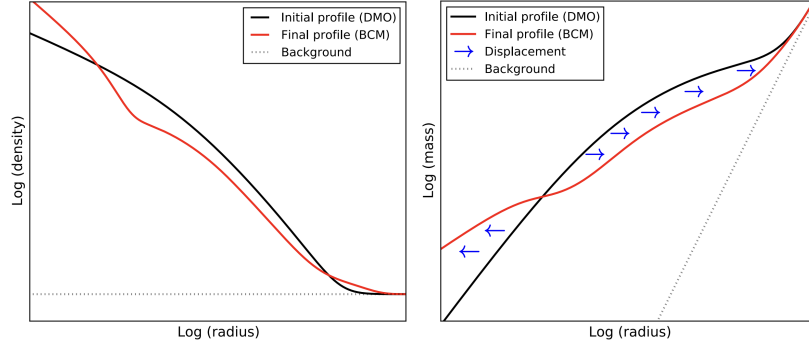


Figure 3.2: A plot showing how the BFC model modifies halo profiles (left) and the cumulative mass (right). At small radii, the profile shifts up to account for star formation processes while it shifts down at larger radii to account for AGN and supernova feedback. The figure is adopted from A. Schneider and Teyssier (2015).

In this thesis, I use the halo gas profiles predicted by the BFC model to obtain a theoretical expectation of the gas power spectrum, which in turn is the electron power spectrum. I derive the 1-halo and the 2-halo terms using the Core Cosmology Library which is discussed in the next section. As an example of how baryonic feedback modifies the power spectrum, Fig. ?? shows how the power spectrum changes with changing values of $\log_{10} M_c$, one of the baryonic feedback parameters.

3.1.2 The Core Cosmology Library (CCL)

The Core Cosmology Library (CCL) (Chisari et al., 2019) is a package that enables fast and efficient calculations of basic cosmological observables with high accuracy. CCL implements a Boltzmann solver in order to obtain a prediction of the linear power spectrum on a grid of redshifts z and scale k . It also provides a bicubic interpolator for accurate calculation of the power spectrum, starting from the initial grid points. In this thesis, I sample 30 grid points in z and 150 points in k . These are enough grid points to ensure sub-percent accuracy as compared to a larger number of grid points. However, since $\log P(k, z)$ is linearly related to $\log k$ for both large and small values of k , it is possible to extrapolate the power spectrum in logarithmic space beyond the ranges specified in Table 3.2. In addition, to obtain the non-linear power spectrum, CCL implements the halo model where the 1-halo and the 2-halo terms are

Parameter	Description	Fixed value
θ_{co}	Core radius of the gas profile relative to r_{200} (Eq. 2.6)	0.3
$\log_{10} M_{\text{c}}$	Pivot mass for the mass dependent slope of the gas profile (Eq. 2.6 & Eq. 2.7)	14
μ	Transition steepness of mass dependent gas parameter (Eq. 2.6 & Eq. 2.7)	0.3
δ	Characterizes the transition from inner to outer part of the gas profile (Eq. 2.6)	7.0
η	Total stellar fraction within a halo (Eq. 2.10)	0.32
$d\eta$	Fraction of central galaxies w.r.t. total stellar fraction (Eq. 2.10)	0.28
N_{star}	Stellar fraction amplitude (Eq. 2.10)	0.040
c_{iga}	Cold gas fraction (see Eq. 2.12)	0.0

Table 3.3: The parameters of the BFC model.

calculated using the halo profiles obtained from the BFC model. The halo mass function and the halo bias are taken from J. Tinker et al. (2008) and J. L. Tinker et al. (2010), respectively. I consider only a linear bias model where the term β from Eq. 2.62 is set to 0. I show a plot of the power spectrum for different redshifts in Fig. 3.3.

In this thesis, I use CCL to obtain a non-linear prediction of the electron power spectrum using the halo model where the integrals in Eqs. 2.70 and 2.71 are calculated using the halo profiles obtained from the BFC model. I use the resultant power spectrum to obtain an expectation of the electron angular power spectrum in bins of redshifts using the Limber approximation, which is described in the next section.

3.1.3 The electron angular power spectrum

To obtain the electron angular power spectra in bins of redshifts, I define the electron density contrast δ_e^i in a redshift bin i as

$$\delta_e^i(\hat{\mathbf{x}}) = \int W^i(z) \delta_e(z, \hat{\mathbf{x}}) dz, \quad (3.7)$$

with $W^i(z)$ the normalized weight function of the redshift bin i . The general form of the weight function is a top-hat function multiplied by a weight factor

$$W_i(z) = \begin{cases} f(z) & \text{if } z_{\text{min}}^i \leq z < z_{\text{max}}^i, \\ 0 & \text{otherwise,} \end{cases} \quad (3.8)$$

where z_{min}^i and z_{max}^i are the minimum and the maximum redshift of a shell i and the form of the function $f(z)$ is chosen arbitrarily so as to account for the expansion of the Universe. I discuss the choice of f in section 3.2.1. The electron angular power spectrum $C_{\delta_e \delta_e}^{(ij)}(\ell)$ between two redshift bins i and j is given

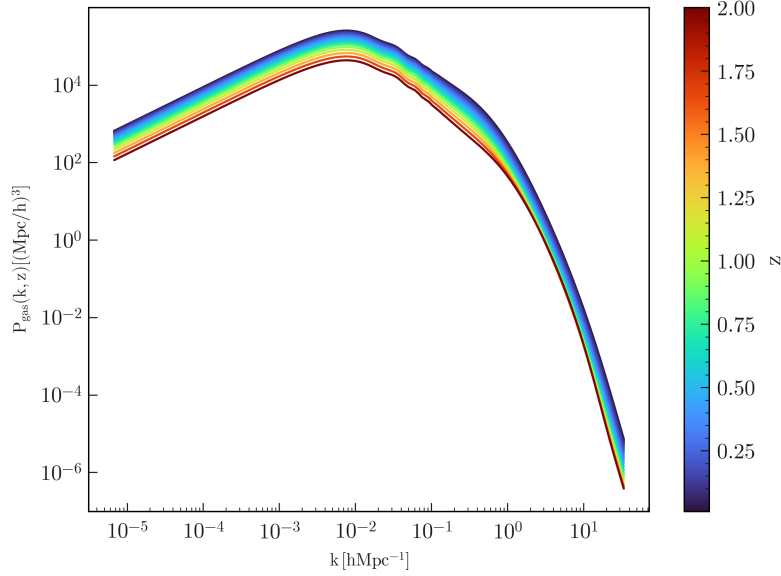


Figure 3.3: A plot showing the power spectrum in logarithmic space. The power spectrum is extrapolated in logarithmic space beyond the range of k and z specified in Table 3.2.

by

$$C_{\delta_e \delta_e}^{(ij)}(\ell) = \frac{2}{\pi} \int d\chi \frac{dz}{d\chi} \frac{dz}{d\chi} W^{(i)}(z(\chi)) \int d\chi' \frac{dz}{d\chi'} W^{(j)}(z(\chi')) \int dk k^2 P_{e, \text{nl}}(k, z(\chi), z(\chi')) j_\ell(k\chi) j_\ell(k\chi'), \quad (3.9)$$

where this derivation follows similarly to the derivation of the covariance matrix in section 2.9. Using the Limber approximation, the expression is simplified into

$$C_{\delta_e \delta_e}^{(ij)}(\ell) = \int d\chi \frac{dz}{d\chi} \frac{dz}{d\chi} \frac{W^{(i)}(z(\chi)) W^{(j)}(z(\chi))}{\chi^2} P_{e, \text{nl}} \left(k = \frac{\ell + 1/2}{\chi}, z(\chi) \right). \quad (3.10)$$

In this thesis, I use the Limber approximation to obtain the angular power spectrum since it is computationally efficient as compared to the non-Limber expression. At large values of ℓ , the non-Limber expression converges to the Limber expression, as can be seen in Fig. 3.4. z_{eff} refers to the effective redshift of the bin which is given by

$$z_{\text{eff}} = \int z W^{(i)}(z) dz \quad (3.11)$$

The choice of 29 redshift bins is validated by looking at the correlation between two neighbouring

shells. I define the ratio

$$R = \frac{C_{\delta_e \delta_e}^{(i,i+1)}(\ell)}{\sqrt{C_{\delta_e \delta_e}^{(i,i)}(\ell) C_{\delta_e \delta_e}^{(i+1,i+1)}(\ell)}}, \quad (3.12)$$

which is a measure of the correlation between two neighbouring shells. I find that for 29 redshift bins, the ratio R is less than 0.1 for all values of ℓ , which means that the correlation between neighbouring shells is small enough to be ignored in this work. I show the comparison in Fig. 3.5. Since I perform the analysis on mock catalogs, I can ignore the correlation between neighbouring shells which is the default when using the Limber approximation. For application on real data, the correlation between neighbouring shells should be taken into consideration by using the non-Limber expression for the angular power spectrum. I show a comparison between the Limber and the non-Limber expressions in Fig. 3.4.

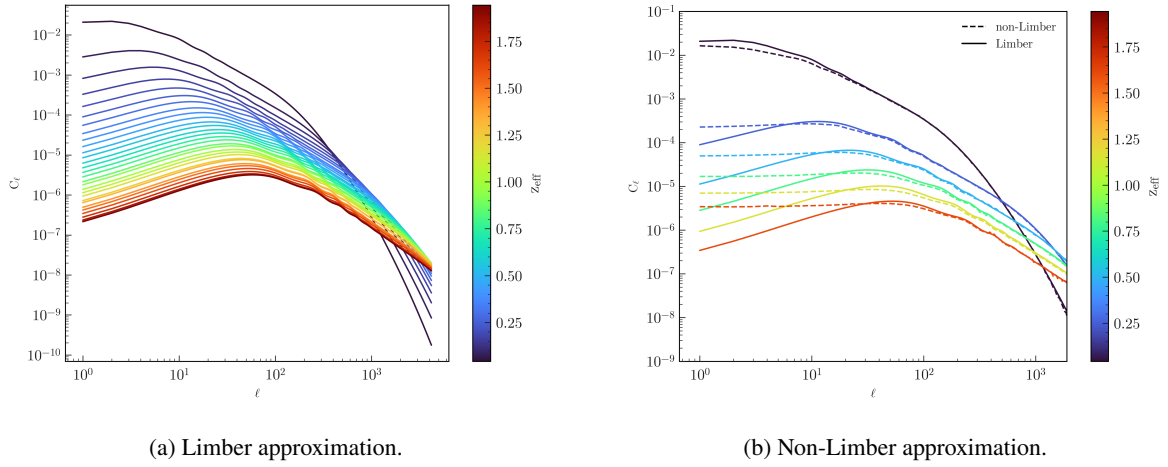


Figure 3.4: (a) The electron power spectrum in the Limber approximation for the redshift bins considered in this thesis. The colour coding represents the effective redshift of every bin as calculated by Eq. 3.11. (b) A comparison between the Limber and the non-Limber approximations for the electron power spectrum. The two calculations agree for large values of ℓ . The non-Limber integral is computed via the Levin approximation using Reischke, 2025.

3.2 Simulating the dispersion measure signal

Table 3.1 and 3.3 contain the fiducial values of the cosmological and baryonic feedback parameters, which are used to obtain an expectation of the power spectrum. In addition, Table 3.2 contains the specifications of the simulation such as the number of redshift bins and the maximum multipole. In particular, the choice of maximum multipole $\ell_{\text{max}} = 4100$ is motivated by the fact that this scale corresponds to a length scale of $k \approx 10 h/\text{Mpc}$ at a redshift of $z = 1$ which is the scale where baryonic feedback modifies the power spectrum (Maraio, Hall and Taylor, 2025).

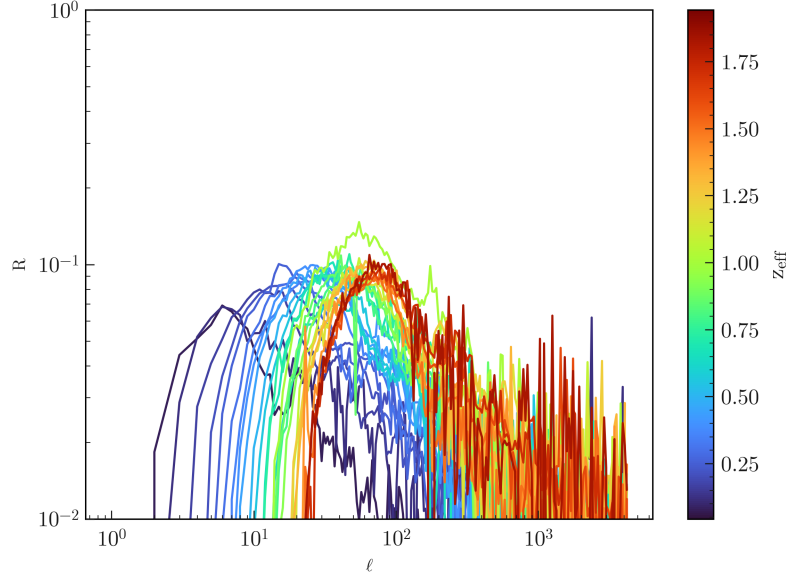


Figure 3.5: A plot showing the ratio R between neighbouring shells for 29 redshift bins. The ratio is less than 0.1 for all values of ℓ , which means that the correlation between neighbouring shells is small enough to be ignored in this work. The spectra are obtained using the non-Limber expression of the power spectrum.

3.2.1 Generating electron density maps

To generate electron density maps, I use the Generator for Large-Scale Structure (GLASS) (Tessore et al., 2023) which is a package that generates lognormal maps of a cosmological field starting from a given angular power spectrum. The GLASS package is used in this thesis to generate maps of electron density in bins of redshift starting from the electron angular power spectrum obtained in section 3.1.3. The GLASS package assumes that the cosmological field follows a lognormal distribution, which is not the case for a real cosmological field. However, the GLASS package preserves the two-point statistics of the cosmological field.

For an arbitrary cosmological field u , the GLASS package assumes that the field follows a lognormal distribution with an underlying Gaussian field g such that

$$u = \lambda (e^g + 1) \equiv g = \ln \left(\frac{u - 1}{\lambda} \right), \quad (3.13)$$

where λ is a parameter that can be chosen by noticing that the lower bound of a Gaussian field that represents density perturbations is -1 , making $\lambda = 1$. The reason behind linking the field u to a field g is that sampling from a Gaussian distribution is computationally efficient as compared to a lognormal distribution.

After feeding the electron power spectrum to the GLASS package, it converts the power spectra to 2-point correlation functions $C(\theta)$ of the field u . Then, it uses the relation between u and g to obtain the 2-point correlation function $G(\theta)$ of the underlying Gaussian field g . The 2-point correlation function of

the Gaussian field g is given by

$$G(\theta) = \ln \left(\frac{C(\theta)}{\lambda^2} + 1 \right). \quad (3.14)$$

After that, $G(\theta)$ is converted to the angular electron power spectrum G_ℓ of the underlying Gaussian field g . Finally, the GLASS package samples from the Gaussian distribution with a power spectrum G_ℓ to obtain a map of the underlying Gaussian field g , which is then converted to a map of the electron density field u using Eq. 3.13. The conversion between the power spectrum and the 2-point correlation function is not trivial since it involves an infinite sum over the multipoles ℓ similar to 2.107, however, the GLASS package implements a solver to perform this conversion efficiently. I refer the reader to section 3.3 of Tessore et al. (2023) for more details about the implementation of the GLASS package.

To use this package, I specify a weight function W as described in section 3.1.3. The choice of the weight function is motivated by the fact that the DM signal is a projection along the line of sight, which means that a comoving volume element at a redshift z contributes to the DM signal with a weight that is proportional to the comoving volume element at that redshift. Therefore, I choose a weight function of the form

$$W^{(i)}(z) = \begin{cases} A^{(i)} \chi^2(z)/E(z) & \text{if } z_{i-1} \leq z < z_i, \\ 0 & \text{otherwise,} \end{cases} \quad (3.15)$$

with $A^{(i)}$ being a normalization constant calculated for each bin i such that

$$\int W^{(i)}(z) dz = 1. \quad (3.16)$$

I choose a maximum redshift of $z_{\max} = 2.$ and a minimum redshift of $z_{\min} = 0.01$ since this is the range

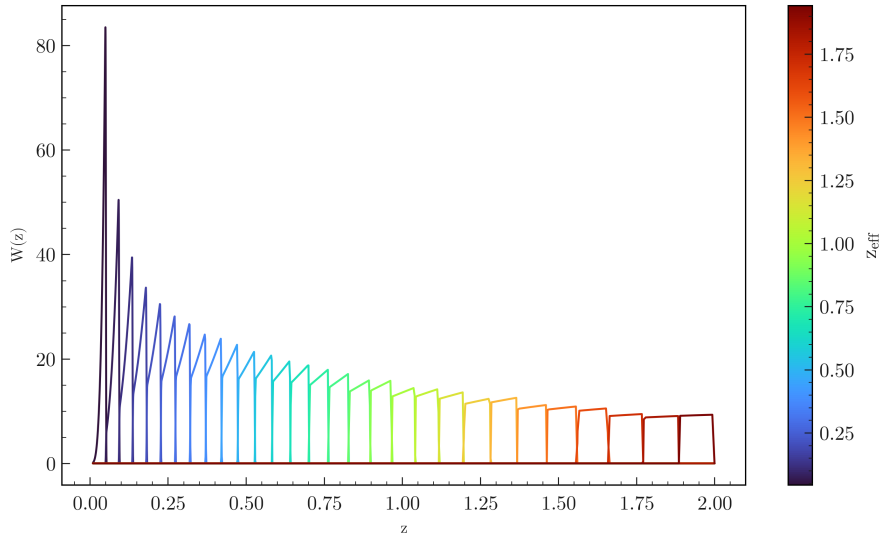


Figure 3.6: A plot showing the normalized weights for the 29 redshift bins. The weights are normalized such that $\int W^{(i)}(z) dz = 1$ for each bin i .

of detected FRBs by current surveys. I convert this redshift range into a range of comoving distance

and divide the comoving distance range into 29 shells; this is chosen such that it is comparable to a similar work by Konar et al. (2025) that used 17 shells for sources up to a maximum $z = 1$. Dividing in comoving distance rather than redshift ensures that the shells are of equal thickness in comoving space, which is more physically motivated since the DM signal is a projection along the line of sight. Fig. 3.7 shows a plot of the resulting shells in redshift space.

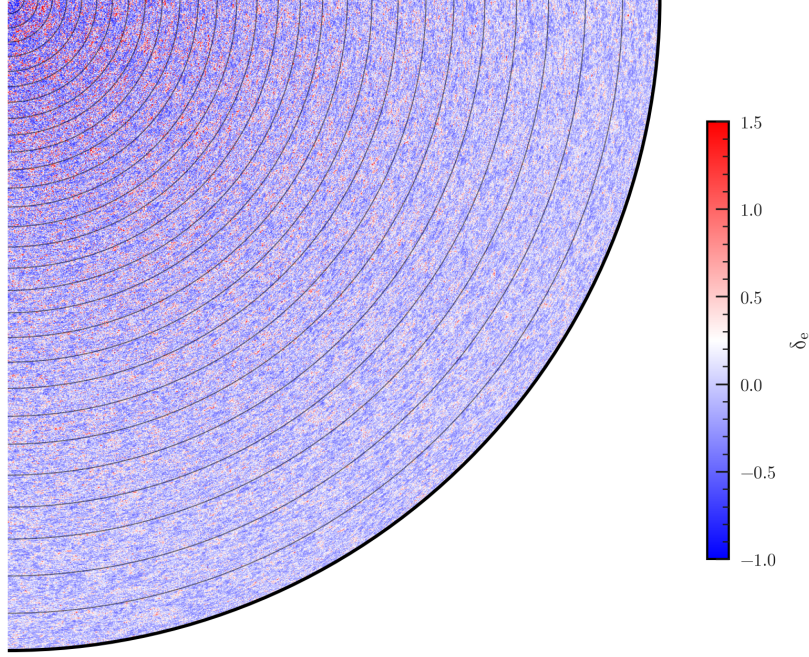


Figure 3.7: A 2D plot showing maps of concentric shells of electron density in redshift space. The maps are obtained via the GLASS package. Notice that the contrast of a shell increases due to structure growth. The innermost shells represent the lowest redshift bin and the outermost shells represent the highest redshift bin.

3.2.2 Discretizing the dispersion measure contributions

To obtain a mock catalog of FRBs, I sample randomly on the sky and according to the source redshift distribution in redshift then sum the contribution from different shells. I use the source redshift distribution predicted by the SKAO (Braun et al., 2015). Fig. 3.8 shows the source redshift distribution used in this thesis. Since the distribution extends beyond the maximum redshift of the shells, I reject the samples that have a redshift higher than 2.0. To sum the contribution from each shell, I discretize Eq. 2.99 by splitting the mean DM contribution from the perturbed contribution such that

$$\text{DM}(\hat{\mathbf{x}}, z) = \langle \text{DM} \rangle(z) + \mathcal{D}(\hat{\mathbf{x}}, z), \quad (3.17)$$

with $\langle \text{DM} \rangle(z)$ given by Eq. 2.97, which can be calculated numerically, and $\mathcal{D}(\hat{\mathbf{x}}, z)$ given by

$$\mathcal{D}(\hat{x}, z) = \mathcal{A} \int_0^z \frac{1+z'}{E(z')} \delta_e(\hat{x}, z') dz'. \quad (3.18)$$

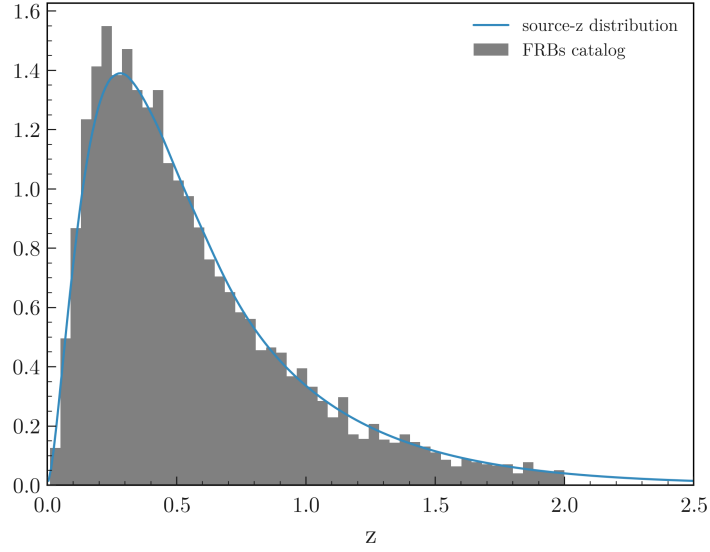


Figure 3.8: A plot showing the normalized source redshift distribution used in this thesis as compared to the sampled distribution of a mock FRB catalog. The distribution is predicted by the SKAO (Braun et al., 2015). The distribution extends beyond the maximum redshift of the shells; therefore, I reject the samples that have a redshift higher than 2.0.

The relationship between $\delta_e(\hat{x}, z)$ and $\delta_e(\hat{x})$ is given by Eq. 3.7. Therefore, the contribution from a GLASS shell i with an effective redshift $\bar{z}_i$ in a direction $\hat{x}$, namely $\delta_e^{(i)}(\hat{x})$, is given in terms of the electron density contrast $\delta_e(\hat{x}, z)$ as

$$\delta_e^{(i)}(\hat{x}) \approx W^{(i)}(\bar{z}_i) \delta_e(\hat{x}, \bar{z}_i) (z_{i+1} - z_i). \quad (3.19)$$

Therefore, by discretizing Eq. 3.18 as

$$\mathcal{D}(\hat{x}, z_{\text{FRB}}) = \mathcal{A} \sum_{i=1}^{n_{\text{shells}}} \frac{1 + \bar{z}_i}{E(\bar{z}_i)} \delta_e(\hat{x}, \bar{z}_i) w_{\text{shell}}^i(z_{\text{FRB}}), \quad (3.20)$$

where w_{shell}^i represents the weight of a shell's contribution depending on the value of z_{FRB} as given by

$$w_{\text{shell}}^i(z_{\text{FRB}}) = \begin{cases} 1 & , \text{ if } z_{\text{FRB}} \geq z_{\text{max}}^i, \\ z_{\text{FRB}} - z_{\text{min}}^i & , \text{ if } z_{\text{min}}^i \leq z_{\text{FRB}} \leq z_{\text{max}}^i \\ 0 & , \text{ if } z_{\text{FRB}} \leq z_{\text{min}}^i \end{cases} \quad (3.21)$$

Combining Eqs. 3.19 and 3.21, the final expression of $\mathcal{D}$ is

$$\mathcal{D}(\hat{x}, z_{\text{FRB}}) = \mathcal{A} \sum_{i=1}^{n_{\text{shells}}} \frac{1 + \bar{z}_i}{E(\bar{z}_i)} \delta_e^{(i)}(\hat{x}) \frac{w_{\text{shell}}^i(z_{\text{FRB}})}{W^{(i)}(\bar{z}_i)} \quad (3.22)$$

where I redefine $w_{\text{shell}}^i(z_{\text{FRB}})$ as

$$w_{\text{shell}}^i(z_{\text{FRB}}) = \begin{cases} 1 & , \text{ if } z_{\text{FRB}} \geq z_{\text{max}}^i, \\ \frac{z_{\text{FRB}} - z_{\text{min}}^i}{z_{\text{max}}^i - z_{\text{min}}^i} & , \text{ if } z_{\text{min}}^i \leq z_{\text{FRB}} \leq z_{\text{max}}^i \\ 0 & , \text{ if } z_{\text{FRB}} \leq z_{\text{min}}^i \end{cases} \quad (3.23)$$

In total, I use the above formalism to sum the contribution from different redshift shells which will eventually yield the large-scale structure contribution DM_l^i of an FRB at redshift z_{FRB}^i and sky position (g_l^i, g_b^i) , yielding a catalog of FRBs.

In the next section, I describe how to add systematic uncertainties on the level of a catalog.

3.3 Simulating the systematic uncertainties

The simulation-based approach which I am using here allows easy integration of systematic uncertainties on the level of a catalog. It allows further use of the simulation pipeline described in Fig. 3.1 in applications of simulation-based inference. In this section, I describe how to contaminate catalogs with systematic uncertainties. Table 3.4 describes the parameters of every systematic effect used in this thesis. Notice that the chosen range of each systematic is selected to mirror a situation between an optimistic and pessimistic scenario. I discuss the choices of the fixed parameters and the chosen ranges below.

Symbol	Description	Range
<i>Redshift Uncertainty</i>		
η	the ratio between a redshift measurement z and its standard deviation	[0.05, 0.4]
<i>The host contribution</i>		
$\text{DM}_{\text{host,mean}}$	The underlying Gaussian mean of the lognormal distribution	[50, 300]
σ_{LN}	The scale of the lognormal distribution	0.76
<i>The Milky Way contribution</i>		
$\text{DM}_{\text{MW}}(\hat{\mathbf{x}})$	The Milky Way contribution as a function of angular position $\hat{\mathbf{x}}$	Given by the NPE2001 model
<i>Selection effects</i>		
DM_{max}	The maximum value of DM that the instrument is able to detect.	[700, 3500]
ϵ	The smoothing scale of the PDF.	50
<i>Missing redshifts</i>		
f	The fraction of FRBs with missing redshift.	[0.05, 0.4]

Table 3.4: A table with the parameters of the systematic uncertainties. The range of the parameters is chosen to be wide enough to cover the expected values for these parameters.

3.3.1 Milky Way contribution

Depending on the angular direction, the measured DM_{tot} receives a contribution from the MW. This contribution varies depending on the angular position of the FRB. Sources that are close to the galactic plane receive a contribution of order $DM_{\text{MW}} \sim 3000 \text{ pc.cm}^{-3}$ with high uncertainty. At larger latitudes, the contribution is as low as $DM_{\text{MW}} \sim 10 \text{ pc.cm}^{-3}$. Therefore, as a way to mitigate the impact of the MW, it is possible to remove any sources that are located within a particular galactic cut. However, due to the current sample size of FRBs, it is not feasible to apply this cut since it will decrease the signal-to-noise ratio due to the reduced sample size. Nonetheless, in this thesis, I explore the possibility of imposing a galactic cut θ_{cut} and study its impact on the resulting DM- z relation. Fig. 2.11 shows a map of DM_{MW} as predicted by the NPE2001 model. I study the impact of imposing a galactic cut for a range $\theta_{\text{cut}} \in [0 \text{ deg}, 60 \text{ deg}]$.

3.3.2 Host contribution

To study the impact of the host contribution, I use a lognormal model of the host contribution as described in section 2.10 with parameters chosen as described in 3.4. The range of mean host contribution $DM_{\text{host,mean}}$ is taken from the work by Theis et al. (2024) on the host contribution using different hydrodynamical simulations. The value for the scale of the lognormal distribution σ_{LN} is the parameter inferred with simulation-based inference from 12 localised FRBs in the work by Konar et al. (2025). In addition to these parameters, I consider a simple model for the evolution of the host contribution with redshift such that

$$DM_{\text{host}}(z) = \frac{DM_{\text{host}}}{1+z} \quad (3.24)$$

with z corresponding to the redshift of the FRB. Assuming that the correlation between the DM and DM_{host} is negligible, the covariance matrix receives an additional contribution

$$\langle DM_{\text{host},i} DM_{\text{host},j} \rangle = \frac{\sigma_{\text{host}}^2}{(1+z_i)(1+z_j)} \quad (3.25)$$

with σ_{host}^2 the variance of the host contribution as defined in section 2.10.

3.3.3 Redshift uncertainty

I simulate the impact of redshift uncertainty by assuming that a redshift measurement z_i is sampled from a Gaussian distribution with mean z_i and standard deviation $\sigma_i = \eta_i z_i$, where η_i is a number between 0 and 1. To propagate the error on the cosmological covariance, consider that $(z_i, z_j) \sim \mathcal{N}((\hat{z}_i, \hat{z}_j), \Sigma)$, with $\mathcal{N}$ the normal distribution and

$$\Sigma = \begin{bmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_j^2 \end{bmatrix} \quad (3.26)$$

To propagate the error in the covariance, I calculate the expectation value $\langle \hat{C}_{ij} \rangle$, given as

$$\langle \hat{C}_{ij} \rangle = \int \mathcal{N}((\hat{z}_i, \hat{z}_j), \Sigma) C_{ij} dz_i dz_j \quad (3.27)$$

With Taylor expansion around $(\hat{z}_i, \hat{z}_j)$ and setting $\Delta_{ij} = (z_i, z_j) - (\hat{z}_i, \hat{z}_j)$, the covariance matrix can be written

$$\langle \hat{C}_{ij} \rangle = \int \mathcal{N} \left(\hat{C}_{ij} + \nabla C_{ij}|_{(\hat{z}_i, \hat{z}_j)} \Delta_{ij} + \frac{1}{2} \Delta_{ij}^T \nabla^2 C_{ij}|_{(\hat{z}_i, \hat{z}_j)} \Delta_{ij} + \dots \right) dz_i dz_j. \quad (3.28)$$

Since the Gaussian distribution has vanishing moments for moments larger than third order, the expression is simplified into

$$\langle \hat{C}_{ij} \rangle = \hat{C}_{ij} + \frac{1}{2} \sigma_i^2 \partial_{ii}^2 C_{ij}|_{(\hat{z}_i, \hat{z}_j)} + \frac{1}{2} \sigma_j^2 \partial_{jj}^2 C_{ij}|_{(\hat{z}_i, \hat{z}_j)}. \quad (3.29)$$

Therefore, it is possible to account for the error in the covariance due to redshift uncertainty using the second derivative of the covariance with respect to redshift z . In chapter 4, I will show the impact of redshift uncertainty relative to the case with the corrected covariance.

3.3.4 Selection effects

The parameters for the selection effect model shown in table 3.4 are well-motivated by the range of $DM_{\max}$ from different surveys as shown in table 2.1. Meanwhile, I choose the smoothing scale to be comparable with the lower bound of the possible values of $DM_{\max}$.

Since the sigmoid function modifies the PDF of the measured DM, the sigmoid scaling can also be used in contaminating a mock catalog with no selection effect in the following way: obtain a weight for each value of the measured DM in a catalog, then generate N_{FRB} random numbers sampled uniformly between 0 and 1, after that, compare the sigmoid weight of each FRB with the random number and remove all FRBs with a random number larger than the sigmoid weight. This sequence of steps reweights the PDF of the observed DM values according to the sigmoid function, yielding an FRB catalog with the desired selection effect.

3.3.5 Unlocalised hosts

I simulate unlocalised FRBs in a fashion similar to what is described in section 2.10. To fill in missing redshifts, I use the fiducial parameters in order to obtain an expectation of the unknown redshift $z_i = z(DM_i)$ from the measured DM_i . To propagate the error to the redshift space, I assume that DM follows a lognormal distribution with underlying Gaussian mean DM_i and variance $\sigma^2(z_i)$, such that

$$DM_i \sim P(DM; \mu^i, \sigma_{\text{LN},i}^2) \quad (3.30)$$

with

$$\mu^i = \ln \left(\frac{DM_i^2}{\sqrt{\sigma^2(z^i) + DM_i^2}} \right), \quad \sigma_{\text{LN},i}^2 = \ln \left(1 + \frac{\sigma^2(z^i)}{DM_i^2} \right) \quad (3.31)$$

The lognormal assumption is valid in the case of this work since the DM contributions from the different shells are lognormal. After that, I draw samples from the distribution, and I use the same mean DM- z relation to map the sampled DM to z -space. After that, I calculate the sample variance of the mapped redshifts σ_i^2 , where $\sigma_i / \sqrt{n_{\text{sample}}}$ is the error on the estimated redshift z_i .

To propagate the error to the covariance, I use the same formalism introduced in section 3.3.3, only as an approximation, since the underlying PDF of the estimated redshift is not a Gaussian distribution. For the values of σ_{ii} , if the FRB has an undefined host, the redshift uncertainty is calculated with the method

introduced above. Meanwhile, if the redshift is exact, then I set the value of $\sigma_{ii} = 0$. In this thesis, I aim to disentangle the different systematic uncertainties. In reality, these effects could be degenerate in the way they impact the measured DM or the covariance. I discuss this further in chapter 5.

3.4 Quantifying the deviation from the mean DM-z relation

This section is based on Statistical Methods in Cosmology lecture given by Licia Verde (Verde, 2010). To quantify the impact of systematic uncertainty on the DM-z relation, I compare a fiducial catalog with a contaminated catalog using the χ^2 test. In particular, the maximum χ^2 estimator is a method to infer model parameters from data by comparing the difference between data d and the mean predicted by the model parameters $\mu(\theta)$ given some weight for every point. In a flat geometry, the χ^2 can be written as

$$\chi^2 = \sum_i w_i (d_i - \mu_i(\theta))^2. \quad (3.32)$$

If data points are correlated, the χ^2 can be written in terms of the weights w_{ij} such that

$$\chi^2 = \sum_{ij} (d_i - \mu_i(\theta)) w_{ij} (d_j - \mu_j(\theta)). \quad (3.33)$$

It can be shown that the minimum variance χ^2 estimator is the estimator with w_{ij} being the inverse of the data covariance matrix C_{ij} , making the final expression of the χ^2

$$\chi^2 = \sum_{ij} (d_i - \mu_i(\theta)) C_{ij}^{-1} (d_j - \mu_j(\theta)). \quad (3.34)$$

The χ^2 can be used to quantify the inconsistency between a model and data in the following way. First, in the neighbourhood of the true model parameters, the data vector follows a Gaussian distribution, and therefore χ^2 follows a χ^2 -distribution with mean $\langle \chi^2 \rangle = \nu$ where ν is the number of degrees of freedom. In this case, $\nu = N_{\text{FRBs}}$ since I do not use the data in inferring any parameter. I define the reduced χ^2 as

$$\chi_\nu^2 = \frac{\chi^2}{\nu} \quad (3.35)$$

Therefore, for a model that describes data, χ_ν^2 should be consistent with 1. In this thesis, I quantify the deviation of χ_ν^2 calculated using a contaminated FRB catalog from $\chi_{\nu, \text{true}}^2$, calculated using the fiducial catalog. For example, it is possible to distinguish between two cases of deviation

- $\chi_\nu^2 > 1$, meaning that the covariance matrix is underestimated or the mean predicted by the model is overestimated.
- $\chi_\nu^2 < 1$, meaning that the covariance matrix is overestimated or the mean predicted by the model is underestimated.

To calculate the χ^2 , the covariance matrix is needed. In the next section, I describe how to efficiently calculate the covariance matrix.

3.4.1 The numerical implementation of the cosmological covariance matrix

The cosmological covariance of FRBs was introduced in section 2.9. For a sample size of 10 000 FRBs, calculating the cosmological covariance is non-trivial with libraries such as `numpy`. Therefore, I use the `jax` library in order to accelerate the calculations of the covariance matrix. `jax` is a python package originally intended for machine learning applications. Nonetheless, `jax` includes parallelization functionalities to parallelize matrix computations on Graphic Processing Units (GPUs) in addition to dynamic for-loops. I exploit the functionalities of the `jax` package to accelerate the calculation of the covariance matrix. The code for the covariance calculation is written as follows

```

1 def compute_covariance_4_vmap(
2     theta_ij, large_angle_filter, z_FRBs,
3     k, z, z_integral, z_divisions,
4     J0, p_chi_k, chi, dz_dchi, w, ell_s, curly_A,
5 ):
6     # --- 1. Log-log P(k,a) interpolator ---
7     z_interp = jnp.geomspace(z[0], z[-1], z_divisions)
8     a = 1.0 / (1.0 + z)
9     sorting = jnp.argsort(a)
10    a, p_k_a = a[sorting], p_chi_k[sorting, :].T
11    log_p_k_a_interp = RegularGridInterpolator(
12        (jnp.log(k), jnp.log(a)), jnp.log(p_k_a), method='linear')
13
14    @jit
15    def p_k_a_jax_interpolator(k_val, a_val):
16        return jnp.exp(log_p_k_a_interp((jnp.log(k_val), jnp.log(a_val))))
17
18    # --- 2. Window outer product ---
19    integration_matrix = w[:, None, :] * w[None, :, :] # (N, N, Nchi)
20    p_k_a_vmapped = vmap(p_k_a_jax_interpolator, in_axes=(0, 0))
21
22    # --- 3. Inner chi-integral (Limber approximation) ---
23    @jit
24    def calculate_limber_covariance_one_ell(ell):
25        power = p_k_a_vmapped((ell + 0.5) / chi, 1.0 / (1.0 +
26            z_integral))
27        integrand = (power[None, None, :] * integration_matrix
28            * dz_dchi[None, None, :] ** 2
29            / chi[None, None, :] ** 2)
30        C = curly_A**2 * jnp.trapezoid(integrand, chi, axis=-1)
31        C = (C + C.T) / 2.0
32        eps = 1e-8 * jnp.trace(C) / C.shape[0]
33        return C + eps * jnp.eye(C.shape[0])
34
35    # --- 4. Interpolate onto FRB redshifts + Bessel weight ---
36    @jit
37    def evaluate_at_ell(ell):
38        C = calculate_limber_covariance_one_ell(ell)
39        interp = RegularGridInterpolator(
40            (z_interp, z_interp), C, method='linear')

```

```

40     C_frb = vmap(vmap(
41         lambda zi, zj: interp(jnp.array([zi, zj]))[0],
42         in_axes=(None, 0)), in_axes=(0, None))(z_FRBs, z_FRBs)
43     C_frb = (C_frb + C_frb.T) / 2.0
44     eps = 1e-8 * jnp.trace(C_frb) / C_frb.shape[0]
45     C_frb = C_frb + eps * jnp.eye(C_frb.shape[0])
46     result = ell * J0(ell * theta_ij) * C_frb / (2.0 * jnp.pi)
47     return (result + result.T) / 2.0
48
49     # --- 5. Outer ell-integral via lax.scan ---
50     @jit
51     def little_integrals(carry, ell1):
52         total_sum, point0, ell0 = carry
53         point1 = evaluate_at_ell(ell1)
54         total_sum = total_sum + 0.5 * (ell1 - ell0) * (point0 + point1)
55         total_sum = (total_sum + total_sum.T) / 2.0
56         return (total_sum, point1, ell1), None
57
58     point0 = evaluate_at_ell(ells[0])
59     carry_init = (jnp.zeros((z_FRBs.size, z_FRBs.size)), point0, ells[0])
60     (final_carry, _) = lax.scan(little_integrals, carry_init, ells[1:])
61
62     # --- 6. Final symmetry, regularisation, large-angle filter ---
63     S = final_carry[0]
64     S = (S + S.T) / 2.0
65     eps = 1e-8 * jnp.trace(S) / S.shape[0]
66     return large_angle_filter * (S + eps * jnp.eye(S.shape[0]))

```

Listing 3.1: Angular covariance via Limber approximation.

Using the `jax.vmap` property it is possible to calculate multiple covariances at a time which is useful when calculating the derivative of the covariance matrix with respect to a model parameter using the forward difference method.

3.5 Simulation-based inference

To forecast the bias in parameter estimation, I use simulation-based inference (SBI). A review of simulation-based inference can be found in Cranmer, Brehmer and Louppe (2020) and in the SBI lecture by Noemi Anau Montel⁴ given at the Postdoc Lecture Series at the Max Planck Institute for Astrophysics. SBI is a method to infer the parameters of a model by comparing an observed data vector d with a data vector obtained from a forward simulation. It is an extension of Bayesian inference in which the parameter posterior is constrained by performing Monte Carlo Markov Chain (MCMC) to obtain the parameter's posterior distribution. If the prior of the model parameter is $P(\theta)$ and the likelihood of data given the parameter is $P(d|\theta)$ then the posterior distribution of the parameters given data $P(\theta|d)$ is written by Bayes' rule

$$P(\theta|d) \sim P(d|\theta) P(\theta). \quad (3.36)$$

⁴ The Postdoc lecture by Noemi Anau Montel

In traditional cosmological applications, there are two considerations. First, by invoking the central limit theorem (CLT)⁵, $P(d|\theta)$ can be assumed to be a multivariate Gaussian. However, this is not necessarily true if the conditions of CLT are not fulfilled, leaving the problem with an intractable likelihood.

To solve this problem, Approximate Bayesian Computation (ABC) can be used to obtain posterior samples without a tractable likelihood and using a forward model. By defining a distance measure between a data vector $\mathbf{x}_0$ and a data vector from the forward model $\mathbf{x}$ such that

$$d(\mathbf{x}, \mathbf{x}_0) = \|\mathbf{x} - \mathbf{x}_0\|. \quad (3.37)$$

The samples from the approximate posterior are drawn with rejection sampling using an acceptance tolerance rate ϵ such that accepted samples satisfy $d(\mathbf{x}, \mathbf{x}_0) < \epsilon$, and the posterior is given by

$$p_{\text{ABC}}(\Theta | \mathbf{x}) = \frac{\int_{d(\mathbf{x}_0, \mathbf{x}) < \epsilon} d\mathbf{x} p(\mathbf{x} | \Theta) p(\Theta)}{\int_{d(\mathbf{x}_0, \mathbf{x}) < \epsilon} d\mathbf{x} p(\mathbf{x})} \quad (3.38)$$

which is guaranteed to converge to the true posterior distribution. However, the sampling efficiency of this method is very low, especially if the problem involves wide priors and a large parameter space. Therefore, machine learning (ML) methods are used to learn the posterior distribution of the observed data. Nonetheless, with high-dimensional data such as that of FRBs, both ML and ABC methods fail to converge quickly. Therefore, a summary statistic that retains the cosmological information is necessary. I describe below how to use the simulation pipeline described in section 3.2 to perform simulation-based inference of the cosmological and baryonic feedback parameters. Table 3.5 shows the free parameters along with their priors. I choose wide enough priors to show the resultant bias in parameter estimation when systematic uncertainties are not taken into consideration in the forward model. In addition, Fig. 3.9 shows a schematic representation of the inference pipeline used. Among the baryonic feedback parameters, I chose $\log_{10} M_c$ since the gas power spectrum is most sensitive to this parameter, as shown in the appendix.

To perform simulation-based inference with the FRB catalog, the observed data vector $\mathbf{x}_0$ is the list of observed DM with dimensionality of 10 000. This is a high-dimensional problem which is computationally prohibitive. Therefore, I use a summary statistic of lower dimension to avoid the curse of dimensionality. In section 3.5.1, I describe the choice for the summary statistic which I use in this thesis.

3.5.1 Data Compression

Following Konar et al. (2025), I use score compression introduced in Alsing and Wandelt (2018) which is a model-dependent compression of a data vector with dimensionality N_{FRBs} into a summary statistic with dimensionality equal to the number of model parameters N_{params} . If the likelihood of data given the parameter is $L(d|\theta)$, then L can be expanded around fiducial parameters θ_* up to second order as

$$L(d|\theta) = L(d|\theta_*) + (d_i - \mu_i(\theta))^T \nabla L|_{\theta_*} + (d_i - \mu_i(\theta))^T \nabla^2 L|_{\theta_*} (d_i - \mu_i(\theta)). \quad (3.39)$$

⁵ The central limit theorem (CLT) states that the probability distribution function of independent and identically distributed random variables approaches a Gaussian distribution as the number of sampled variables approaches infinity if the true probability distribution function has a finite variance.

Parameter	Description	Prior
Ω_c	Cold dark matter density parameter	U[0.2, 0.5]
Ω_b	Baryon density parameter	U[0.02, 0.07]
h	Dimensionless Hubble parameter	U[0.5, 0.8]
σ_8	Amplitude of matter fluctuations	U[0.6, 1.0]
$\log_{10} M_c$	Pivot mass for the mass dependent slope of the gas profile	U[11.0, 15.0]

Table 3.5: Prior distributions for the free parameters.

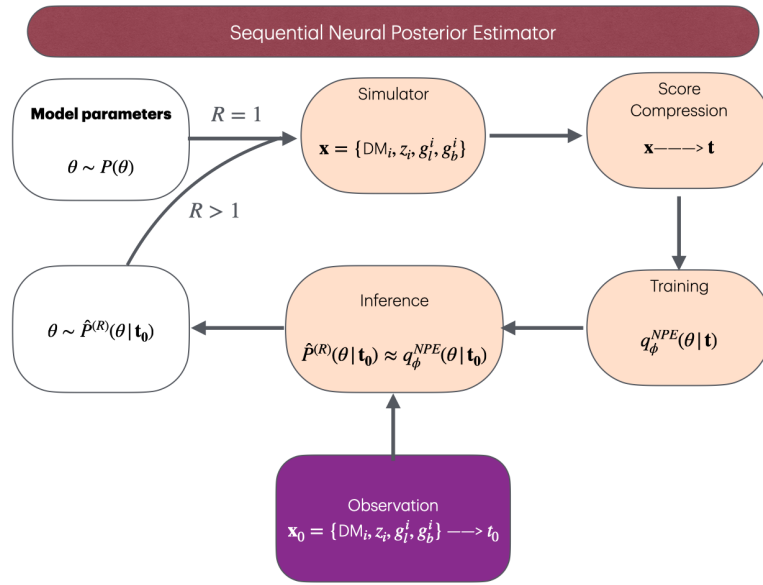


Figure 3.9: A flowchart of the inference pipeline with the sequential neural posterior estimator (SNPE) method. The details of this method are described in section 3.5.2. In total, the inference consists of multiple inference rounds with the prior used in the first round corresponding to the initial parameter prior, while the prior used for later rounds is the learned posterior distribution from the preceding round. This concentrates the samples around the area of high posterior with each round.

The second term in this expression is the first term that is data-dependent through the gradient of the likelihood $\nabla L|_{\theta_*}$; therefore, this quantity can be used as a summary statistic. This quantity is called the score and is given by

$$\mathbf{t} = \nabla L|_{\theta_*}. \quad (3.40)$$

The score is a linear compression scheme since it represents the first order of the likelihood's Taylor expansion. The full expression of the score is given by

$$\mathbf{t} = \nabla \mu_*^T \mathbf{C}_*^{-1} (\mathbf{d} - \mu_*) + \frac{1}{2} (\mathbf{d} - \mu_*)^T \mathbf{C}_*^{-1} \nabla \mathbf{C}_* \mathbf{C}_*^{-1} (\mathbf{d} - \mu_*) - \frac{1}{2} \text{tr}(\mathbf{C}_*^{-1} \nabla \mathbf{C}_*).$$

It can be proven that the covariance matrix associated with $\mathbf{t}$ is exactly the Fisher information matrix

$$\text{cov}_*(\mathbf{t}, \mathbf{t}) = \mathbf{F}_* \quad (3.41)$$

Therefore, the score $\mathbf{t}$ satisfies the Cramér-Rao bound for a lossless compression (RAO, 1945). Even though the score compression is lossless, it is not necessarily sensitive to all the properties in the original data vector.

3.5.2 Neural Posterior Estimator

Instead of using ABC for inference, training a neural network to learn the posterior distribution of model parameters is a more efficient and computationally affordable approach for SBI (Papamakarios and Murray, 2016). There are multiple versions of the inference algorithm but they all reduce to producing a large number of simulation pairs $\{\theta, \mathbf{x}\}$ and training a neural network to learn the conditional probability distribution of the model parameter θ given an observation with a compression statistic $\mathbf{t}(\mathbf{x})$ as $P(\theta|\mathbf{t})$, then obtaining the posterior distribution given a measurement $\mathbf{t}_0 = \mathbf{t}(\mathbf{x}_0)$ as $P(\theta|\mathbf{t} = \mathbf{t}_0)$. A posterior estimator that uses a neural network as an embedding to learn the posterior distribution of model parameters is called a neural posterior estimator (NPE) $q_{\Phi}^{\text{NPE}}(\theta|\mathbf{x})$, with Φ being the parameters of the neural network. One of the most important considerations in constructing an NPE is that this estimator should eventually have the same properties of a probability distribution. Therefore, one possible way of formulating an NPE is to map a random variable z with $z \sim \mathcal{N}(z; 0, 1)$ to a point in the parameter space θ where this map is bijective, differentiable, and parameterized with a set of hyperparameters Φ such that

$$\theta = f_{\Phi}(z). \quad (3.42)$$

The map f_{Φ} can take many forms, and the ensemble of these forms are labeled as normalizing flows (Rezende and Mohamed, 2015) since it maps a normal (Gaussian) distribution to the target distribution. In this thesis, I use Masked Autoregressive Flows (Papamakarios, Pavlakou and Murray, 2017), an advanced version of normalizing flows, which will be discussed in section 3.5.2.

The purpose of training in general is to reduce the dissimilarity between the true posterior and the estimated posterior. This is equivalent to minimizing the expectation value of the Kullback-Leibler divergence defined as

$$D_{\text{KL}}(P(x)||q(x)) = \int dx P(x) \ln \left(\frac{P(x)}{q(x)} \right). \quad (3.43)$$

By taking the expectation value of the expression above, the loss function of the inference problem is given by

$$\mathcal{L}^{\text{NPE}}[\Phi] = - \mathbb{E}_{\mathbf{x}, \Theta \sim p(\mathbf{x}, \Theta)} \ln q_{\Phi}^{\text{NPE}}(\Theta | \mathbf{x}) \quad (3.44)$$

Masked Autoregressive Flows & Mixed Density Network

In this thesis, the choice of the architecture and depth of the neural network is hypertuned through multiple training runs which leads to a converging training chain. After many tests, I use Masked Autoregressive Flows (MAF) which is a type of normalizing flows where an initial simple distribution (a Gaussian distribution for example) is mapped into a more complicated distribution through a sequence of bijective mappings in the following way. If x_i is a random variable whose probability distribution

given its predecessors $\mathbf{x}_{1:i-1}$ is given by

$$p(x_i | \mathbf{x}_{1:i-1}) = \mathcal{N}(x_i | \mu_i, (\exp \alpha_i)^2) \quad \text{where} \quad \mu_i = f_{\mu_i}(\mathbf{x}_{1:i-1}) \quad \text{and} \quad \alpha_i = f_{\alpha_i}(\mathbf{x}_{1:i-1}), \quad (3.45)$$

where instead of mapping one variable to another through a bijective map as in Eq. 3.42, I map the preceding variable to the distribution parameters of the succeeding variable μ_i and α_i . f_{μ_i} and f_{α_i} are unconstrained scalar functions that compute the mean and the log-standard deviation of the distribution which the training process aims to constrain. Eq. 3.45 describes the flow on the level of the distribution. On the level of the random variables, it is possible to sample from the distribution by following the sequence

$$x_i = u_i \exp \alpha_i + \mu_i \quad \text{where} \quad \mu_i = f_{\mu_i}(\mathbf{x}_{1:i-1}), \quad \alpha_i = f_{\alpha_i}(\mathbf{x}_{1:i-1}) \quad \text{and} \quad u_i \sim \mathcal{N}(0, 1). \quad (3.46)$$

The functions f_{μ_i} and f_{α_i} can take any form that is bijective and differentiable with a degree of non-linearity to capture complicated features in data. The sequence of mappings in Eq. 3.46 are called transformations. In addition, each map f has a number of hidden features which corresponds to the width of the neural network and a number of blocks which corresponds to the depth of the neural network. In thesis, I use MAF to infer the model parameters from a contaminated catalog with missing redshifts and from the fiducial catalog.

Mixed Density Netowrks (MDN) applied to neural posterior estimation was introduced by Papamakarios and Murray (2016). The architecture of this netwrok is simpler that MAF where the posterior is estimated as a sum of normal distribution, each with a mean and standard deviation which the learning process optimizes. In this thesis, I use MDN in inferring the model paremeters for the case of redshift uncertainty

The truncated sequential neural posterior estimator

In practice, it is possible to sample a large number of points in the parameter space and forward model the corresponding observation. If the prior range is wide, the number of required simulations for inference is large, causing a computational bottleneck especially if performing one simulation is computationally expensive as in the case of this thesis. To reduce the number of required simulations, I use the sequential neural posterior estimator (SNPE) whose algorithm is described by the schematic plot in Fig. 3.9. The main improvement in this algorithm consists of learning the posterior distribution sequentially by shrinking the prior range from one round to another where each new round uses the high-density regions of the inferred posterior from the preceding round as a prior. This method is called the truncated sequential neural posterior estimator (TSNPE) (Deistler, Goncalves and Macke, 2022) since the original prior is truncated at the high posterior region. In this thesis, the posterior is learned by iterating for a number of rounds that corresponds to 6000 simulations in total.

The coverage tests

Given the nature of the neural posterior estimator, the complexity and architecture of the neural network is arbitrary. Therefore, there is a possibility that the inferred posteior be underfitted, signalling that the model is not complex enough, overfitted, signalling that the model is more complex than needed, or calibrated, signalling a successfull inference. The underfitted and overfitted posterior appear repectively as wide or narrow poseriors. In this work, I perform the Tests of Accuracy with Random Points (TARP) (Lemos et al., 2023). The test aims to assess the coverage of the posterior in the parameter space by

drawing many parameter samples θ^* from the prior and forward model these samples to obtain a set of observations $\mathbf{x}^*$. Then, the observations are invoked in the trained posterior. After that, N points are drawn from each posterior yielding a set of points $\{\theta_i\}$. By comparing the distance between the parameters sampled from the prior and the inferred parameters, it is possible to assess whether the neural model is calibrated, overconfident, or underconfident.

Rigourously, this is done by calculating the credibility level α of each observation defined As

$$\alpha = \frac{1}{N} \sum_{i=1}^N \mathbf{1} [d(\theta_i, \theta_r) < d(\theta^*, \theta_r)] \quad (3.47)$$

where θ_r is a point randomly sampled from the prior, $d(., .)$ is the distance between two points in the parameter space, and $\mathbf{1}[\cdot]$ is the indicator function, taking the value 1 if $(\cdot)$ is true and the value 0 if $(\cdot)$ is false. Furthermore, the set of values α from each prior sample are binned into M bins to calculate the expected coverage probability (ECP) such that

$$\text{ECP}(\alpha) = \frac{1}{N} \sum_{j=1}^M \mathbf{1} [\alpha_j \leq \alpha] \quad (3.48)$$

By plotting α against ECP, the following can be concluded

- if the ECP – α curve is above the curve where ECP = α , the model is underconfident, and the constrained posterior is wider than the true posterior.
- if the ECP – α curve is below the curve where ECP = α , the model is overconfident, and the constrained posterior is narrower than the true posterior.
- if the ECP – α curve is consistent with the curve where ECP = α , the model is calibrated.

Results & Discussion

In this chapter I present the results of the χ^2 test and the SBI. In addition, I discuss these results and explain the observed patterns. I plot the data points of DM- z from the true and contaminated catalogs in the DM- z plane and show the mean in a line and the corresponding 1σ as a shaded area, as will be shown in the following sections.

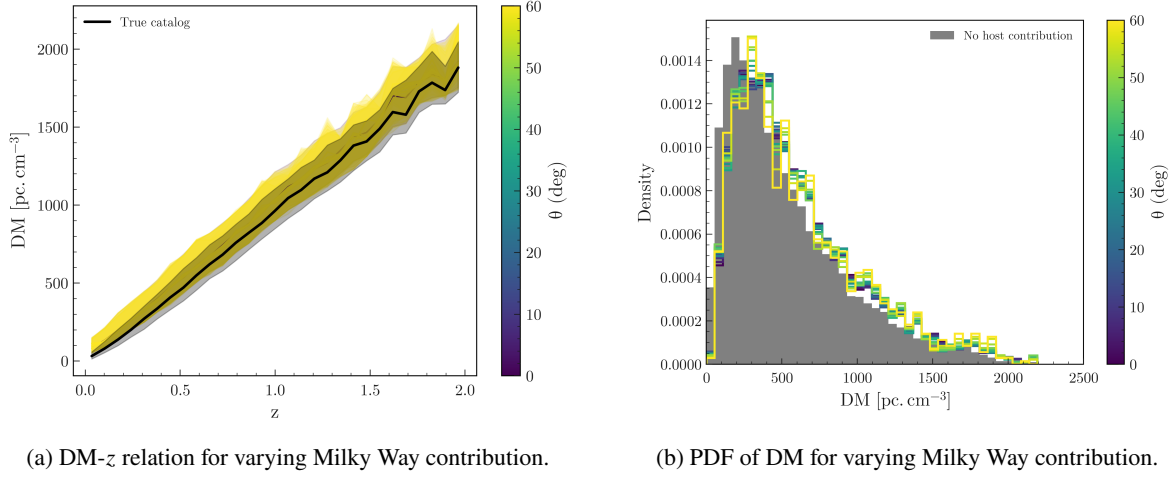
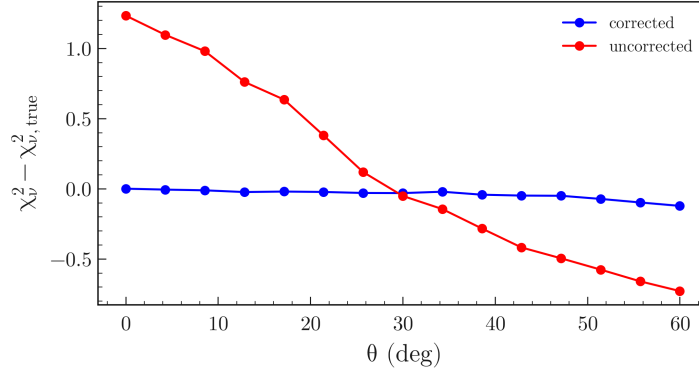
In general, the systematic scenarios considered in this thesis are simple, therefore, they provide a lower limit on the bias caused by each of these uncertainties.

4.1 Results from the Chi-Squared Test

In this section, I present the results of the χ^2 test for all the 5 systematic uncertainties that I am tackling in this thesis.

4.1.1 Milky Way Contribution

The MW uncertainty is added by taking a true mock catalog and adding the corresponding MW contribution as predicted by the NPE2001 model. As can be seen from Fig. 2.11, the largest DM_{MW} corresponds to sources that are located near the galactic plane. Therefore, Figs. 4.1 shows the variation of the PDF and the DM- z relation with changing galactic cut. In general, DM_{MW} adds a component to DM_{tot} which ultimately modifies the mean DM and adds a component to the variance. This can be seen from the left-hand side of Fig. 4.1. By looking at the right-hand side of Fig. 4.1, the PDF changes when adding the DM_{tot} with overall shift to the right. This corresponds to a changing mean for the overall PDF. Meanwhile, there is no obvious pattern for changing the variance. The variation of χ^2 with galactic cut are shown in Fig. 4.2. For galactic cuts below 30 deg, the χ^2 is larger than the true value, and it gets close to the true value around 30 deg and then decreases for larger galactic cuts. Therefore, it is possible to choose an appropriate galactic cut in order to mitigate the impact of MW contribution on the DM- z relation. This result requires further verification by studying the variance and the mean of sky area that could be masked out due to having large DM_{MW} .


 Figure 4.1: DM- z relation and PDF for the Milky Way contribution systematic.

 Figure 4.2: χ^2 as a function of the Milky Way contribution parameters. The uncorrected line in red is the χ^2 without accounting for the MW contribution on the level of the covariance. The corrected line refers to the χ^2 with accounting for the MW contribution on the level of the covariance.

4.1.2 Host Galaxy Contribution

The host galaxy contribution is added by sampling from a lognormal distribution with a mean $DM_{\text{host,mean}}$ and a lognormal scale σ_{LN} . I vary $DM_{\text{host,mean}}$ and study its impact on the mean DM- z relation. The left hand side of Fig. 4.3 shows that for large values of $DM_{\text{host,mean}}$, the variance of the DM- z relation deviates from the true variance at low redshifts and leads to less deviation at higher redshift. This is due to having a redshift-dependence of the host contribution in the contaminated catalog as in Eq. 3.24. The right-hand side plot shows the variation of the PDF with $DM_{\text{host,mean}}$. Increasing $DM_{\text{host,mean}}$ corresponds to increasing the mean and the variance of the PDF relative to a true catalog without a host contribution.

Fig. 4.4 shows the variation of χ^2 relative to the value of the reduced fiducial catalog. The red line shows that the χ^2 increases with increasing value of $DM_{\text{host,mean}}$ which means that the covariance is underestimated. The blue line corresponds to correcting the covariance matrix by adding the variance as in Eq. 3.25. In total, the host contribution leads to biased DM- z relation by larger than 1σ and up to 4σ

depending on $DM_{\text{host,mean}}$.

The host contribution model considered in this thesis is motivated by results from analyzing hydrodynamical simulations. Nonetheless, the scatter of DM_{host} in this model is overestimated since the model implicitly marginalizes over the different variables which controls the host contribution such as the halo mass of the host and the position of an FRB in the host. A more complicated and physically motivated model was introduced by Reischke, Kovač et al. (2024) where the host galaxy contribution is modelled probabilistically, however, with explicit marginalizations over halo masses and the positions of FRBs in the halo. Therefore, future work should explore whether the lognormal model is able to constrain host contribution if the true underlying contribution is sampled from a more complicated model such as the one introduced by Reischke, Kovač et al. (2024).

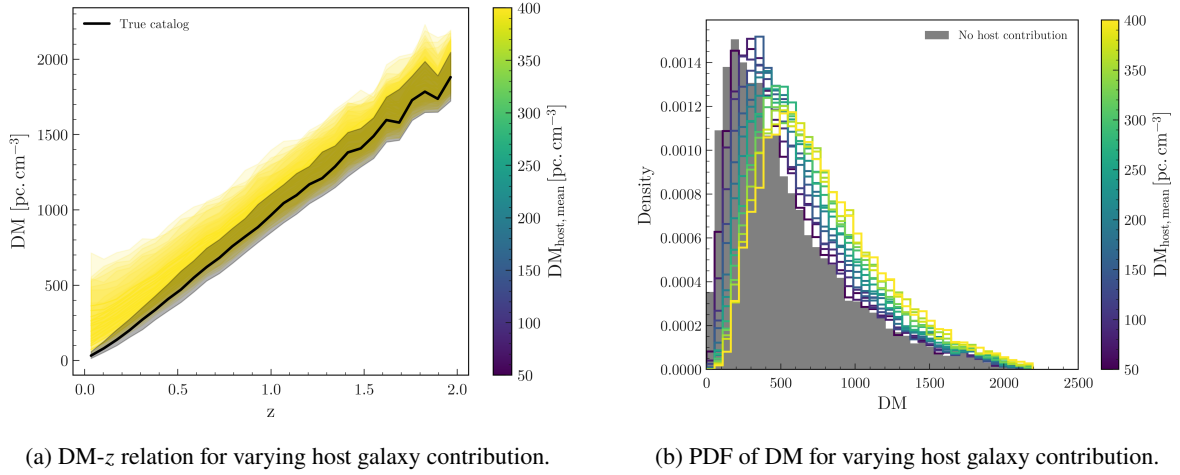


Figure 4.3: DM- z relation and PDF for the host galaxy contribution systematic.

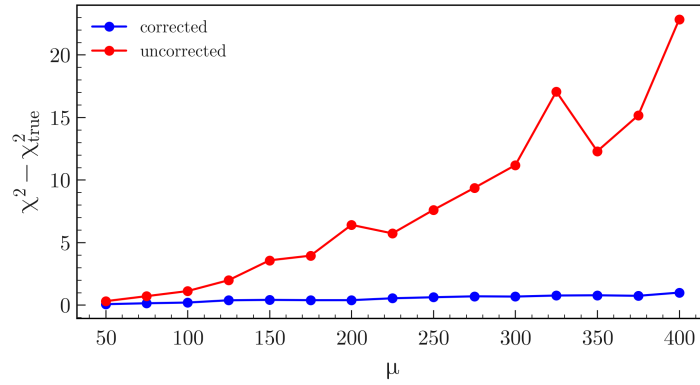


Figure 4.4: χ^2 as a function of the host galaxy contribution parameters.

4.1.3 Redshift Uncertainty

The impact of redshift uncertainty on the DM- z relation is shown on the left-hand side of Fig. 4.5. By increasing the error ratio η , the mean DM- z relation is consistent with the mean DM- z relation for low values of η with only the variance getting larger than that of the true catalog. For larger η , both the mean and variance are different from the true catalog. This is further reflected in the right-hand side of Fig. 4.5 where the variance of the PDF increases with increasing η while the peak of the distribution shifts to the left, corresponding to a lower mean.

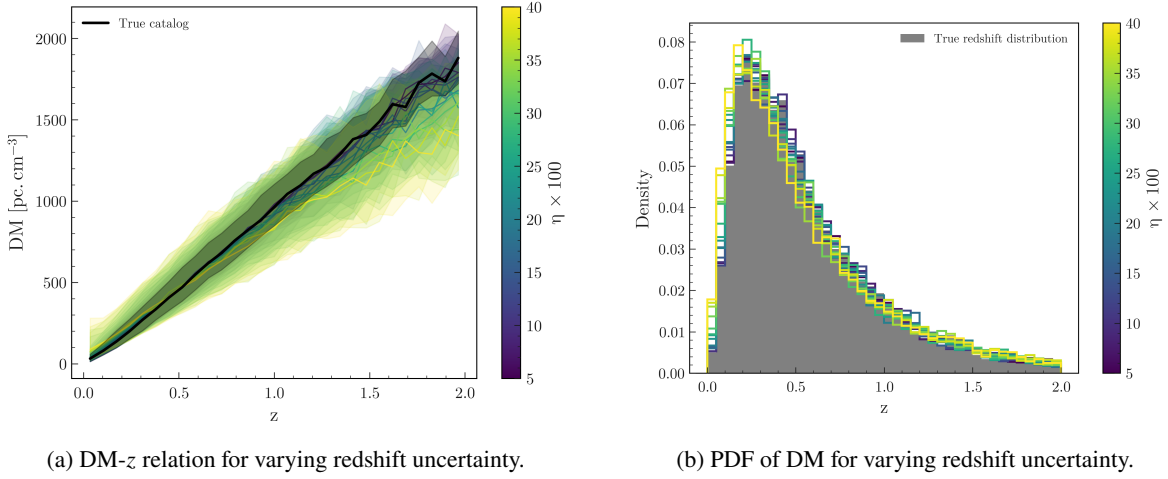


Figure 4.5: DM- z relation and PDF for the redshift uncertainty systematic.

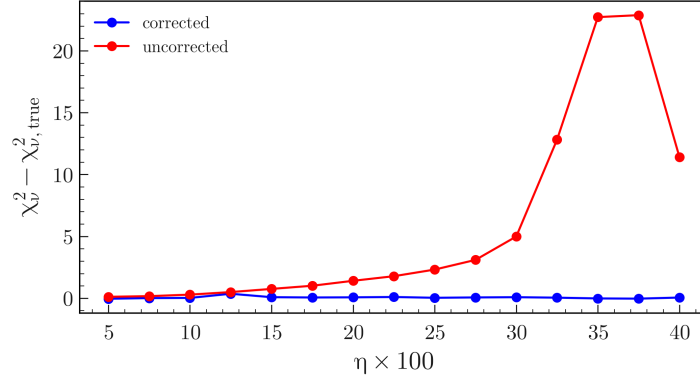
The deviation in χ^2 is shown in Fig. 4.6. The corrected line corresponds to propagating the error to the covariance as described by Eq. 3.29. Meanwhile, the deviation from the fiducial catalog crosses the 1σ level for $\eta \approx 15\%$ and gets up to 4.5σ for $\eta \approx 35\%$. The χ^2 decreases afterwards since the corresponding mean DM- z of the contaminated catalog decreases. Therefore, for a simple scenario of redshift uncertainty, accounting for the redshift uncertainty is important to constrain the DM- z relation and the corresponding parameters.

The redshift uncertainty model discussed in this thesis is a simple model of uncertainty which assumes symmetric error. In other scenarios, it is possible that the redshift error is asymmetric, and the error propagation in this case is nontrivial. Nonetheless, with the simulation based approach, it is possible to integrate any scenario of redshift uncertainty. This is a crucial property when solving inference problem with FRBs detected by different surveys and localised in different methods.

As mentioned in section 2.10, it is possible to localise FRBs by comparing their galactic coordinates with wide galaxy surveys to find possible candidates for a host galaxy with each host candidate having a certain probability to be the host. In this case, propagating the error is not a trivial task either and the redshift uncertainty can be simulated as being sampled from a weighted sum of Gaussian distributions where each distribution represents a candidate for a host galaxy.

4.1.4 Selection Effects

Fig. 4.7 shows the impact of reweighting the PDF with a sigmoid function that represents the telescope selection effects. The sigmoid function modifies the shape of the PDF at large values of DM in addition


 Figure 4.6: χ^2 as a function of the redshift uncertainty parameter η .

to changing the normalisation. The impact is negligible for larger values of DM_{max} since the range of DM values considered in this work are below $\text{DM} \approx 2000 \text{ pc.cm}^{-3}$ since the simulation is limited to sources up to redshifts 2.

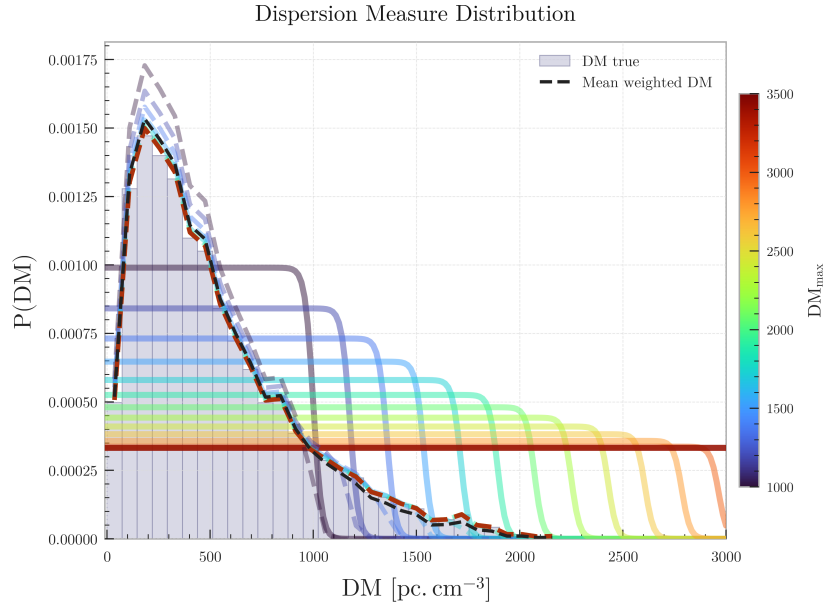


Figure 4.7: PDF of DM for varying selection effect parameters.

As a consequence, the χ^2 shown in Fig. 4.8 shows less than 1σ deviation from the true catalog. By considering sources at higher redshift this effect has more impact on the inferred DM- z relation. Furthermore, I have limited the study to the expected source-redshift distribution expected for the SKOA, however, for a full cosmological analysis the source redshift distribution is modelled and parameterized. The parameters are freed and then marginalized to obtain constraints on the cosmological parameters (Hoyle et al., 2018).

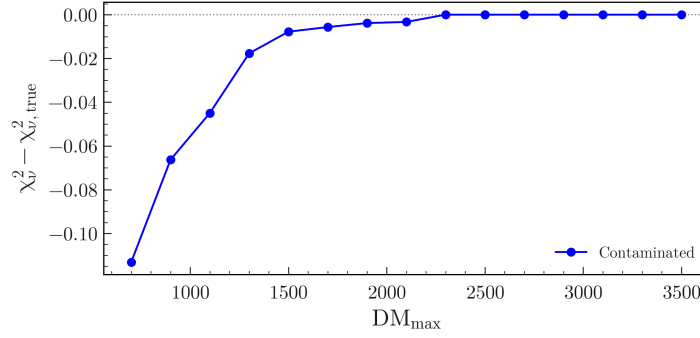
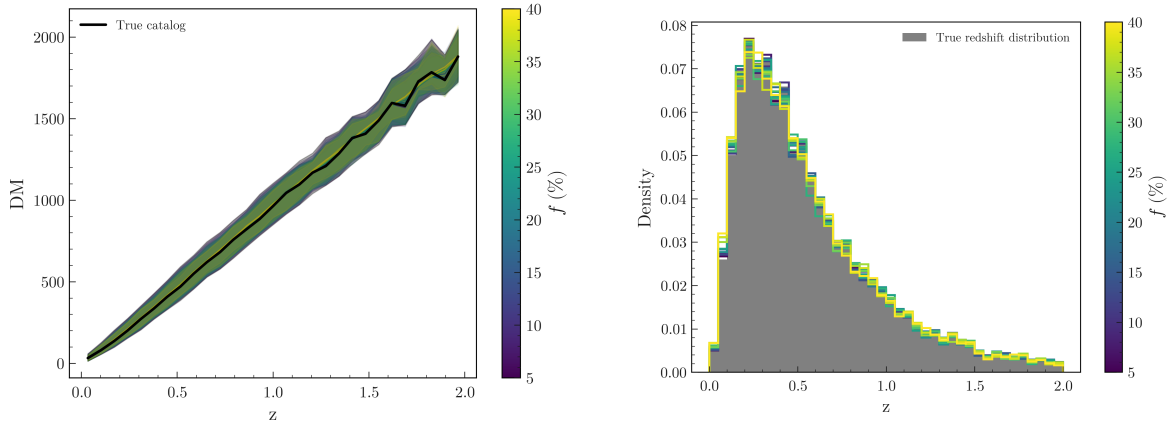


Figure 4.8: χ^2 as a function of the selection effect parameters ($DM_{\max}$, ϵ).

4.1.5 Unlocalised Hosts

In this section, I present the impact of using unlocalised FRBs in a cosmological analysis for three cases: randomly missing redshifts, a fraction of high redshift sources is missing, and a fraction of low redshift sources is missing.

The left-hand side of Fig 4.9 shows the resulting DM- z relation for a varied fraction of unlocalised FRBs f . The mean DM- z relation is consistent with that of the true catalog, but the variance is larger than the true catalog for the entire range of f at higher redshifts. The PDF displayed on the right-hand side of Fig. 4.9 shows a slight change in the width of the PDF which is reflected by the change in the variance of the DM- z relation.



(a) DM- z relation for a random subset of unlocalised hosts.

(b) PDF of DM for a random subset of unlocalised hosts.

Figure 4.9: DM- z relation and PDF for the case of a random subset of unlocalised hosts.

The χ^2 value shown in Fig. 4.10 fluctuates for different values of f since the process of choosing unlocalised FRBs is random. Nonetheless, the deviation from the true catalog ranges between 1σ and 2.5σ .

Fig. 4.11 shows the variation of the DM- z relation and the PDF for a fraction of unlocalised hosts with high redshifts. The left-hand side of the figure shows the mean of the DM- z relation is consistent with the fiducial catalog, but the variance is underestimated. This is due to mapping a measured DM in

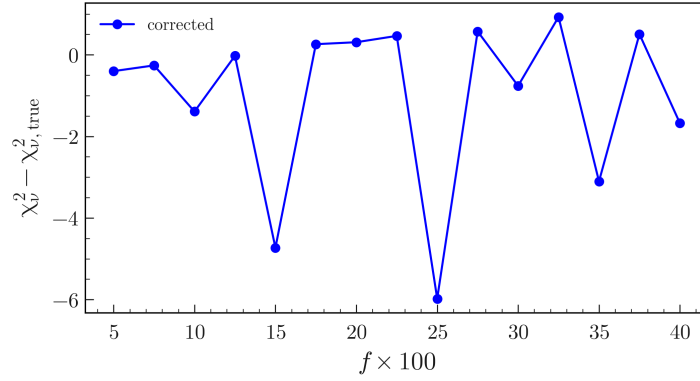
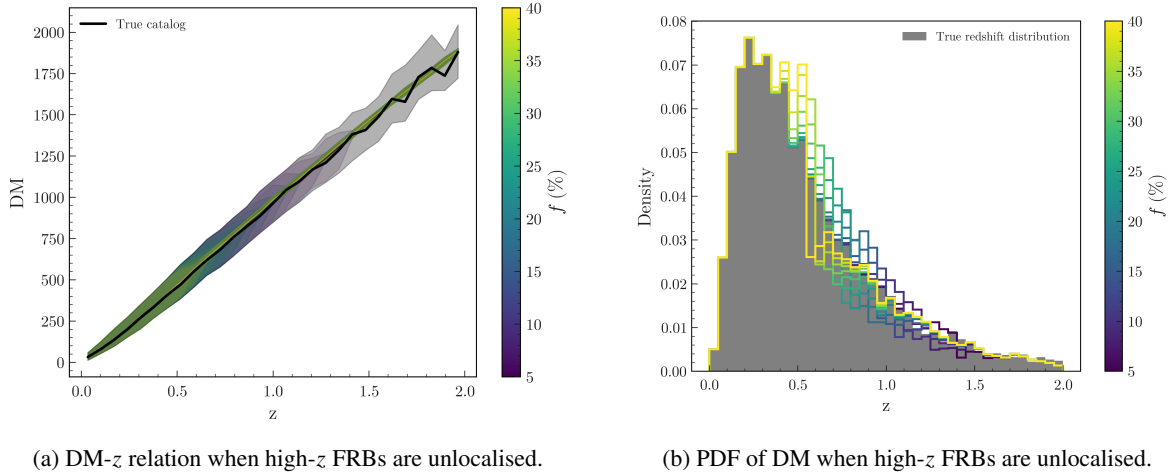


Figure 4.10: χ^2 as a function of the fraction of unlocalised hosts for the random subset case.

the contaminated catalog to the corresponding redshift according to the fiducial mean DM- z relation. This is reflected on the right-hand side plot with the shape of the PDF changing for large values of DM since this corresponds to unlocalised sources at high z .



(a) DM- z relation when high- z FRBs are unlocalised.

(b) PDF of DM when high- z FRBs are unlocalised.

Figure 4.11: DM- z relation and PDF for the case where high-redshift FRBs are unlocalised.

Fig. 4.12 shows a significant deviation from the true value of χ^2 . The deviation reaches 5σ for larger fraction of missing redshifts, but the deviation is only 1σ for $f \approx 25\%$. The deviation is larger compared to the case of randomly unlocalised hosts since sources at high redshifts have a stronger signal since DM is an integrated effect along the line of sight and FRBs from higher redshift correspond to larger DM values.

The left-hand side of Fig. 4.13 shows the variation in the DM- z relation when a fraction of low redshift FRBs has unlocalised hosts. The left-hand side shows a slight deviation of the variance from the variance of the fiducial catalog at low redshift while the mean of the contaminated catalog is consistent with that of the fiducial catalog for the entirety of the redshift range and for varying fraction of the unlocalised hosts. The PDF on the other hand shows a deviation of the peak with larger values of f . In addition the shape of the PDF changes at low values of z . However, the width of the PDF does not change, reflecting

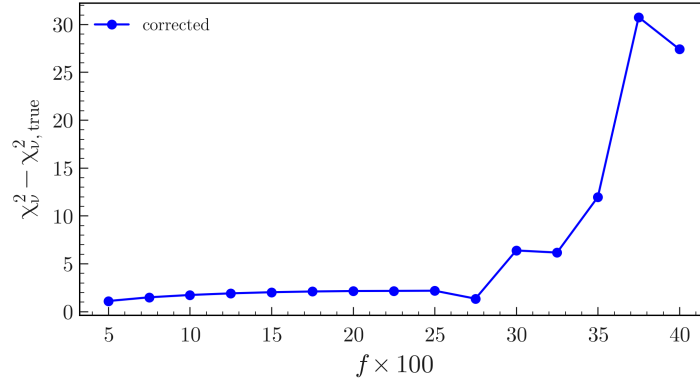
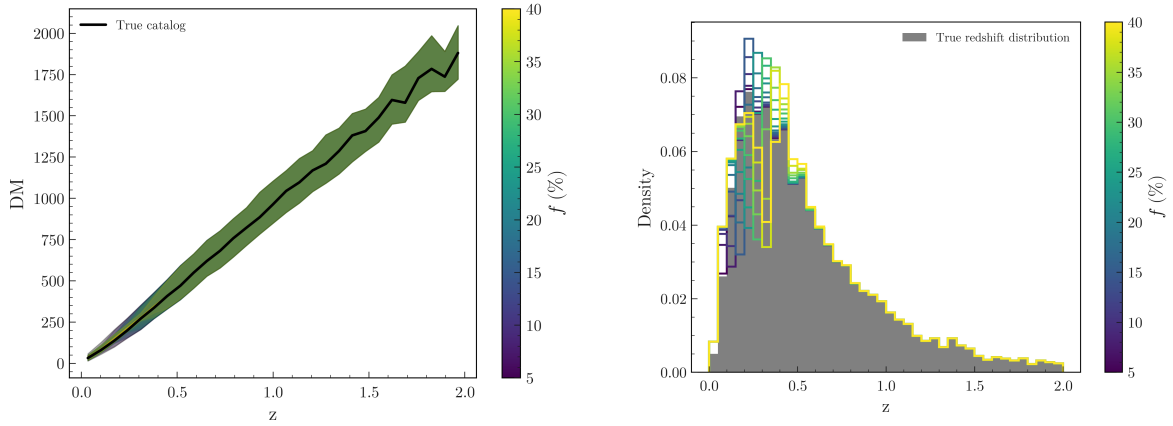


Figure 4.12: χ^2 as a function of the fraction of unlocalised hosts for the high-redshift subset case.

the unchanging variance from the left-hand side plot.



(a) DM- z relation when low- z FRBs are unlocalised.

(b) PDF of DM when low- z FRBs are unlocalised.

Figure 4.13: DM- z relation and PDF for the case where low-redshift FRBs are unlocalised.

The variation of the χ^2 in Fig. 4.14 is consistent with the variation observed from the left-hand side plot of Fig. 4.13 where the deviation due to unlocalised hosts at low redshift does not exceed 1σ for the range of f considered in this study. This can be corroborated by noticing that low redshift sources have a lower signal since the physical distance that the pulse covers is small as compared to high redshift sources.

In total, using unlocalised hosts at high redshifts yields the largest deviation in the DM- z relation while random unlocalised hosts also yields a significant deviation but less than the case for high redshifts. In practice, this can inform a strategy for optical follow-up observation where high redshift sources are more 'rewarding' to observe as compared to low redshift sources. Nonetheless, the redshifts of the sources are not known a priori, however, it can be estimated from the measured value of DM whether the source is at higher or low redshift where larger values of DM indicate that the FRB is more probable to originate at high redshifts.

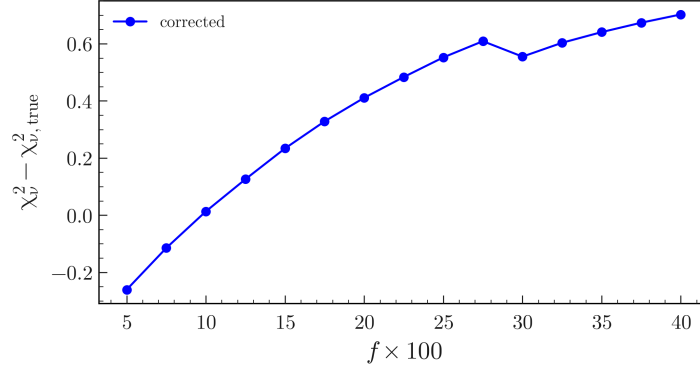


Figure 4.14: χ^2 as a function of the fraction of unlocalised hosts for the low-redshift subset case.

4.2 Results from Simulation Based Inference

In this section, I present the results from SBI for randomly unlocalised FRBs and redshift uncertainty. For the redshift uncertainty I choose a value of $\eta = 0.25$ to contaminate the catalog. For randomly unlocalised hosts, I choose $f = 0.25$. Both numbers are chosen to be intermediate values of the considered range of uncertainty parameters.

Fig. 4.15 shows the parameter constraints using SBI for the case of randomly unlocalised hosts. The posterior is trained for 4 rounds with a total of 6000 simulations. I used MAF architecture with 10 transformations and 40 hidden features with 2 blocks for each individual map. The TARP test was not performed for this case since the resulting posterior is not constraining of the model parameters where for each contour plots the resulting posterior is prior dominated except that of $\Omega_b - h$ where the direction of the degeneracy is recovered. Nonetheless, it is possible to see that the obtained constraints on the rest of the parameters are biased with respect to the fiducial model parameters. This is directly related to what can be observed in the left-hand side of Fig. 4.9 where the variance of the DM- z relation is different from the fiducial, and the bias propagates to the corresponding parameters that control the variance, in particular σ_8 and $\log_{10} M_c$.

Similarly, Fig. 4.16 shows parameter constraints using SBI using a catalog contaminated with redshift uncertainty. In this case, I use MDN with 30 hidden features and 5 components. By looking at the contours of $\Omega_b - h$, the degeneracy between these two parameters is recovered since the redshift uncertainty with $\eta = 0.25$ does not affect the mean DM- z relation as per Fig. 4.5. Meanwhile, since the variance is different as compared to fiducial catalog, the constraints in the $\sigma_8 - \log_{10} M_c$ are biased with respect to the fiducial parameters. In addition, there is a positive degeneracy between σ_8 and $\log_{10} M_c$ since both parameters control the amplitude of the gas power spectrum and therefore the overall amplitude of the variance.

To understand the reasons behind prior domination and unconstrained Ω_c , I ran the After many tests, the optimal architecture to be used in this thesis turned out to be 10 transformations, 50 hidden features, and 2 blocks for each individual map. The parameter constraints are shown in Fig. 4.17. Upon performing the TARP test, the inferred posterior is slightly overconfident as compared to the true parameter posterior as can be seen from Fig. 4.18, but these constraints can be used to improve certain aspects of the inference pipeline in the following way. The parameter degeneracies are recovered both in $\Omega_b - h$ and $\sigma_8 - \log_{10} M_c$. However, the constraints on Ω_c are prior dominated. The reason can be

TSNPE posterior — MAF — Round(s): 0, 1, 2, 3, 4

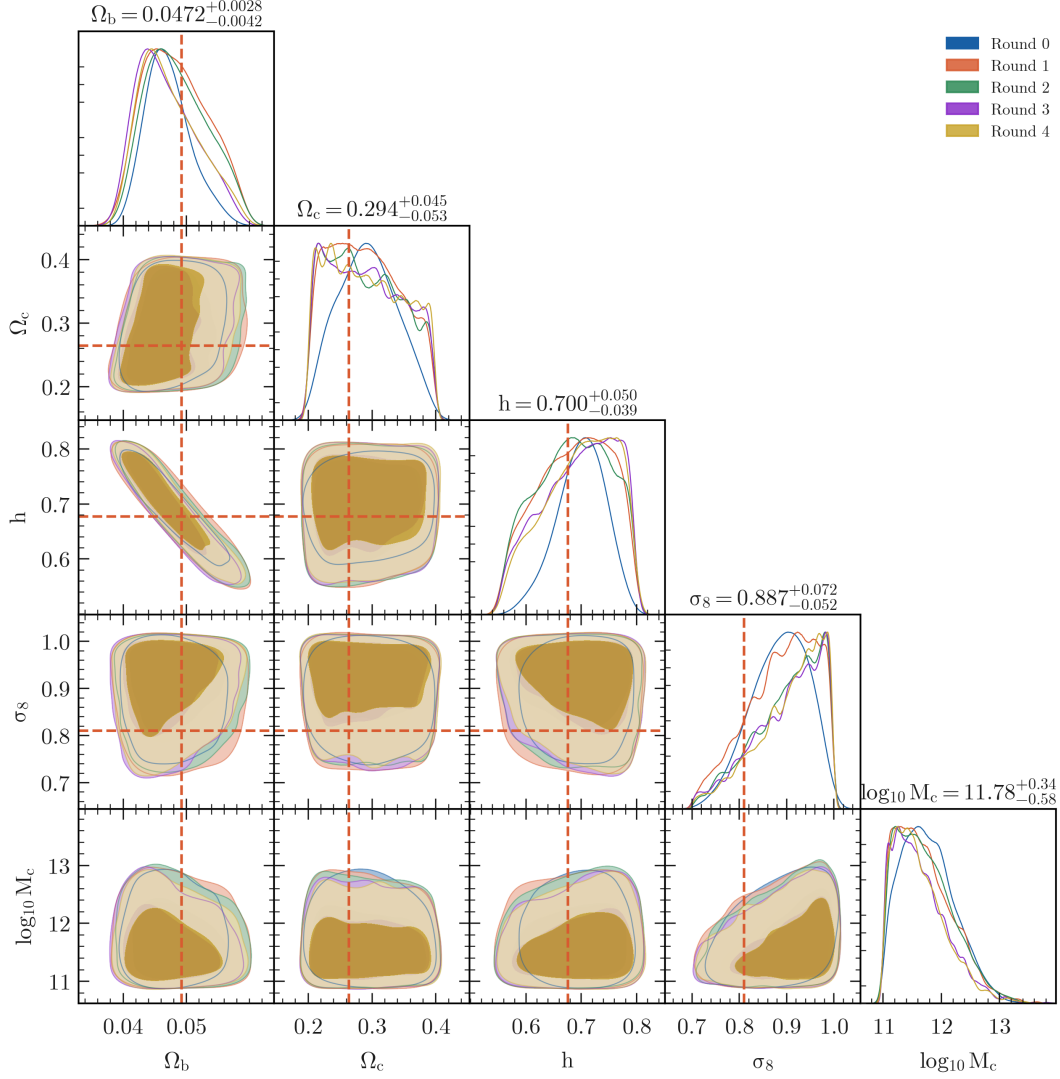


Figure 4.15: Posterior distributions from SBI for the missing redshift scenario.

traced back to the choice of summary statistic where the score compression is only sensitive to the mean which scales with the product $\Omega_b h$ and the scale-independent variance which scales with the amplitude of the power spectrum, with the amplitude of the power spectrum. In the appendix, it is possible to see the variation of power spectrum amplitude with both $\log_{10} M_c$.

In total, the score compression statistic is not sensitive to scale-dependent information. To perform inference with the simulation pipeline, the chosen summary statistic should include scale-dependent information such as the angular power spectrum. With the results given above, it is not possible to give a

TSNPE posterior — MDN — Round(s): 0, 1, 2

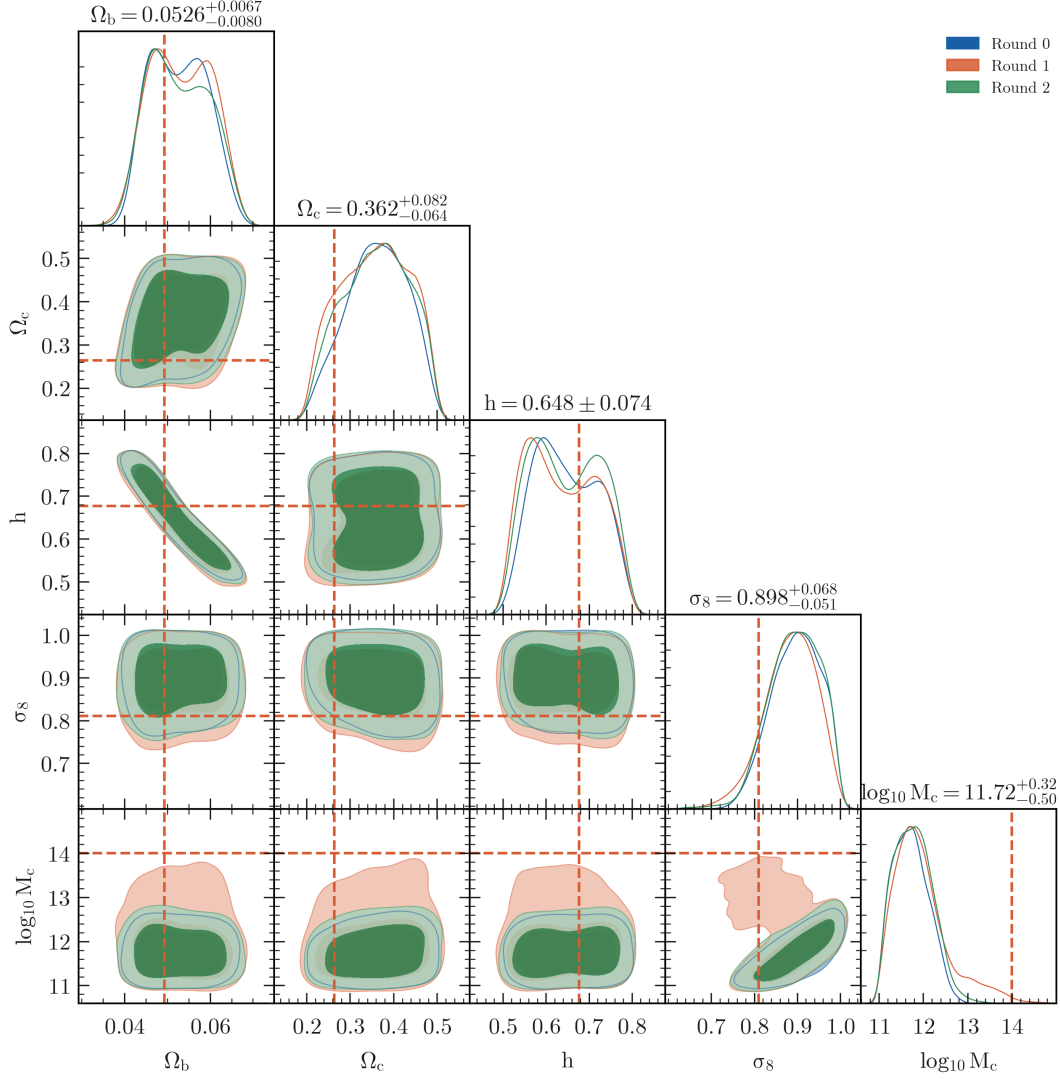


Figure 4.16: Posterior distributions from SBI for the redshift uncertainty scenario.

conclusive answer on the parameter bias that a systematic uncertainty can have. The first step to do that is to choose an appropriate compression statistic which could ultimately have a different dimension as compared to the score compression, leading to a different optimal neural network architecture to obtain a trustworthy posterior. Another choice of a summary statistic is the PDF, binned into a certain number of bins such that to reduce the noise level of the PDF.

TSNPE posterior — MAF — Round(s): 0, 1, 2

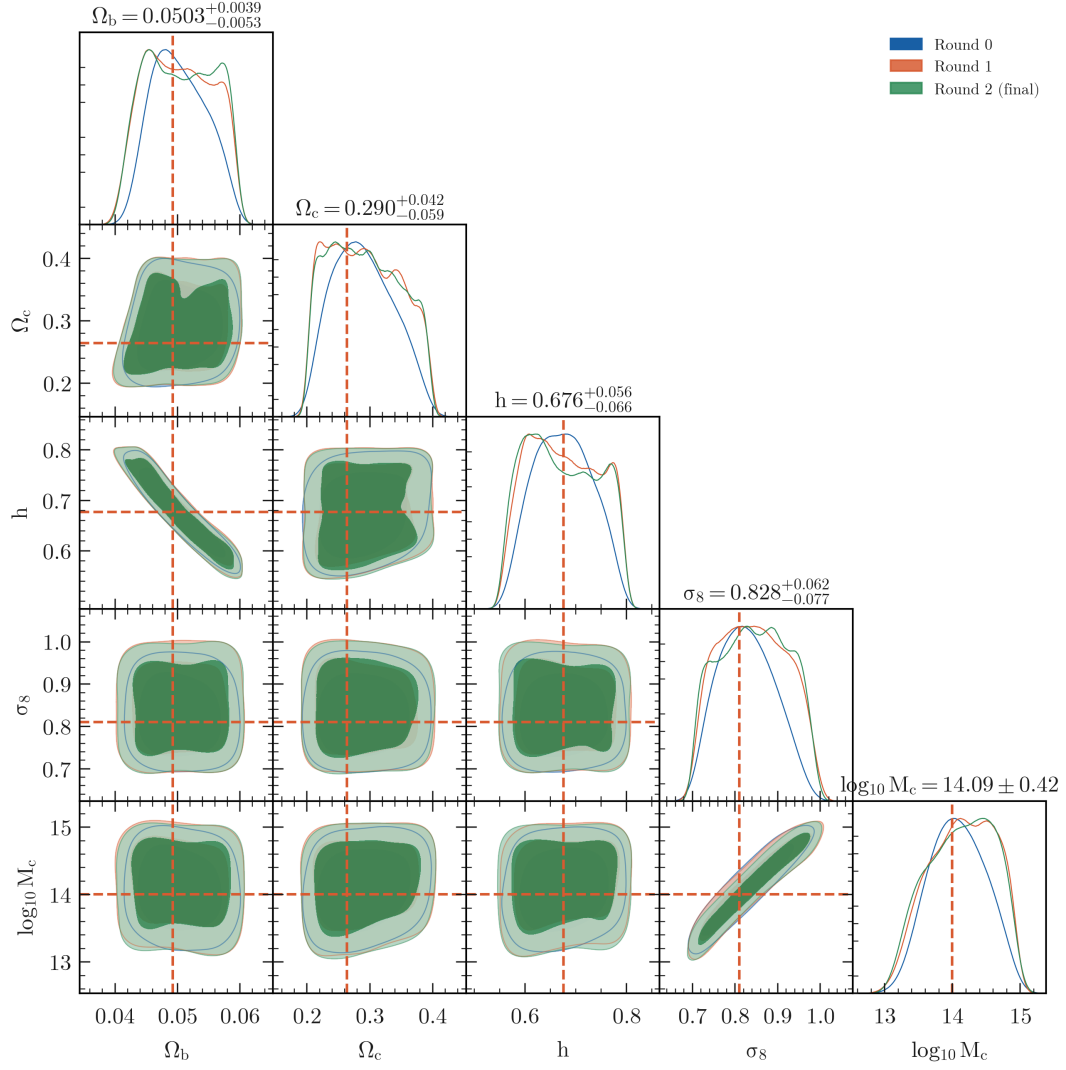


Figure 4.17: Posterior distributions from SBI for the fiducial scenario.

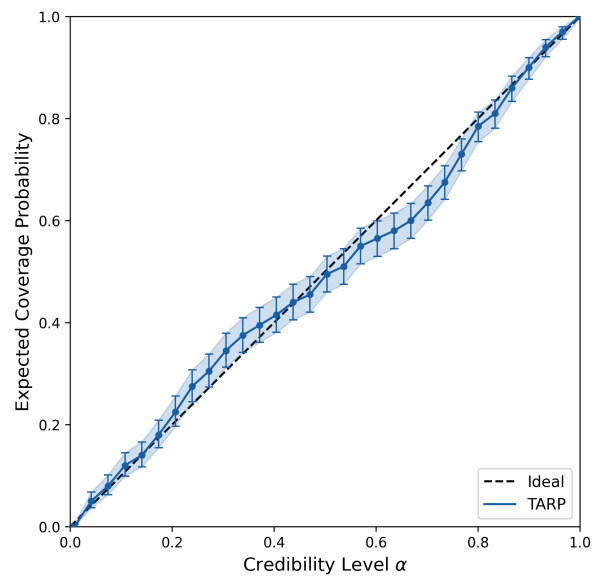


Figure 4.18: TARP coverage test for the fiducial SBI posterior.

Conclusion

talk about how different systematic uncertainties could be degenerate.

talk about more complicated systematic scenarios.

talk about the additional systematic including the Web-halo model.

speeak about how this thesis explore optimistic scenarios of systematic uncertainties.

future work should focus on the studying a combination of systematics all together since there is a degeneracy in

.1 Useful information

- Fourier transform conventions. - Plane wave decomposition into spherical harmonics in the derivation of the cosmological covariance. Look at equation 13 on these lecture notes <https://scipp-legacy.phsci.ucsc.edu/haber/ph215/PlaneWaveExpansion.pdf>. - Definition of the Legendre polynomial.
- Virial theorem. - spherical collapse model and the critical density. - The Dirac delta function. - The Limber approximation. - The 4 velocity of massless particles (necessary for the derivation of the dispersion measure expression)

Sensitivity of the Gas Power Spectrum to Parameter Variations

I present the sensitivity to the four cosmological parameters: the baryon density Ω_b , the cold dark matter density Ω_c , the amplitude of matter fluctuations σ_8 , and the reduced Hubble constant h , followed by the baryonic feedback parameters.

Cosmological Parameters

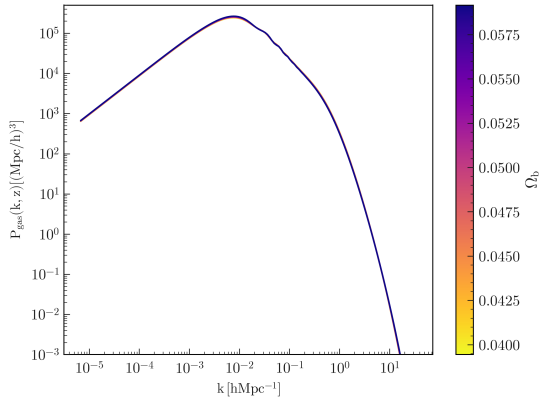


Figure A.1: Sensitivity of the gas power spectrum to the baryon density parameter Ω_b . Higher baryon fractions increase the overall gas content, boosting power on all scales.

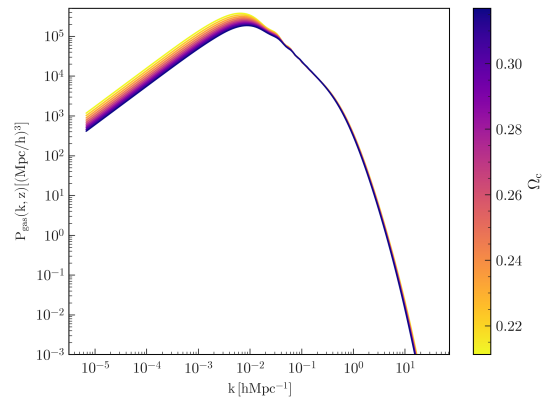


Figure A.2: Sensitivity of the gas power spectrum to the cold dark matter density Ω_c . Changes in Ω_c alter the depth of the gravitational potential wells that drive gas clustering.

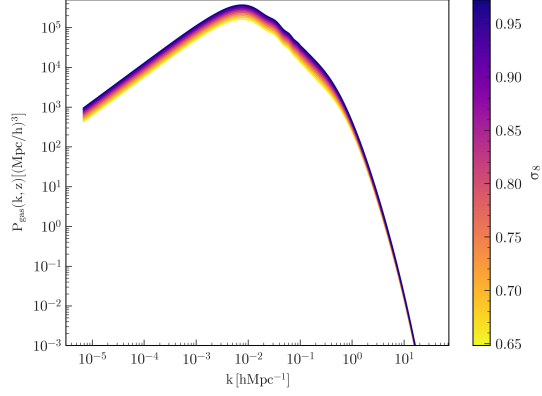


Figure A.3: Sensitivity of the gas power spectrum to σ_8 , the RMS amplitude of linear matter fluctuations on $8 h^{-1}$ Mpc scales. Larger σ_8 enhances structure formation and increases power at small scales.

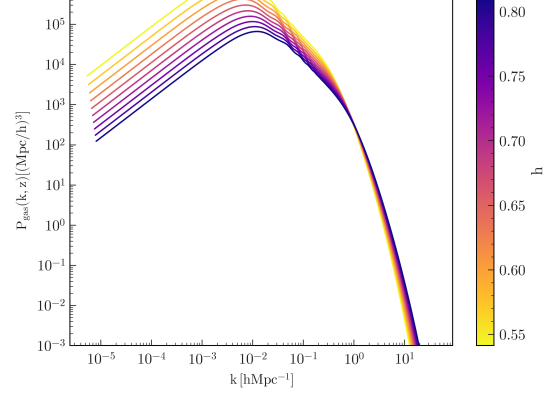


Figure A.4: Sensitivity of the gas power spectrum to the reduced Hubble constant h . Variations in h shift the characteristic scales of the power spectrum and affect the conversion between comoving distances and redshifts.

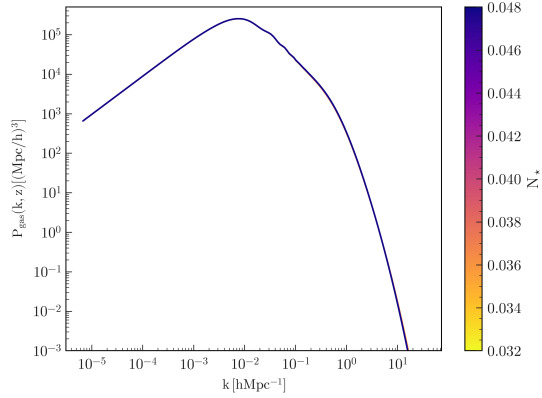


Figure A.5: Sensitivity of the gas power spectrum to the stellar feedback parameter $N_\star$, which controls the efficiency of star formation and the associated stellar-driven gas ejection.

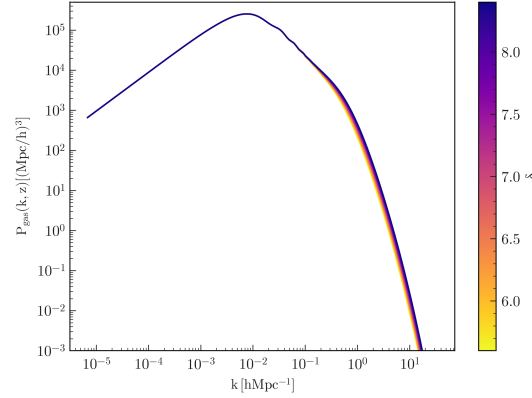


Figure A.6: Sensitivity of the gas power spectrum to the slope parameter δ , which governs the inner profile of the gas distribution within haloes.

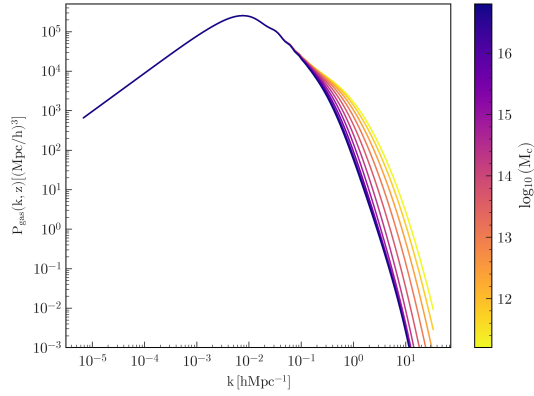


Figure A.7: Sensitivity of the gas power spectrum to $\log_{10} M_c$, the characteristic halo mass scale for AGN feedback. Haloes above M_c expel their gas more efficiently, suppressing power at small scales.

Astrophysical Parameters

A.1 Numerical derivative stability of the covariance matrix

A.2 The Sensitivity of compressed statistics to model parameters

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